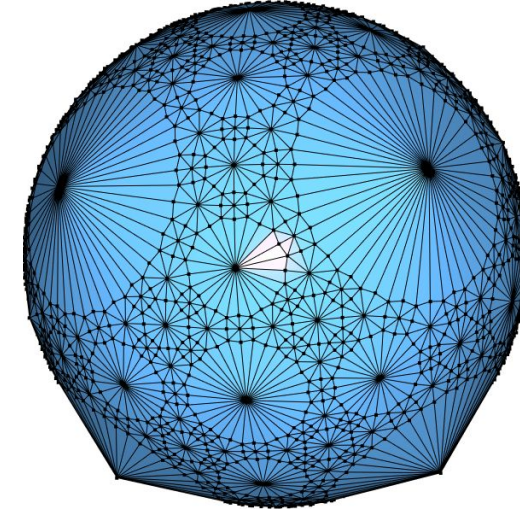
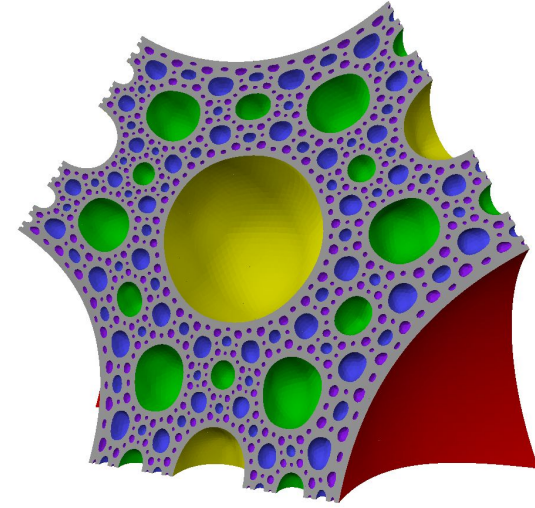
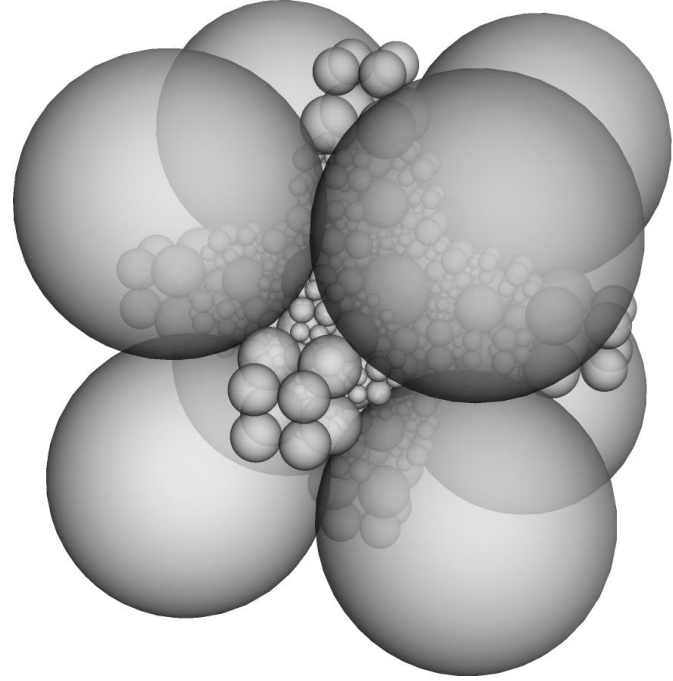




Conception de topologies fractales à partir d'empilements de sphères

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Boris Bordeaux¹, Iván Rasskin², Christian Gentil¹, Lionel Garnier¹

¹*LIB – Université de Bourgogne*

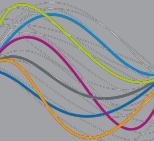
²*amU – Aix-Marseille Université*

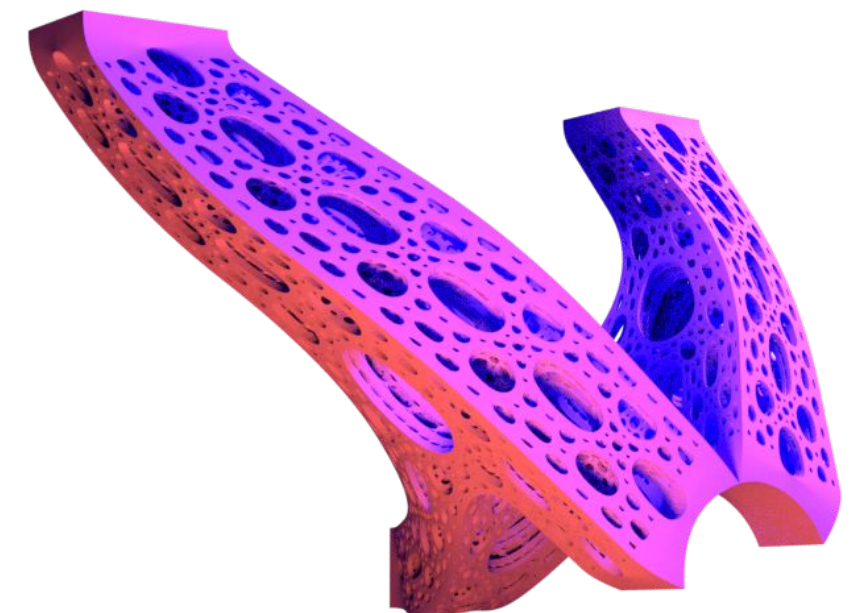
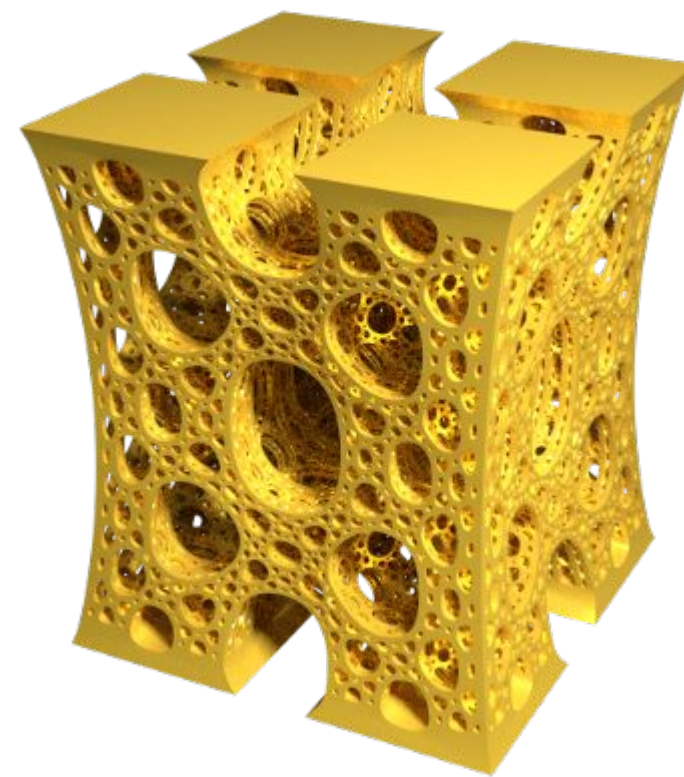
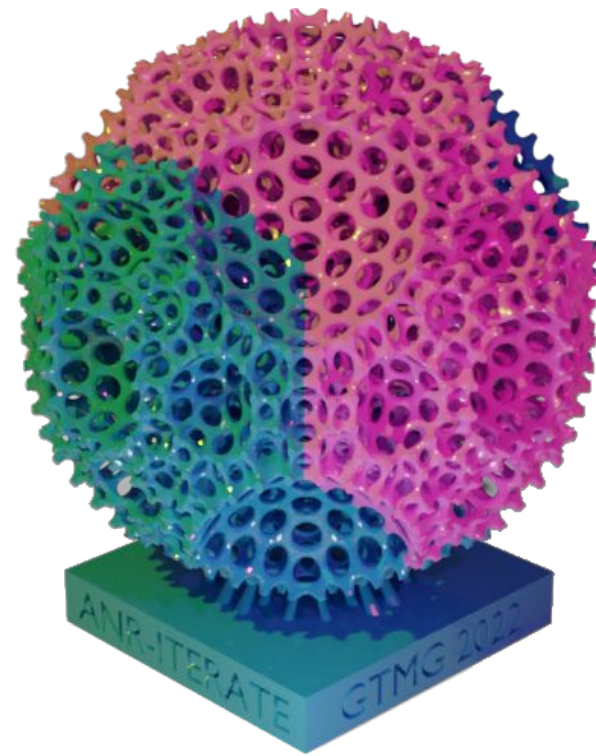
Projet ANR ITERATE – ANR-20-CE46-0011 (2021-2025)

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24 Octobre 2024, Geometry & Computing (GTMG)

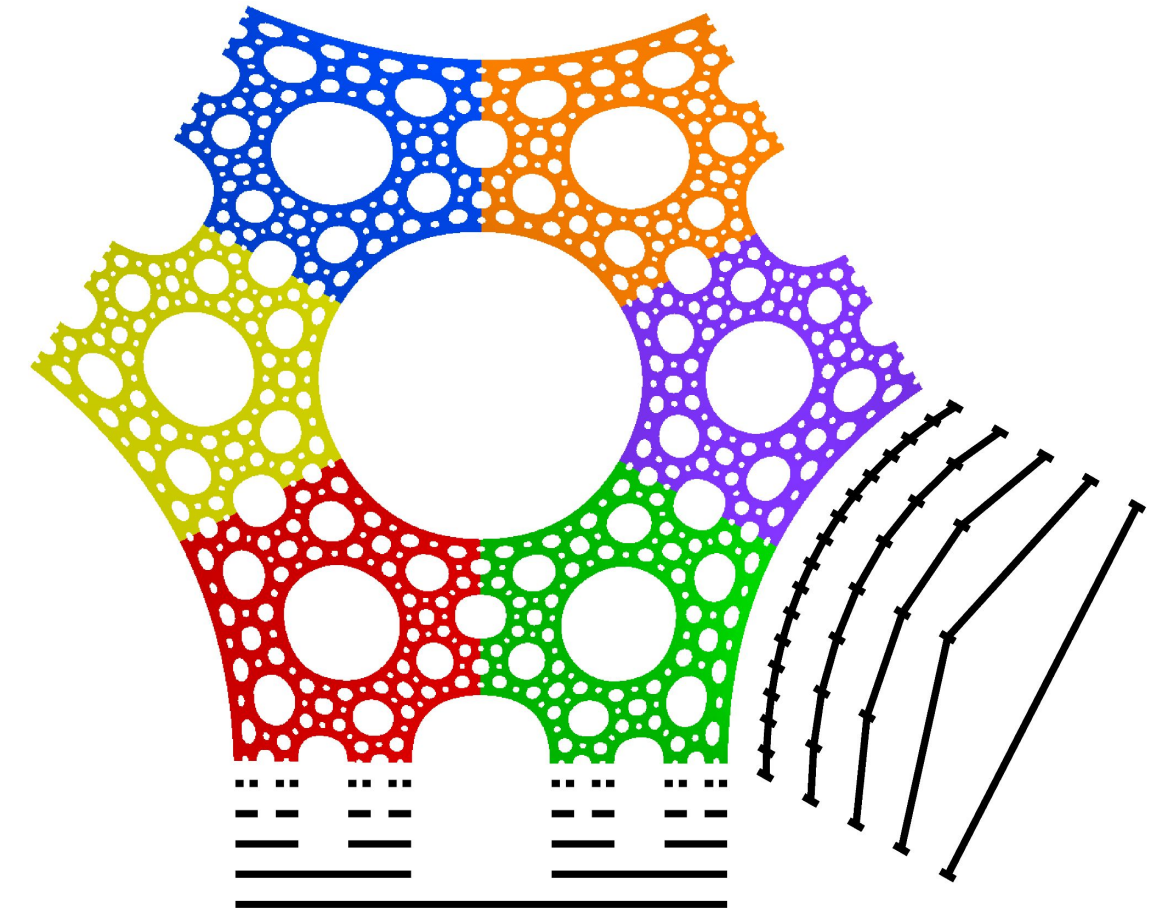
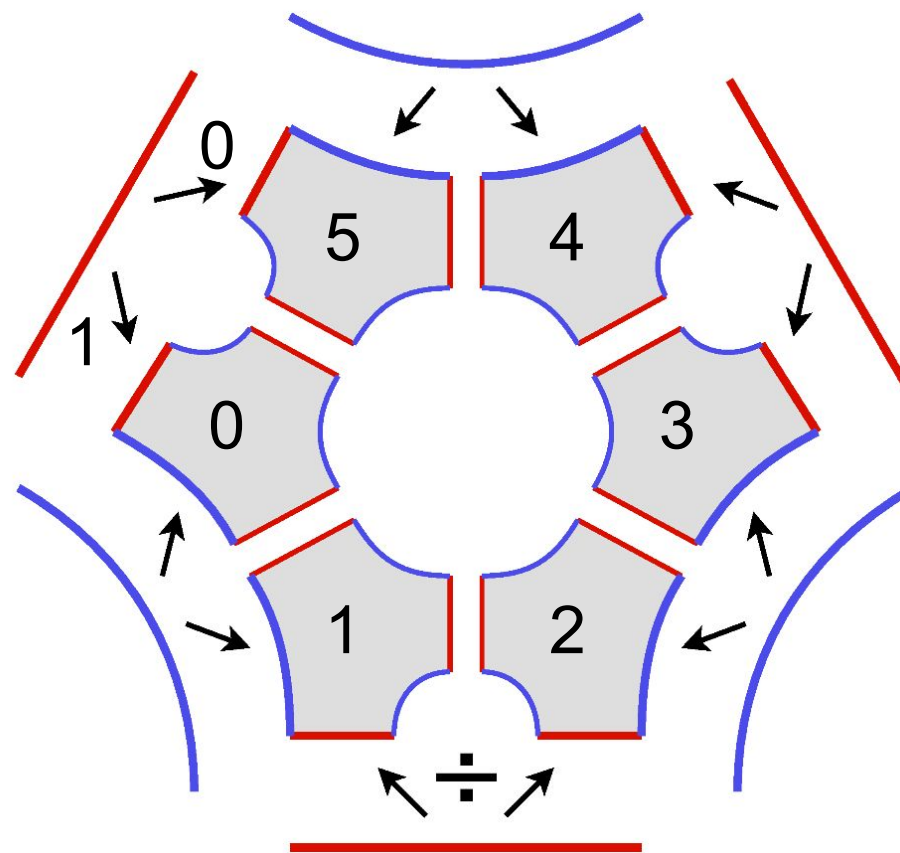
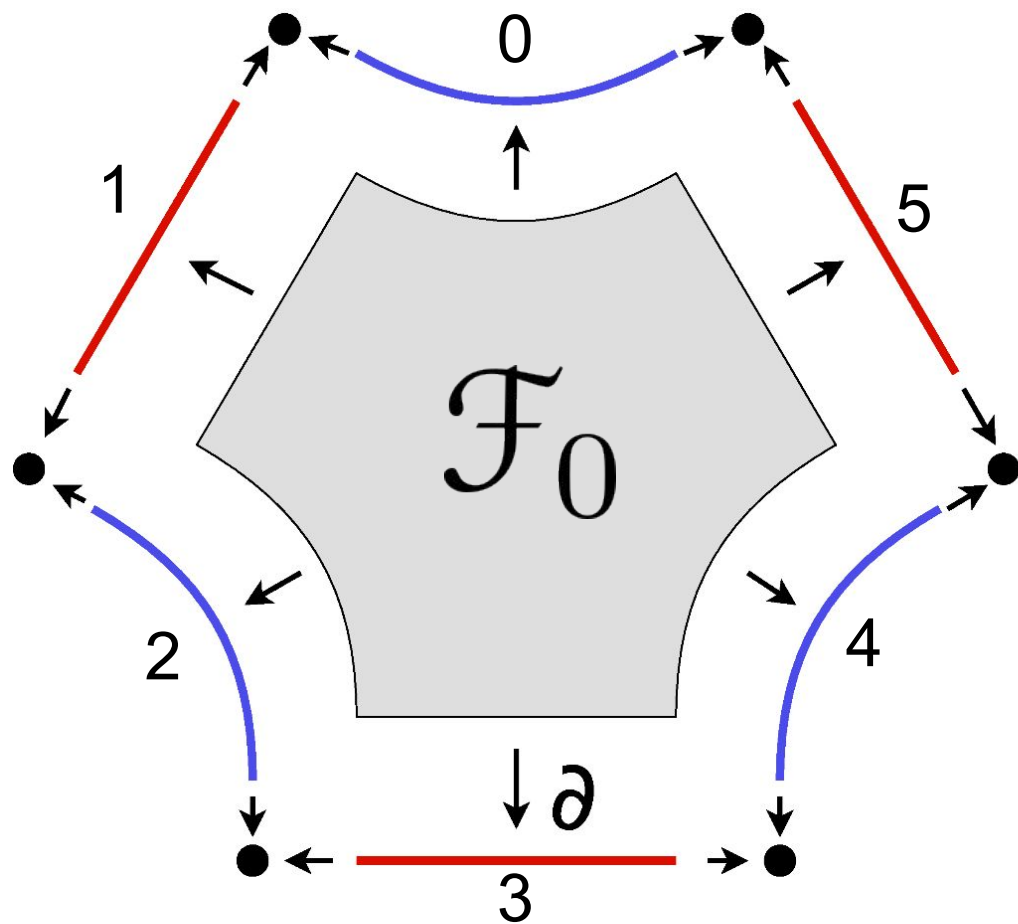

COLLÈGE DOCTORAL
VOTRE DOCTORAT
EN BOURGOGNE-
FRANCHE-COMTÉ

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BOURGOGNE FRANCHE-COMTÉ 



■ BC-IFS

- Décomposition cellulaire
- Contrôle de la topologie fractale
- Topologie cohérente

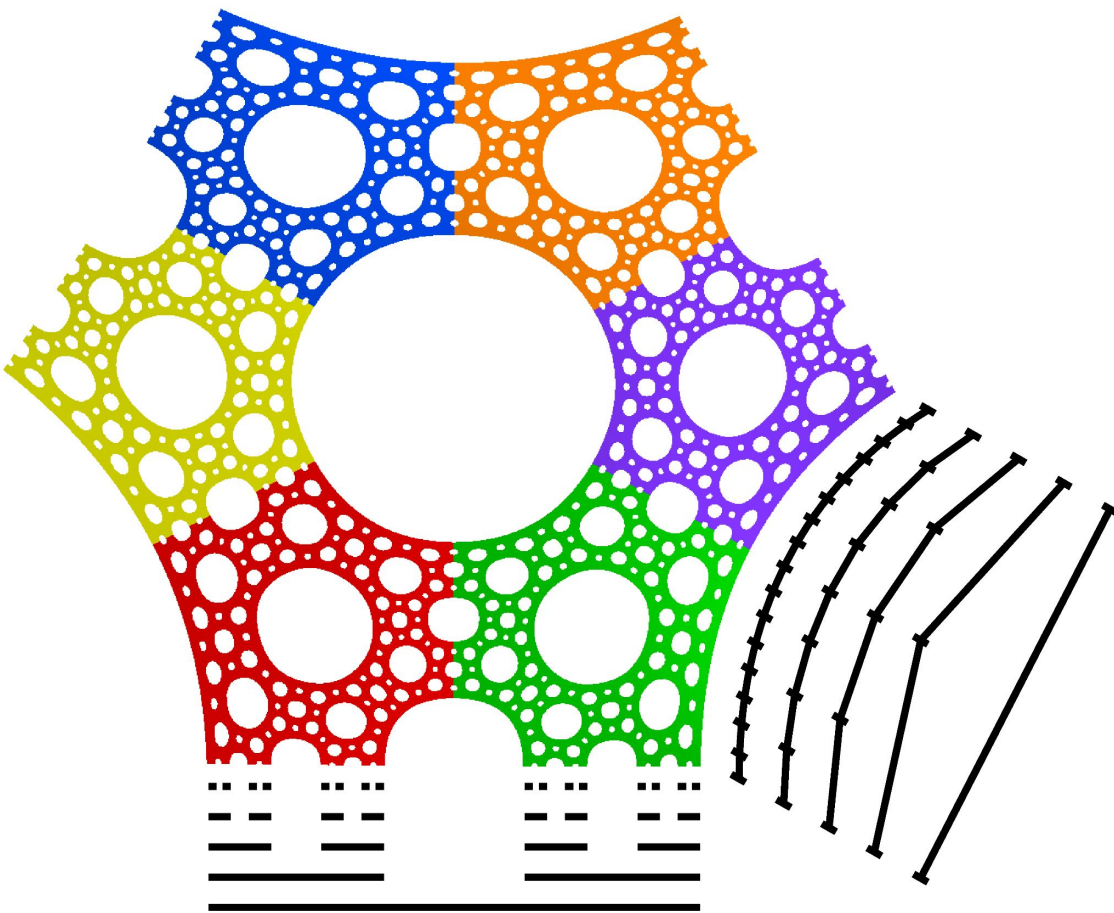
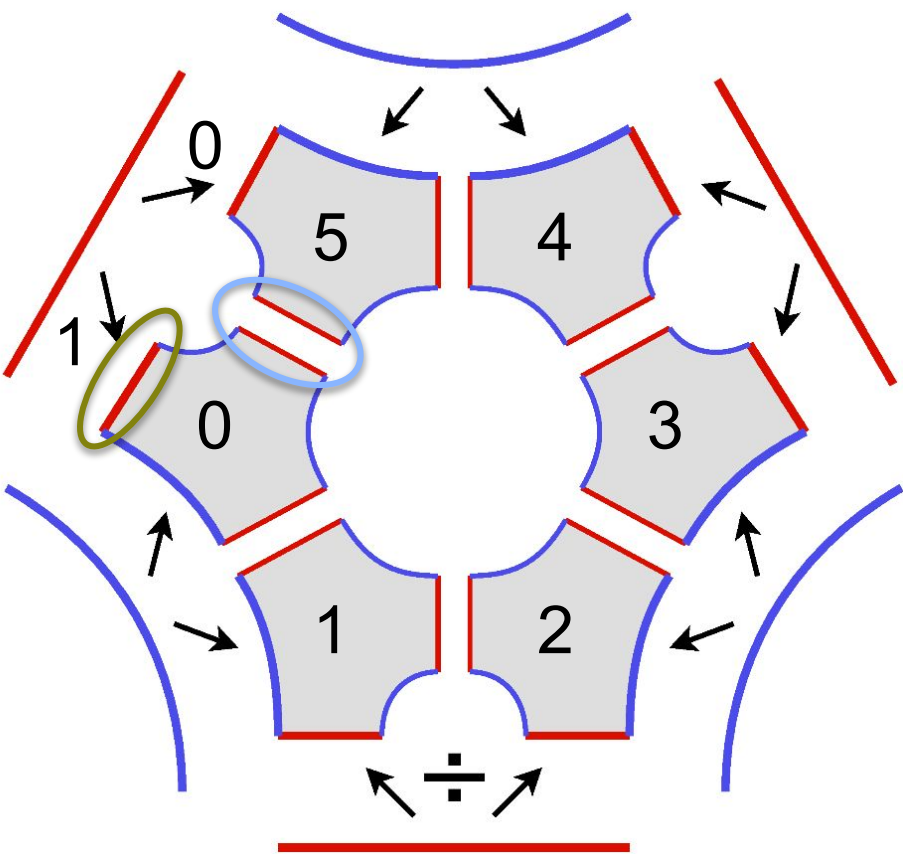
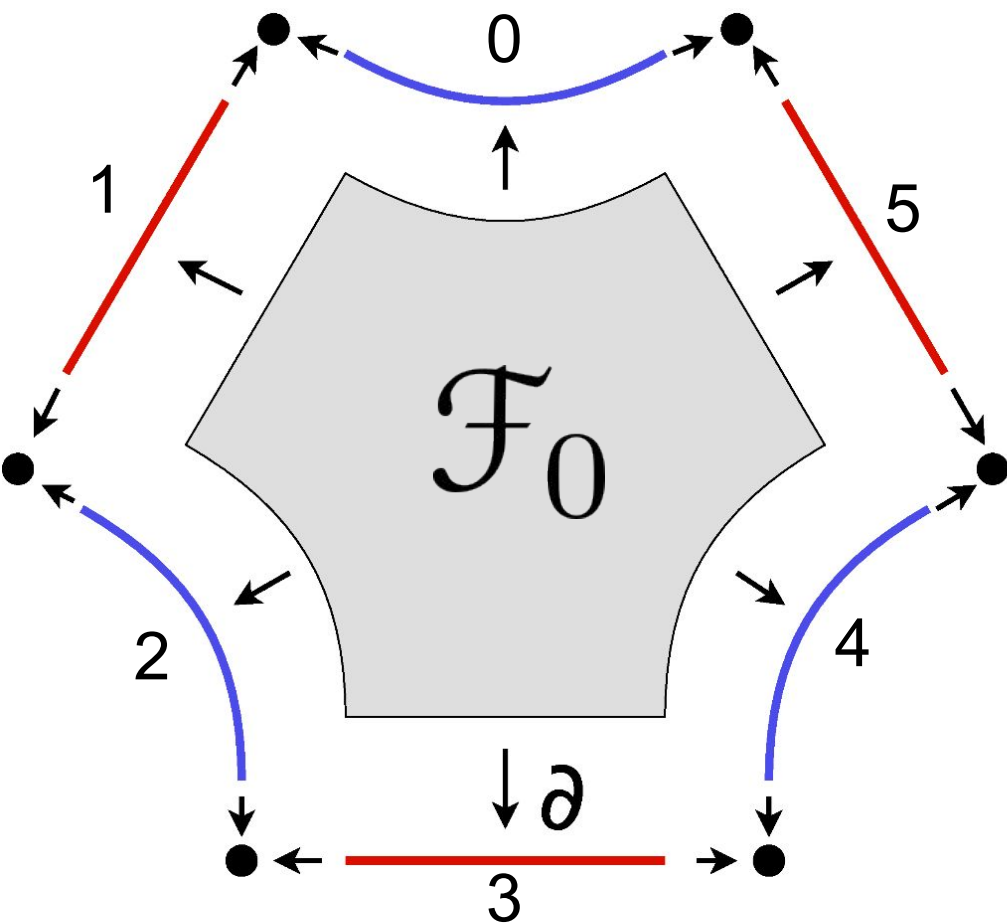


■ BC-IFS

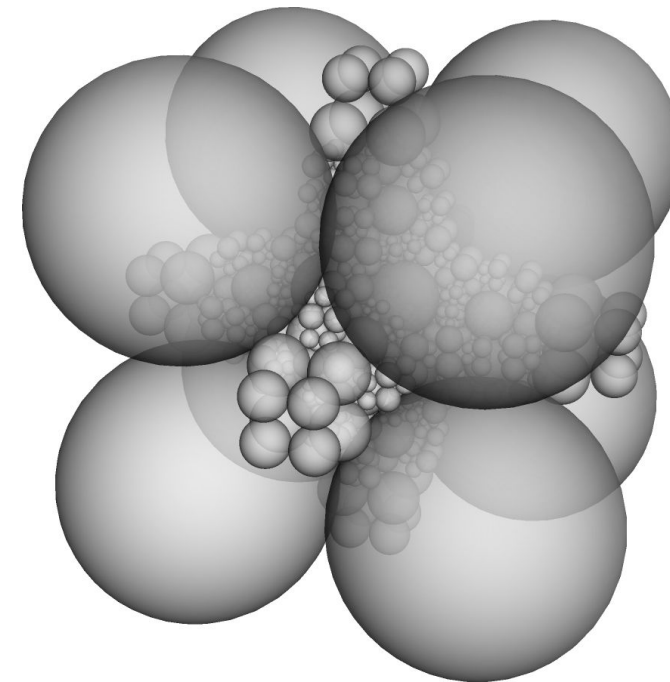
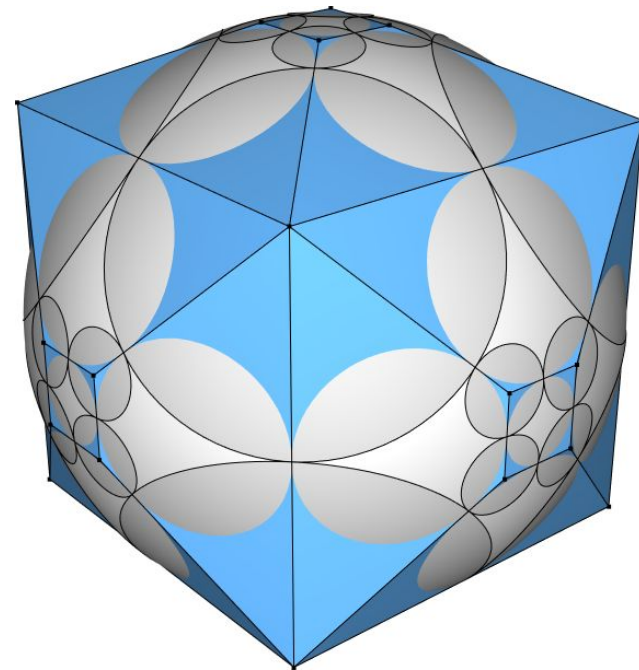
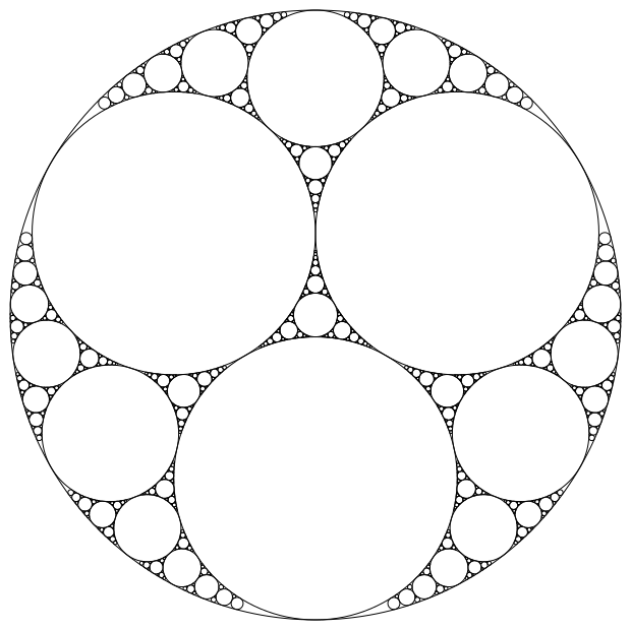
- Décomposition cellulaire
- Contrôle de la topologie fractale
- Topologie cohérente

$$\div_0^{\mathcal{F}_0} \partial_5^{\mathcal{F}_0} = \div_5^{\mathcal{F}_0} \partial_3^{\mathcal{F}_0}$$

$$\div_0^{\mathcal{F}_0} \partial_1^{\mathcal{F}_0} = \partial_1^{\mathcal{F}_0} \div_1^{C_2}$$



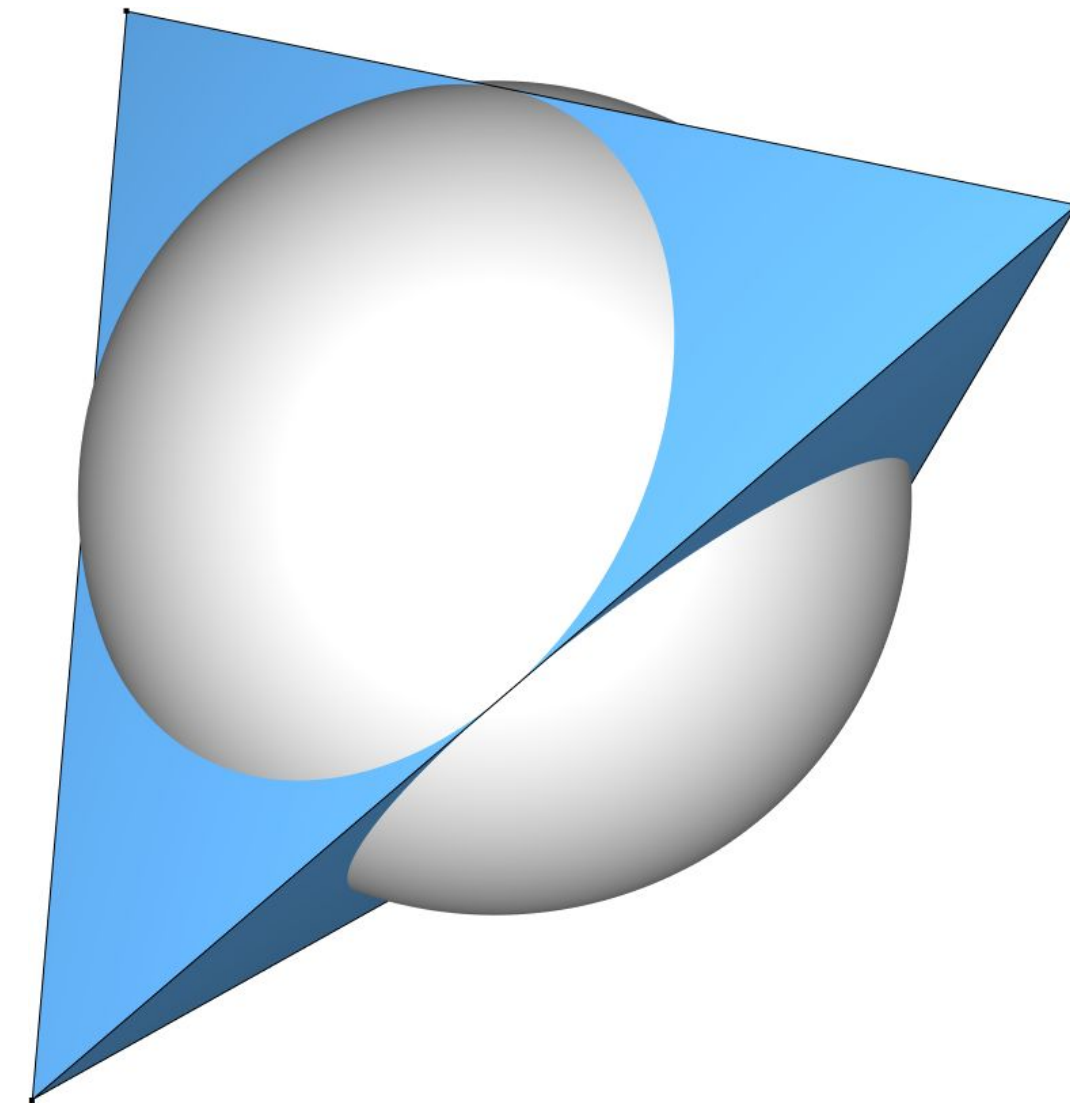
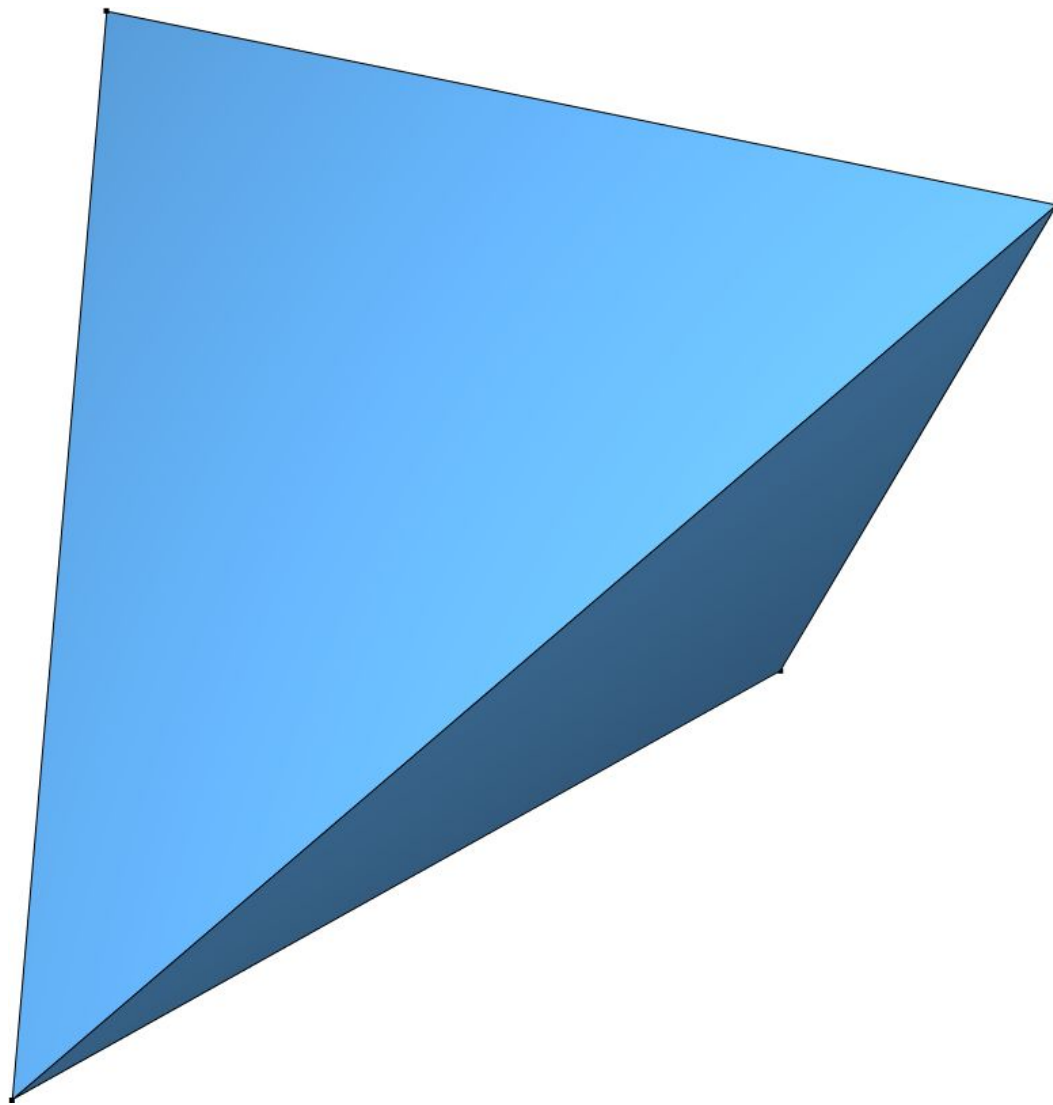
Exploitation des empilements polytopaux Apolloniens de sphères



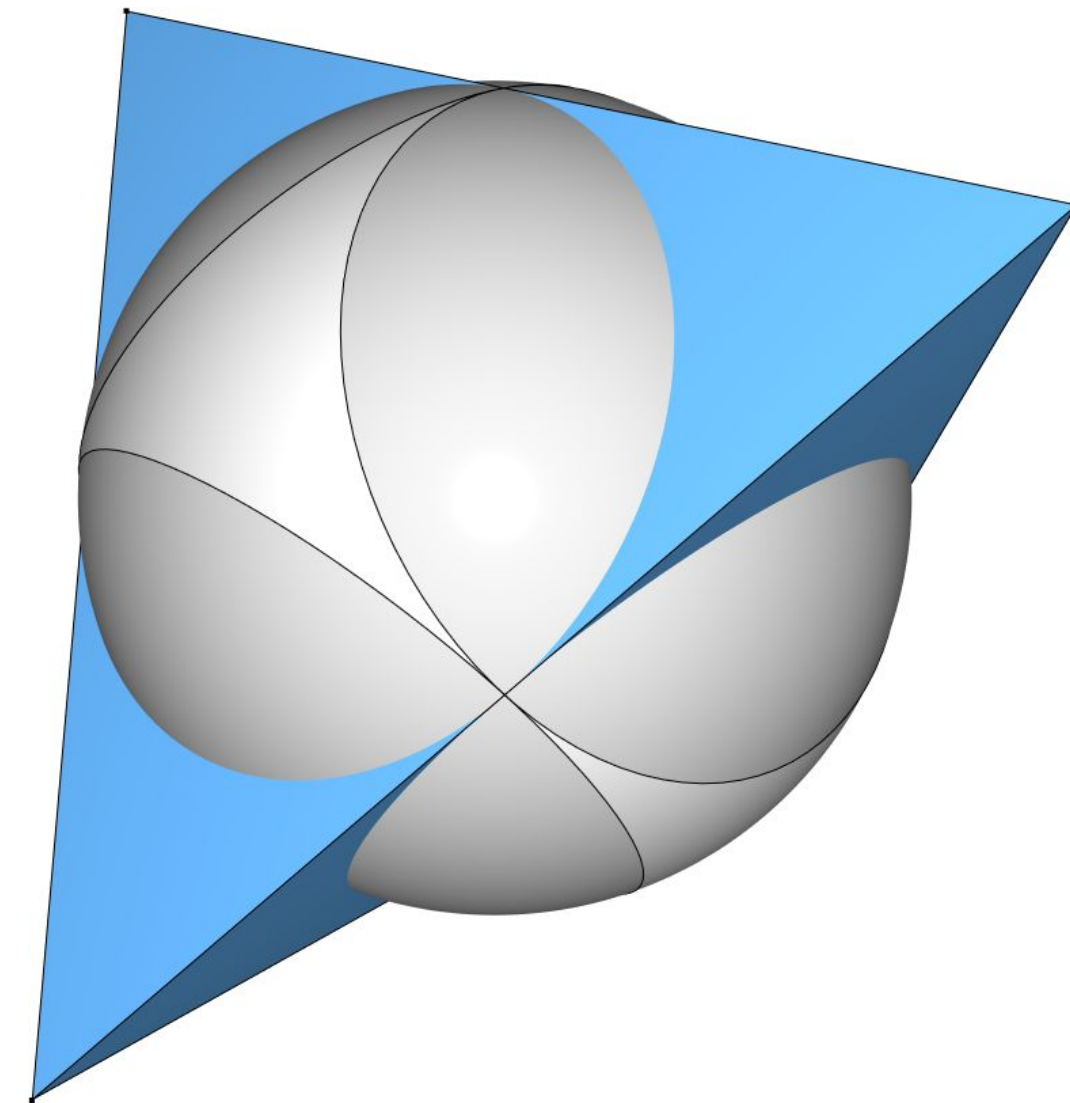
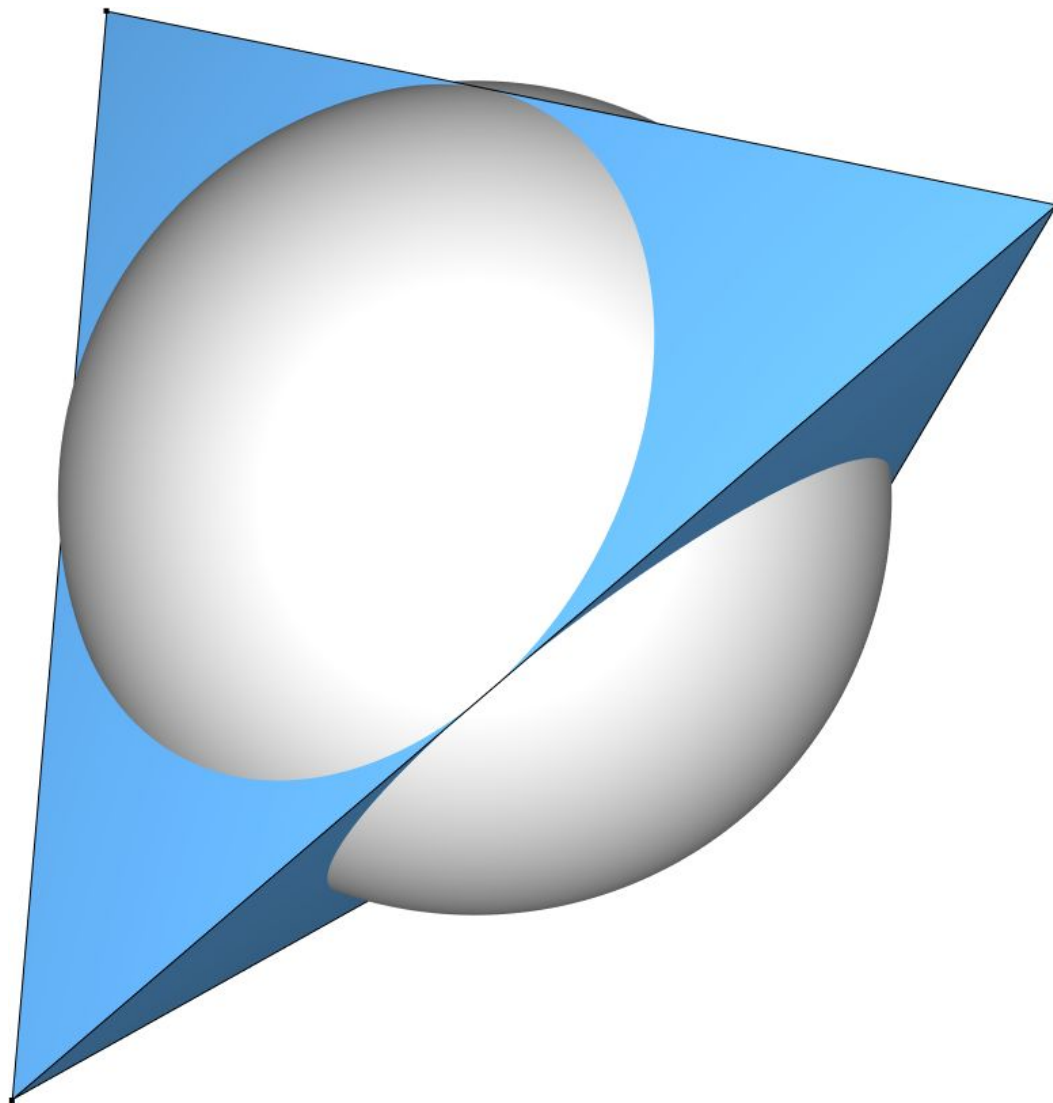
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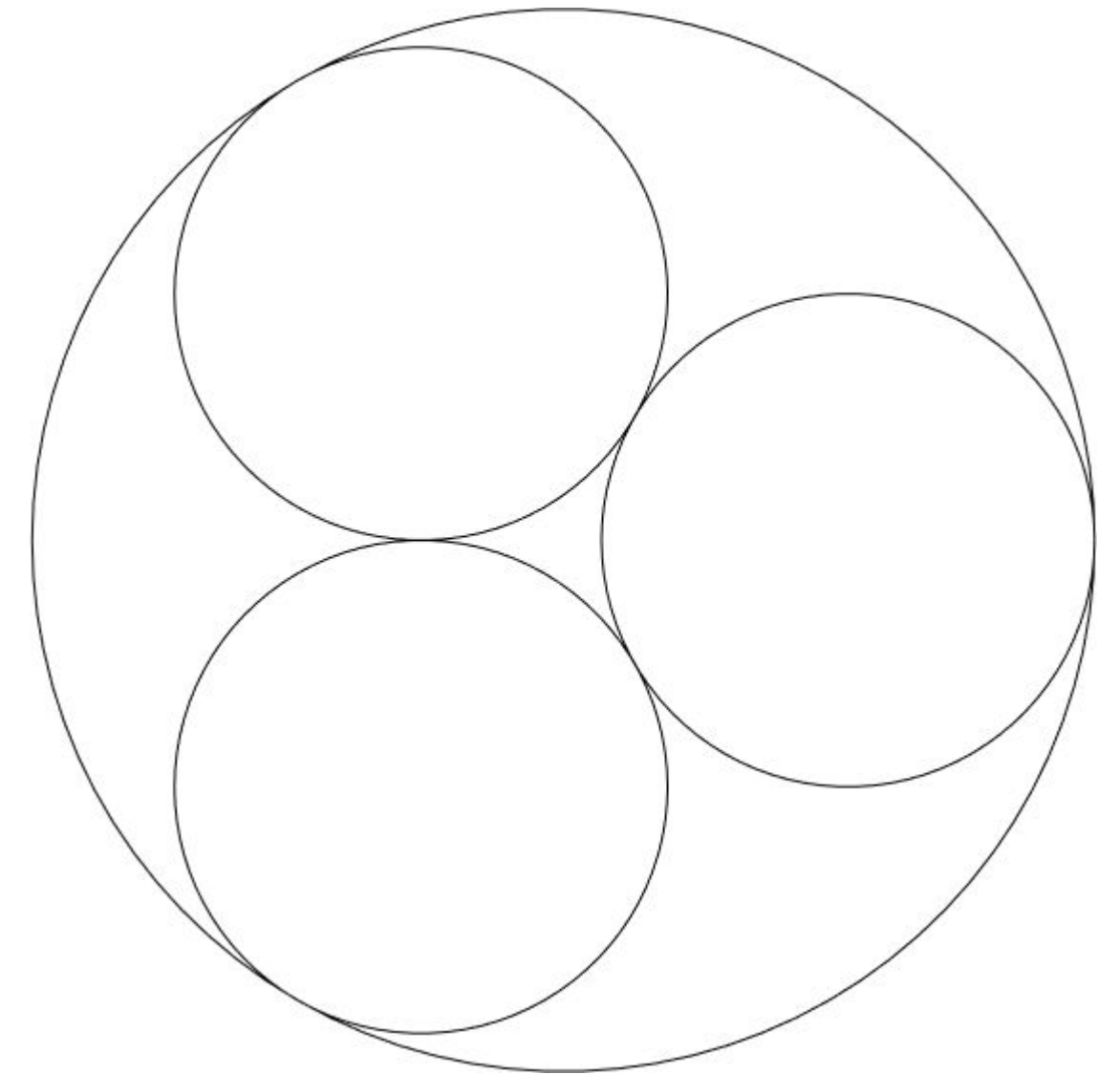
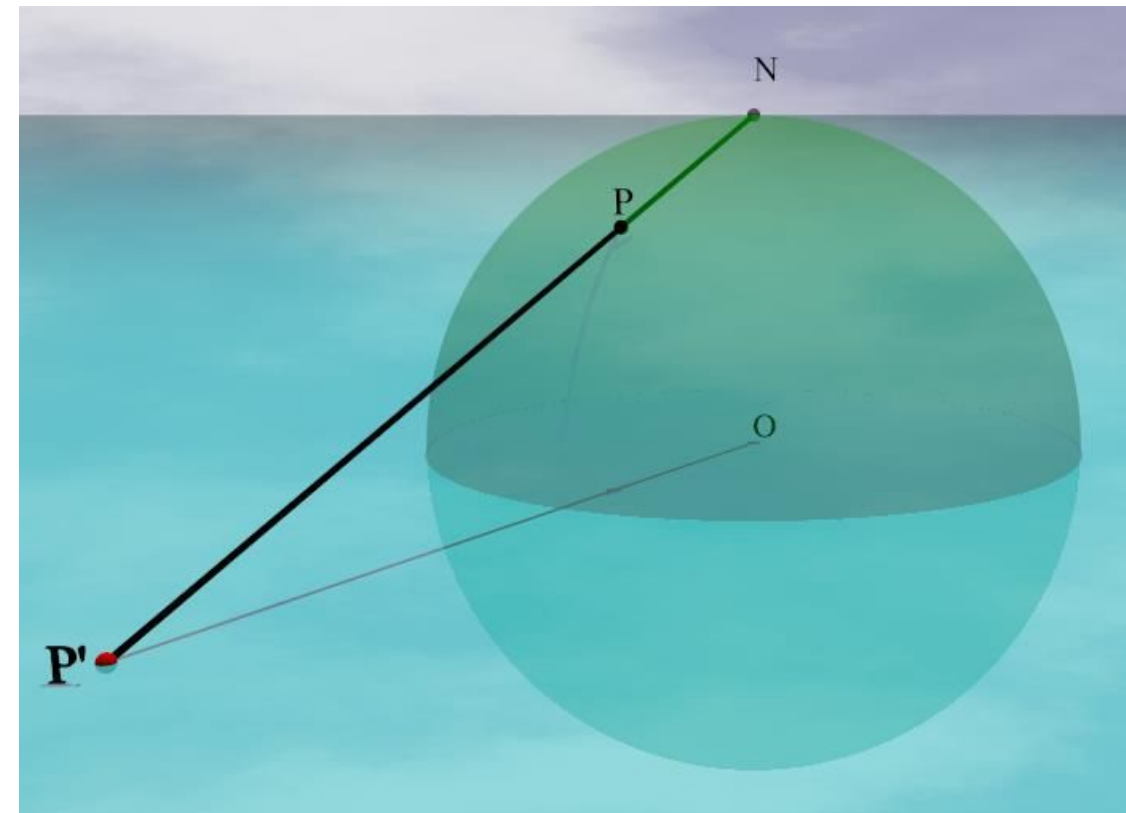
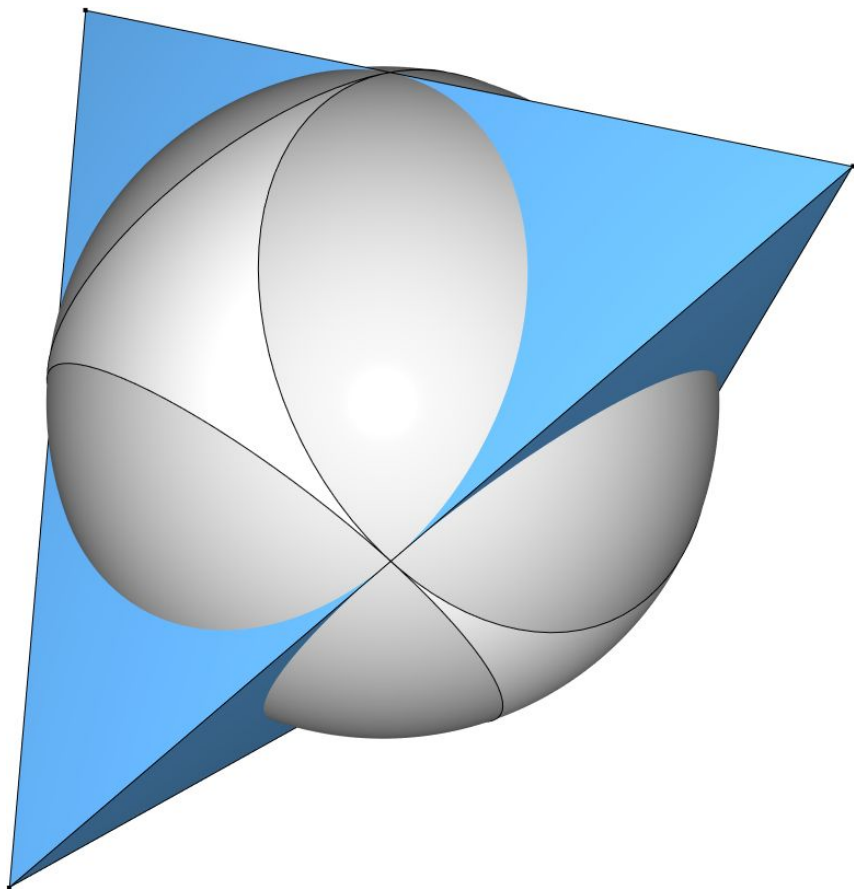
- Polyèdre arête-inscrit
 - Tangent à la sphère unité en toutes ses arêtes



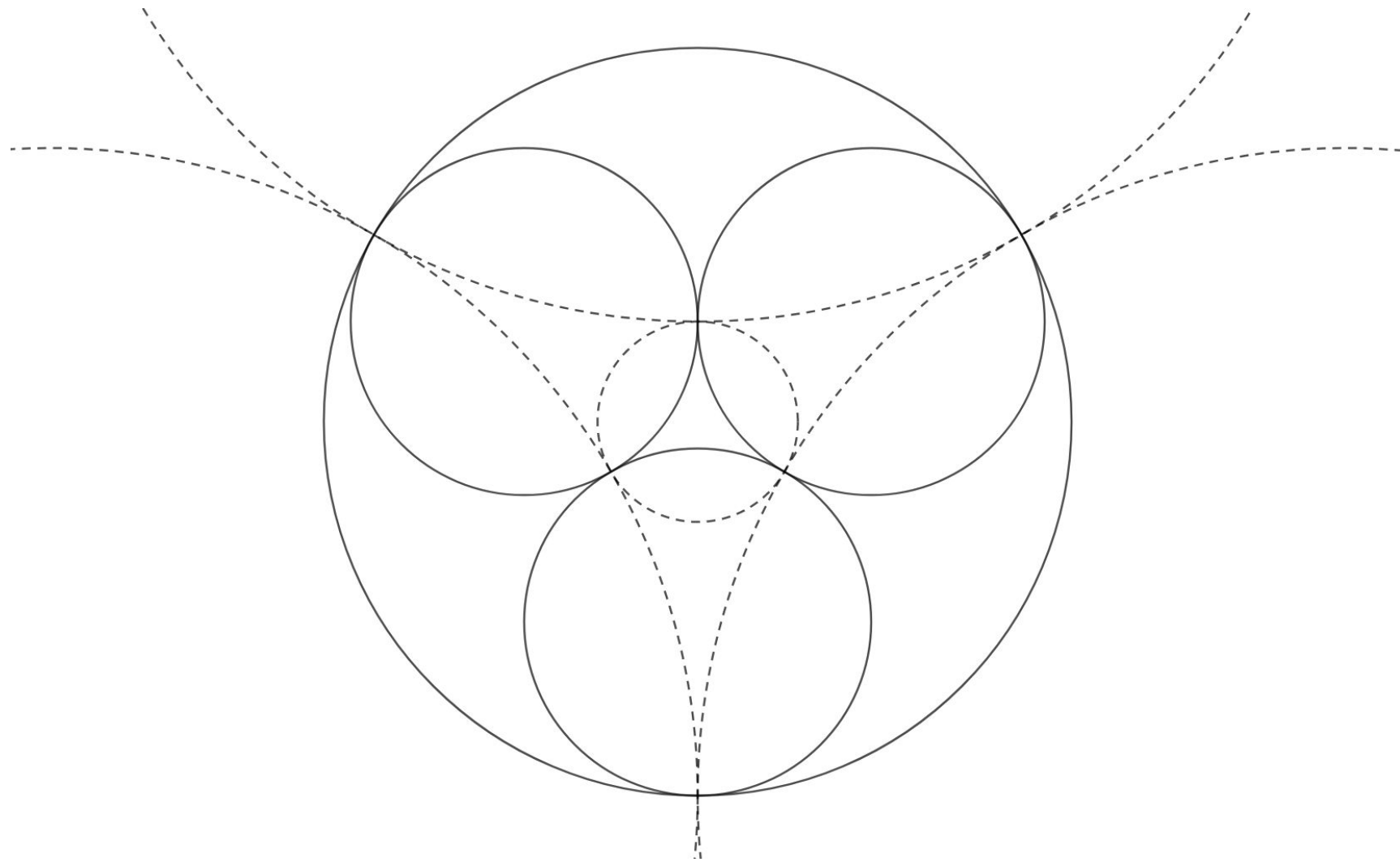
- Zones illuminées
 - Éclairées par des lumières aux sommets du polyèdre



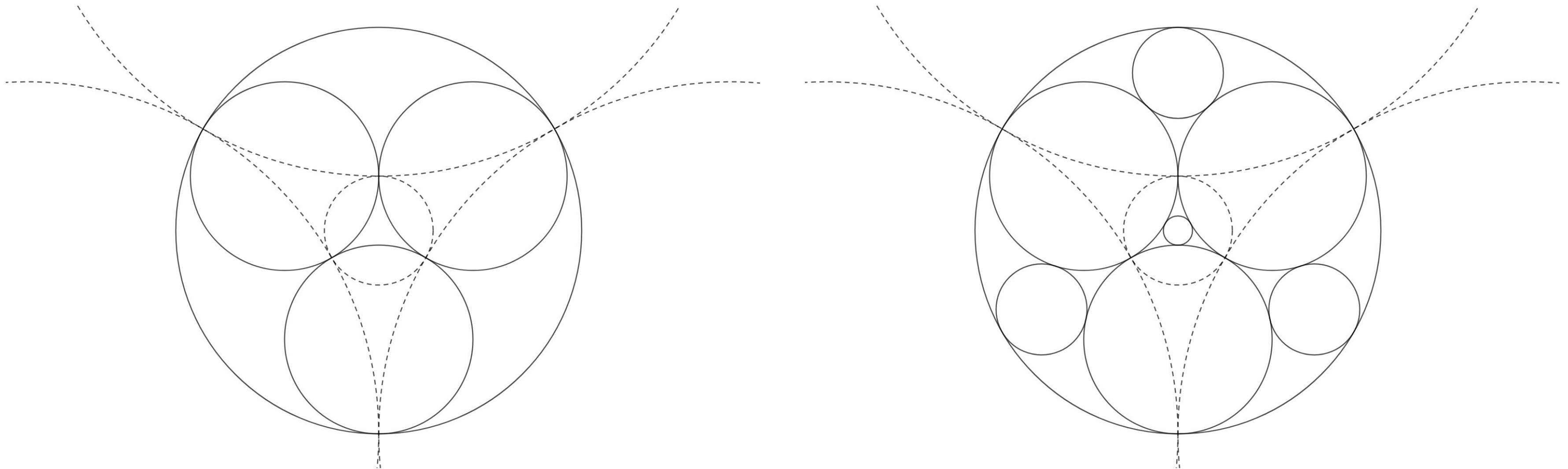
- Projection stéréographique
 - Projection des cercles sur le plan
 - Projection conforme



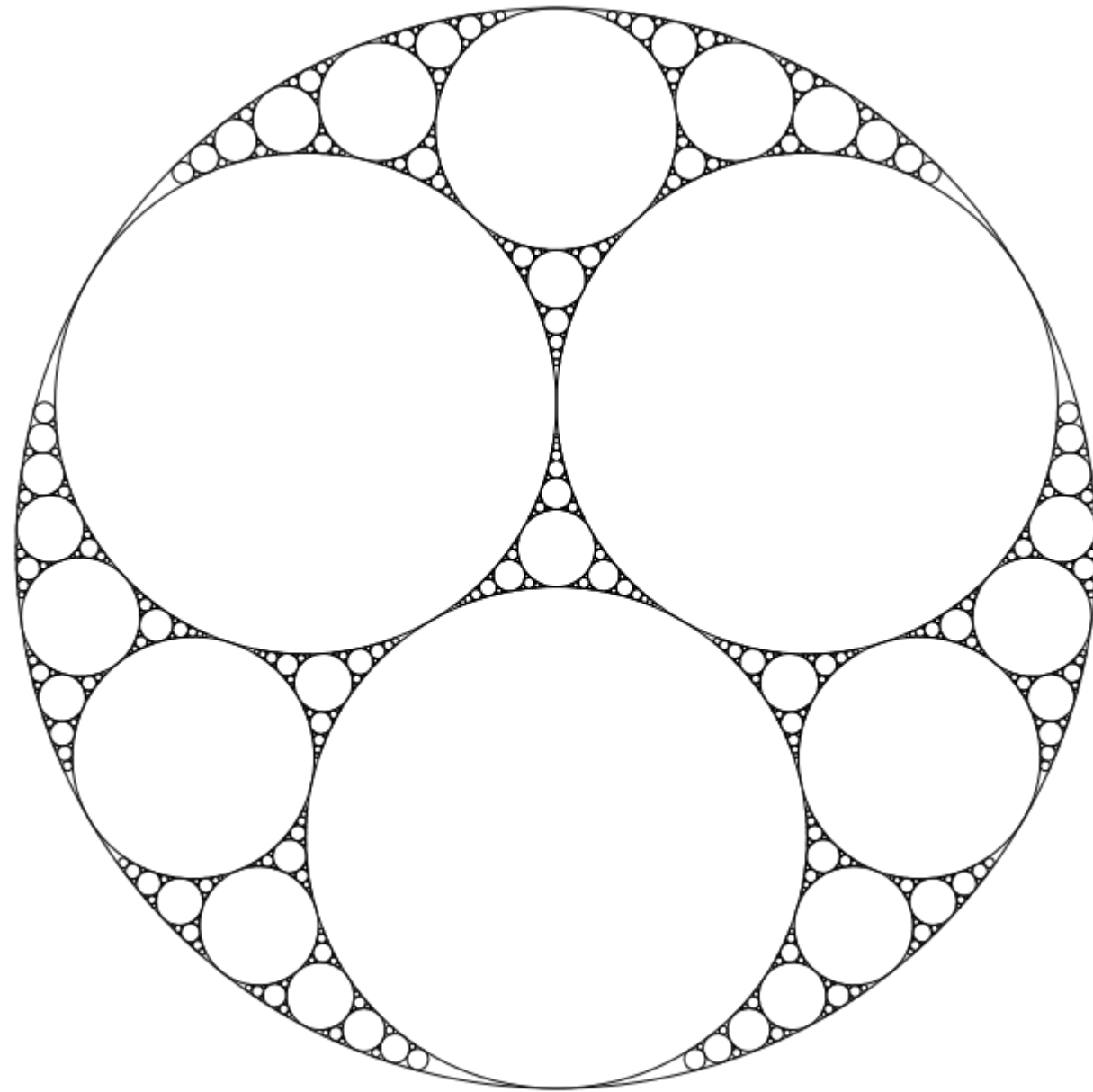
- On fait pareil avec le dual du polyèdre
 - Le dual étant aussi arête-inscrit



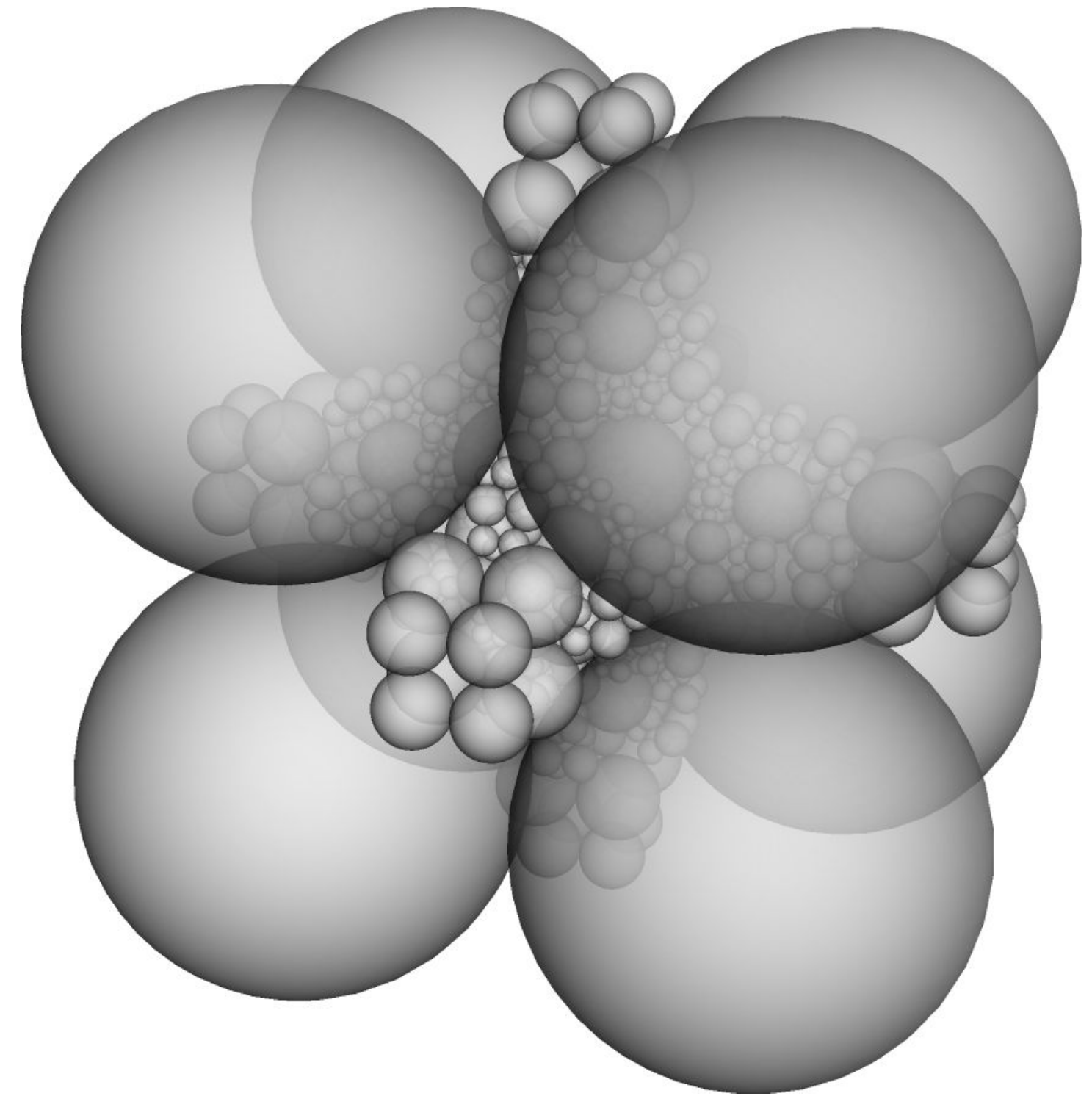
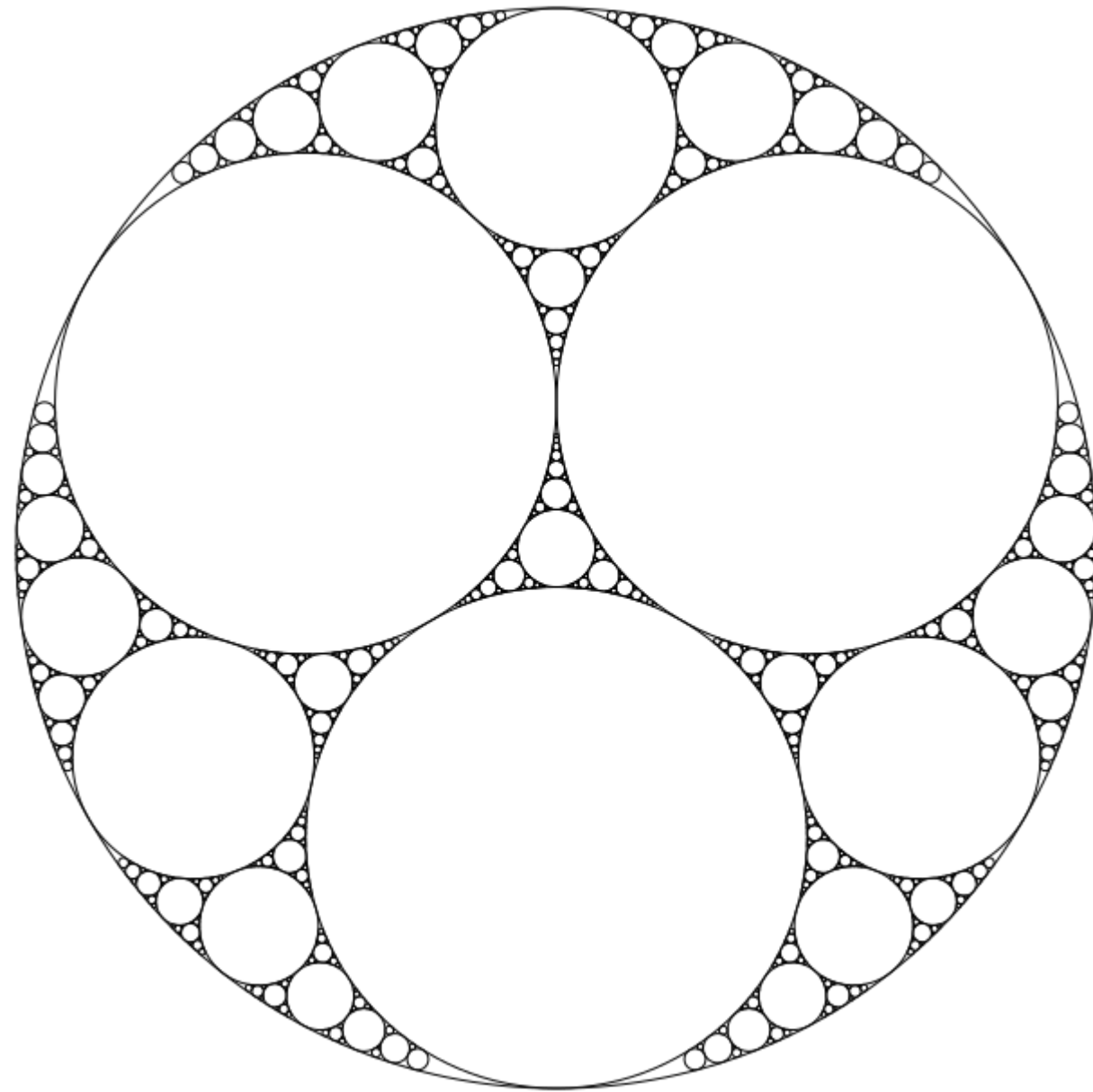
- Inversions des cercles du polyèdre par ceux du dual
 - On obtient de nouveaux cercles



- On répète les inversions
 - Forme fractale

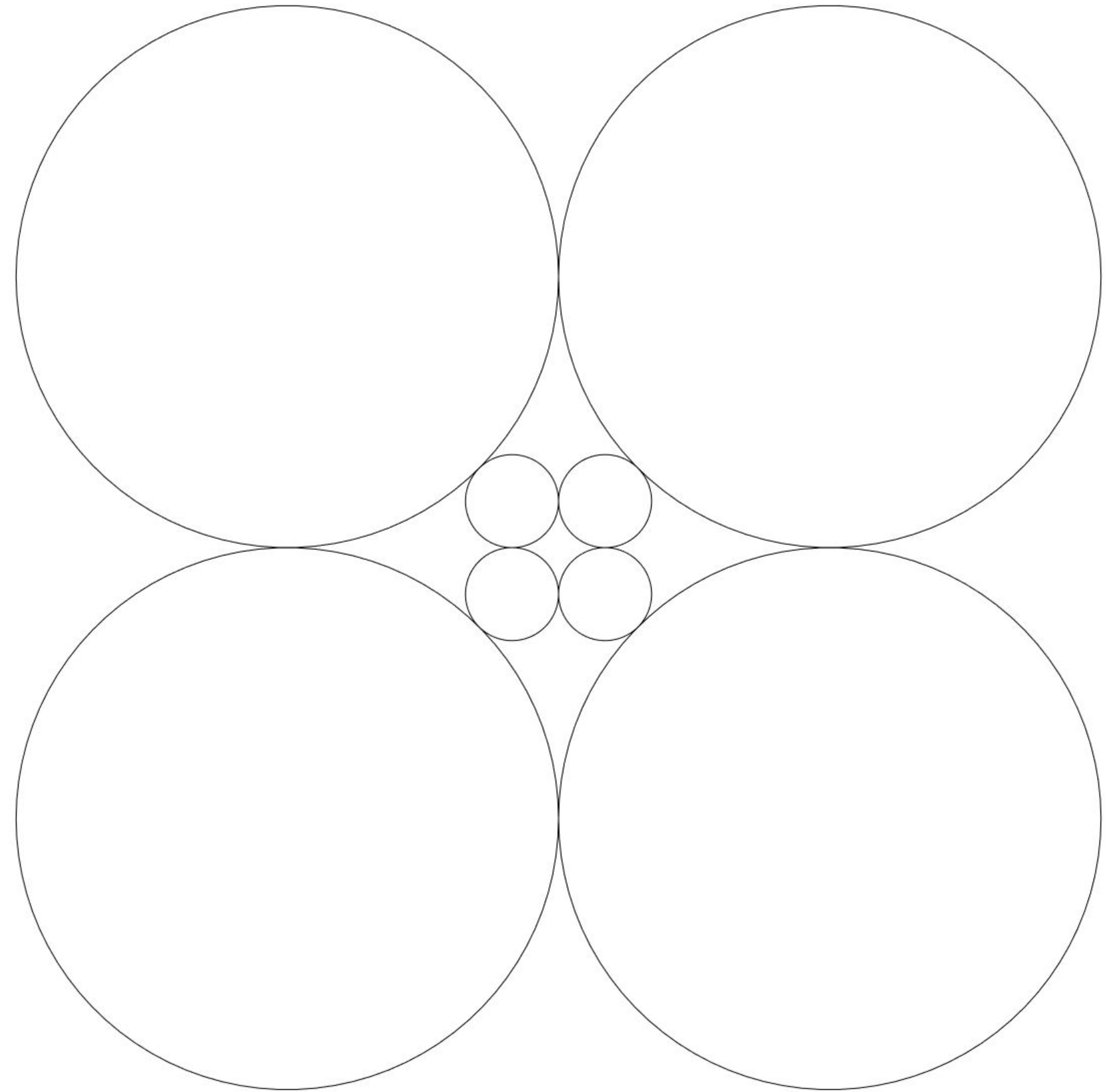
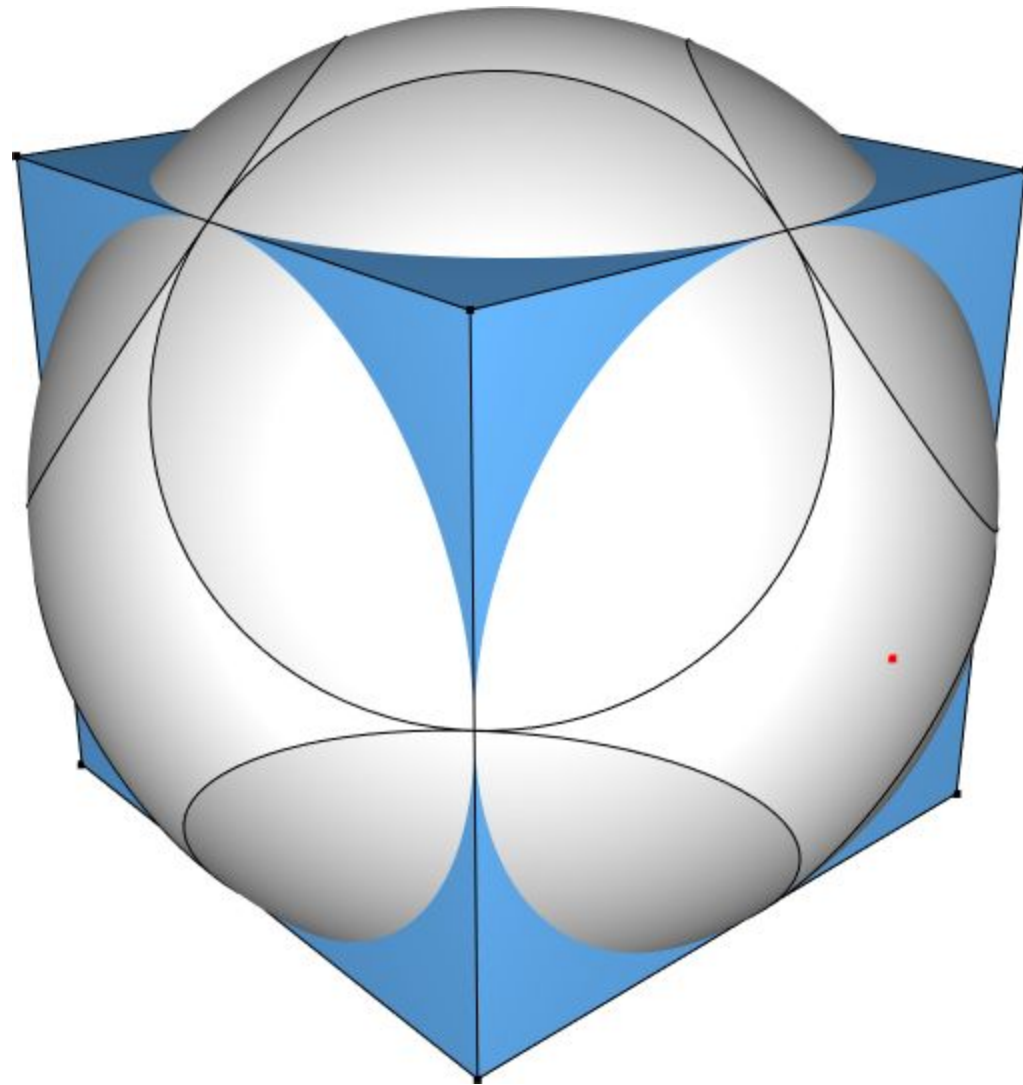


- Fonctionne aussi en 3D
 - Empilements de sphères

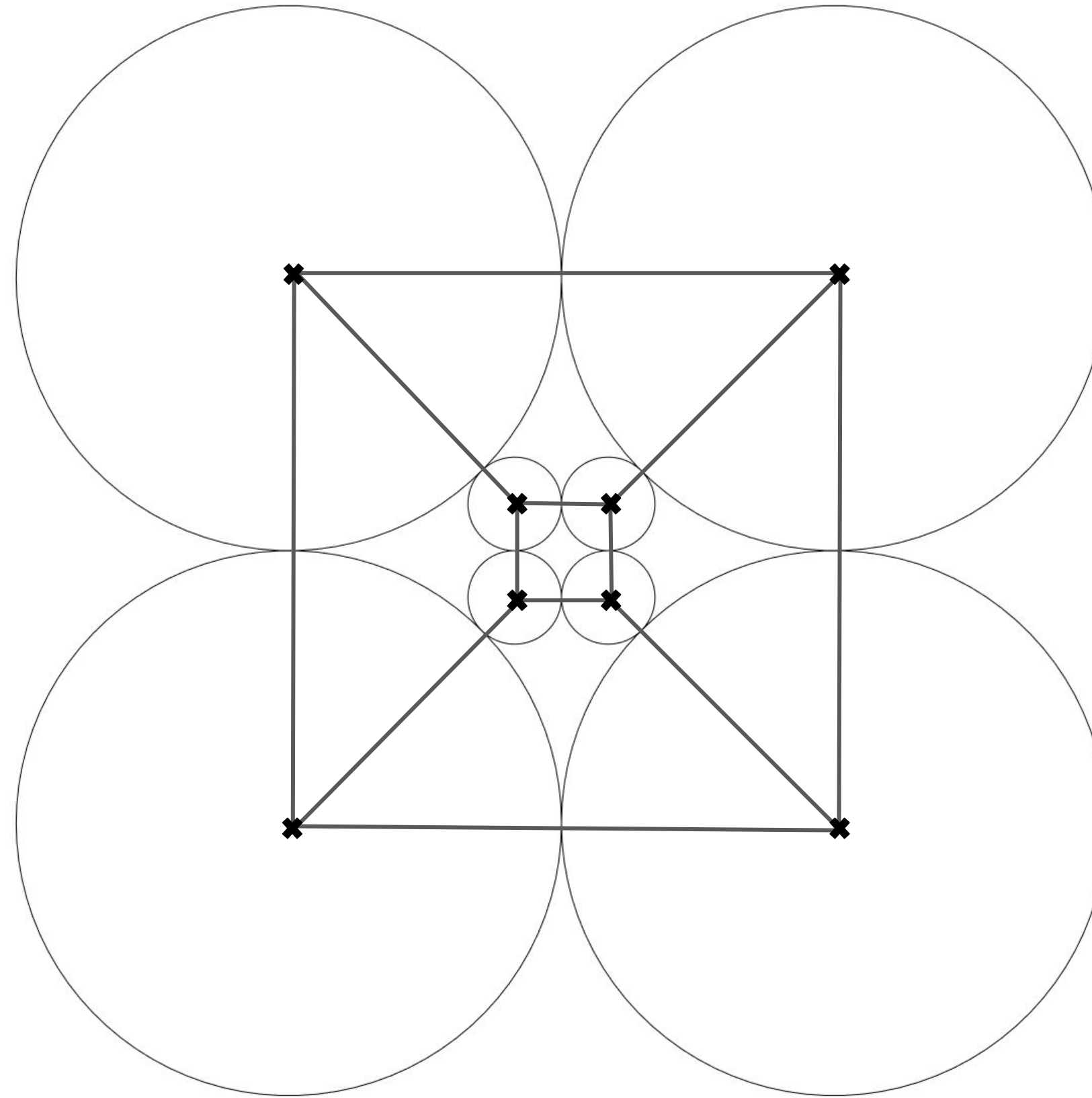


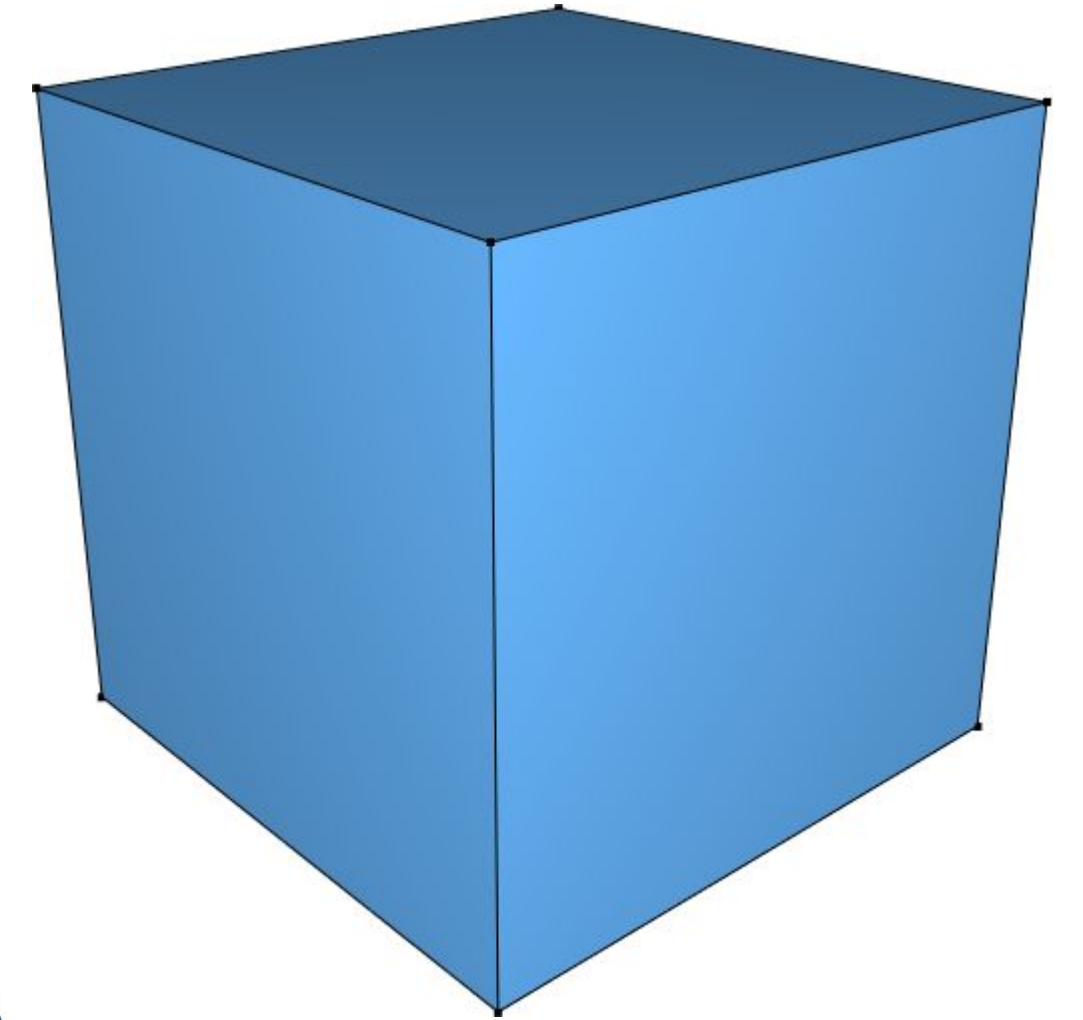
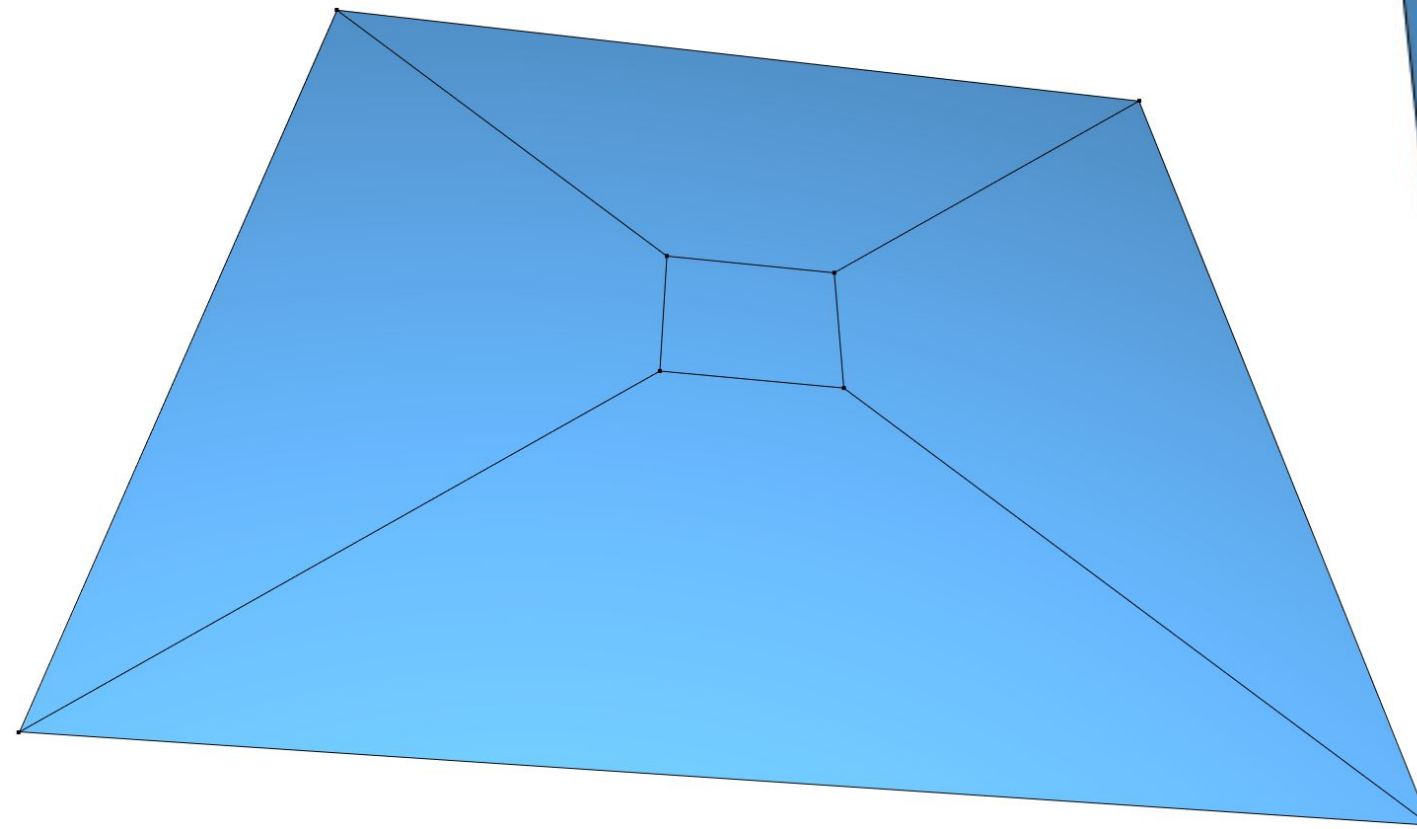
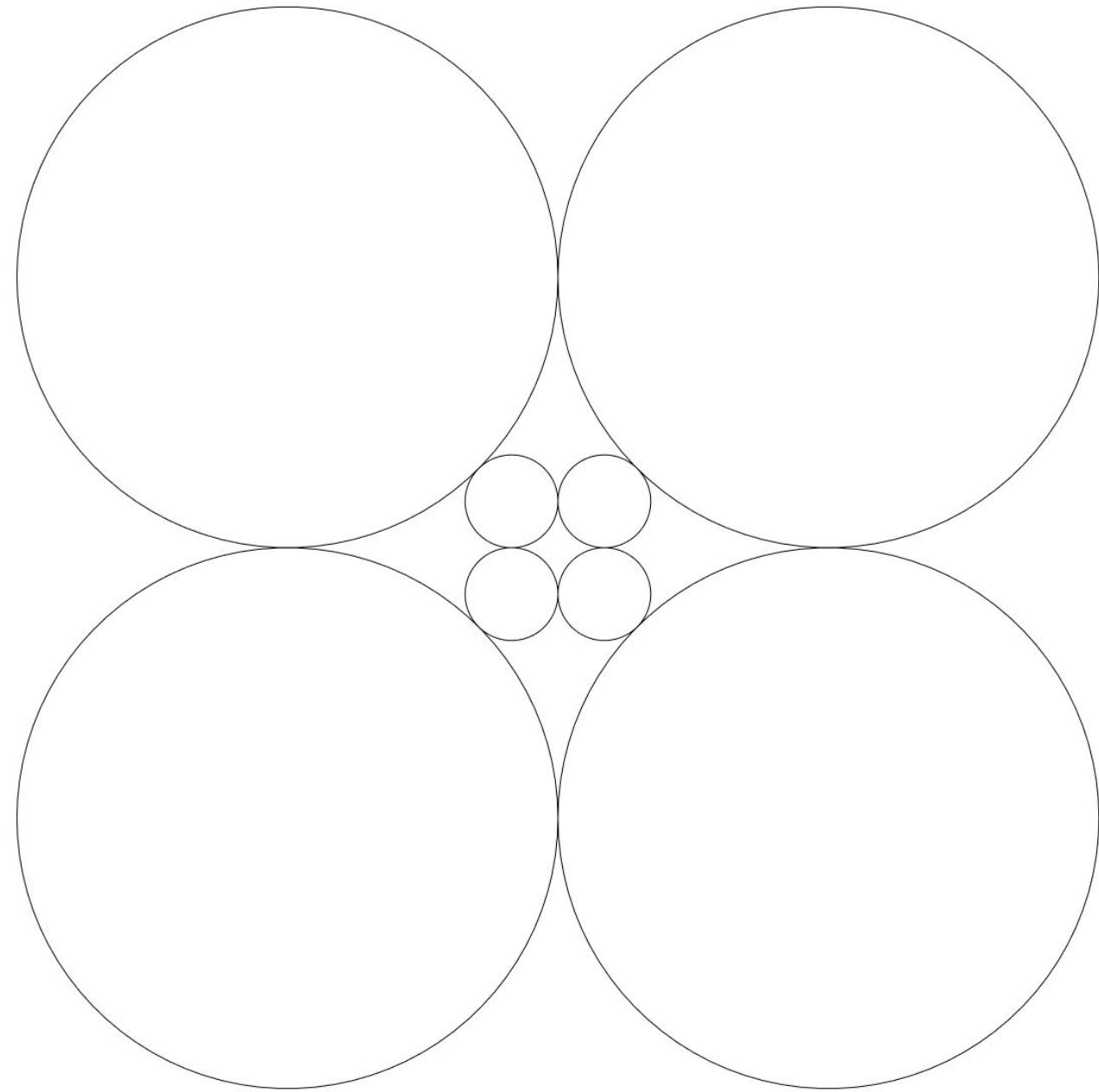
Passage d'un empilement de cercles à un polyèdre

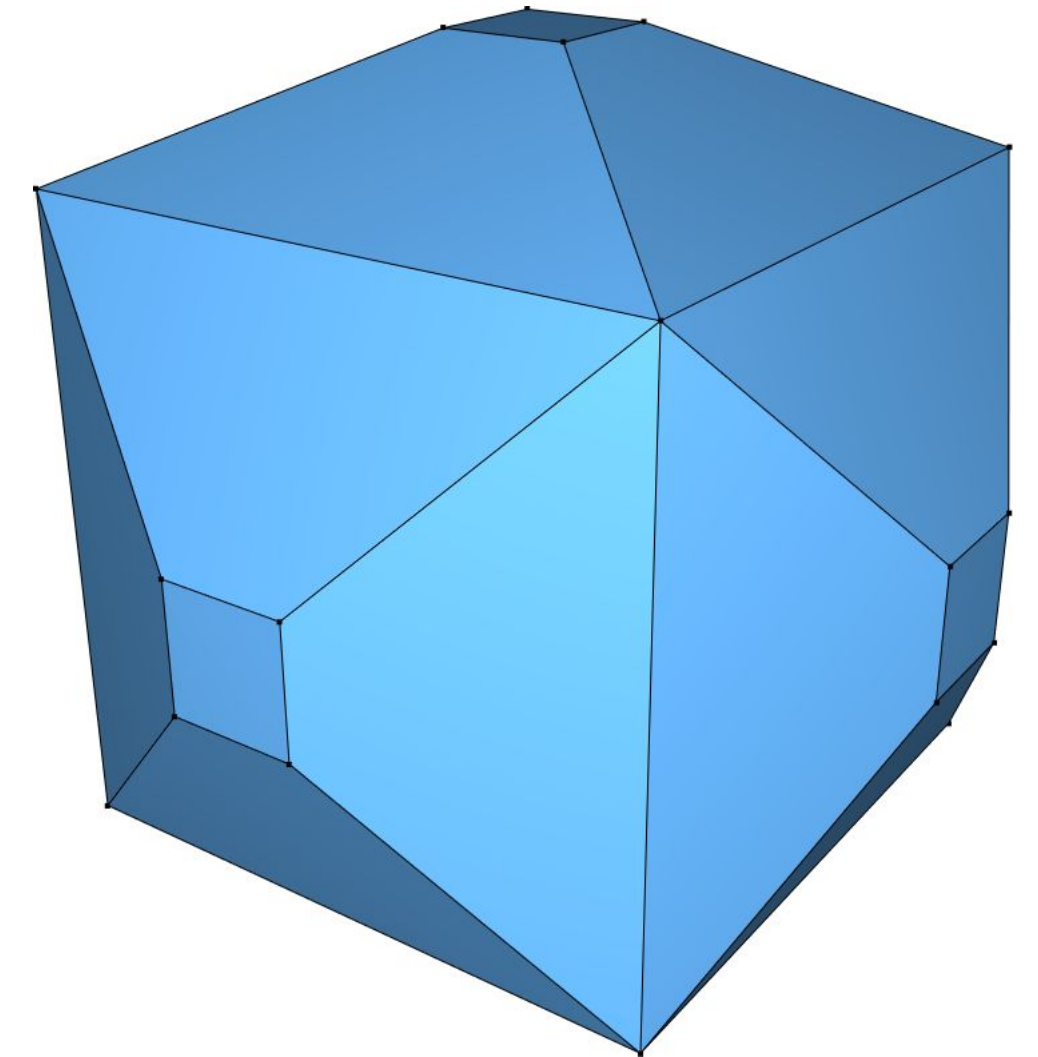
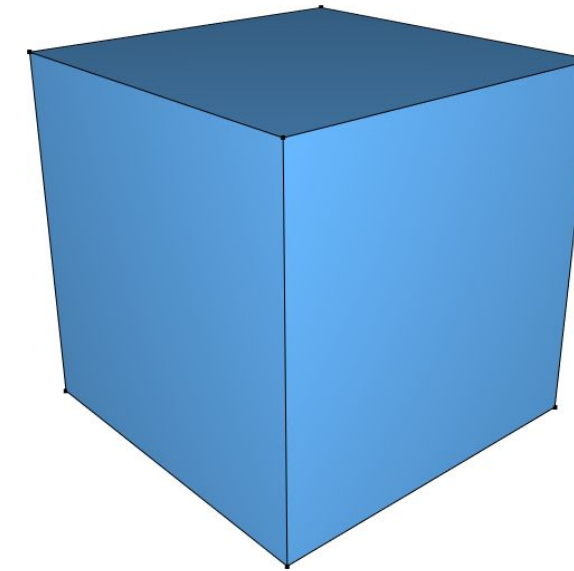
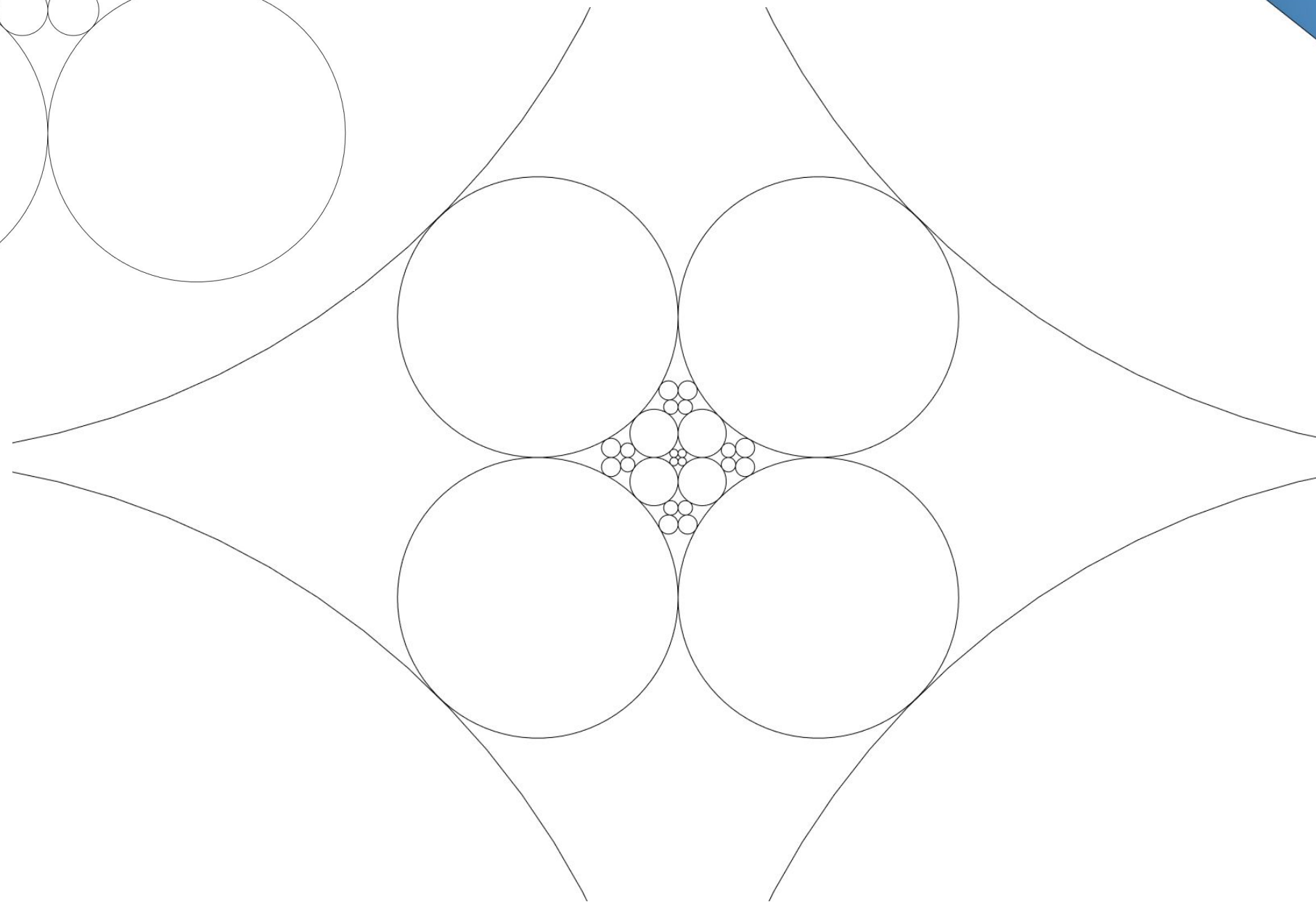
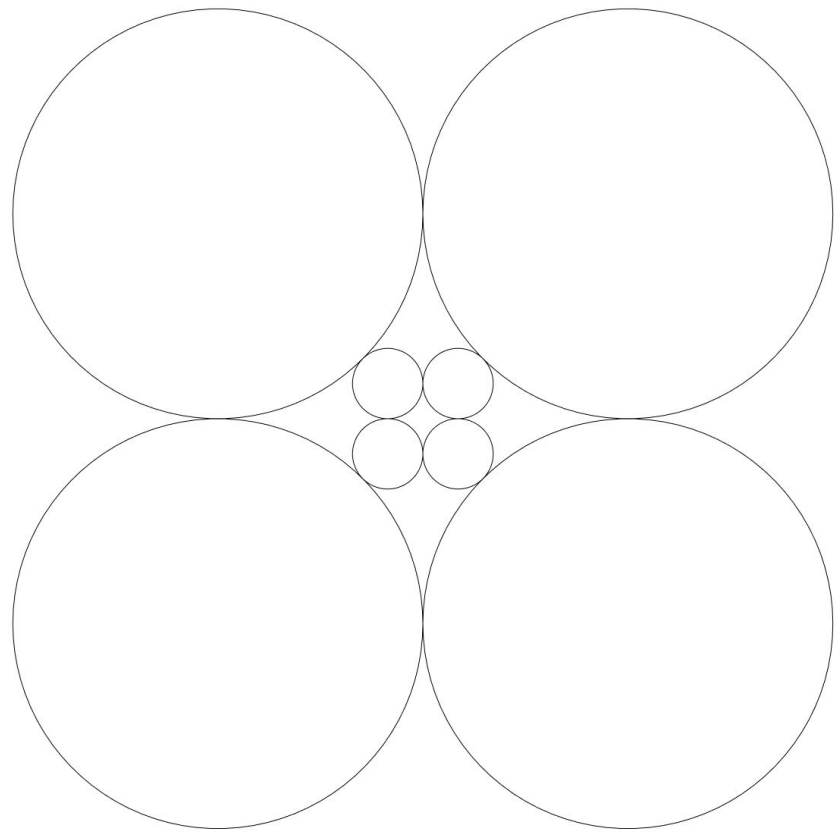




- Sommets
 - Centre des cercles
- Arêtes
 - Adjacences
- Faces
 - Cycles



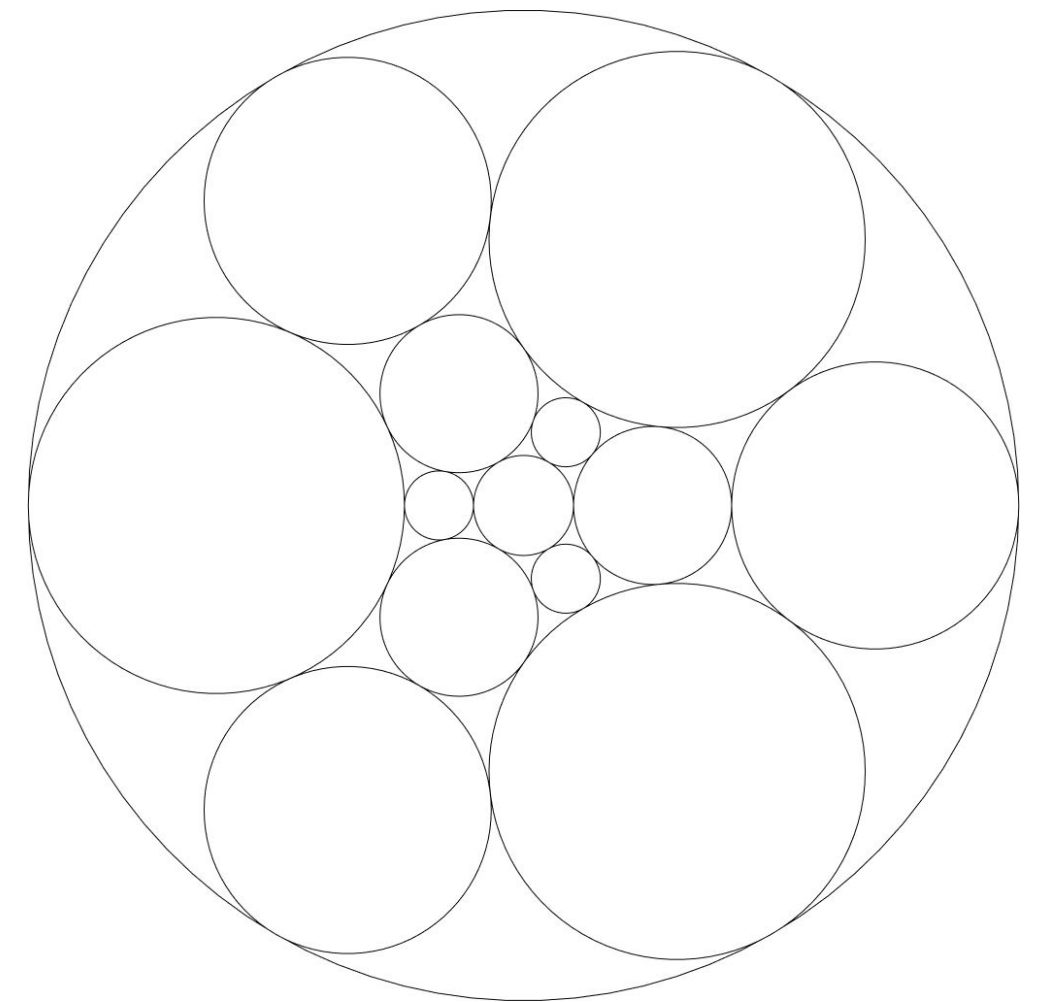
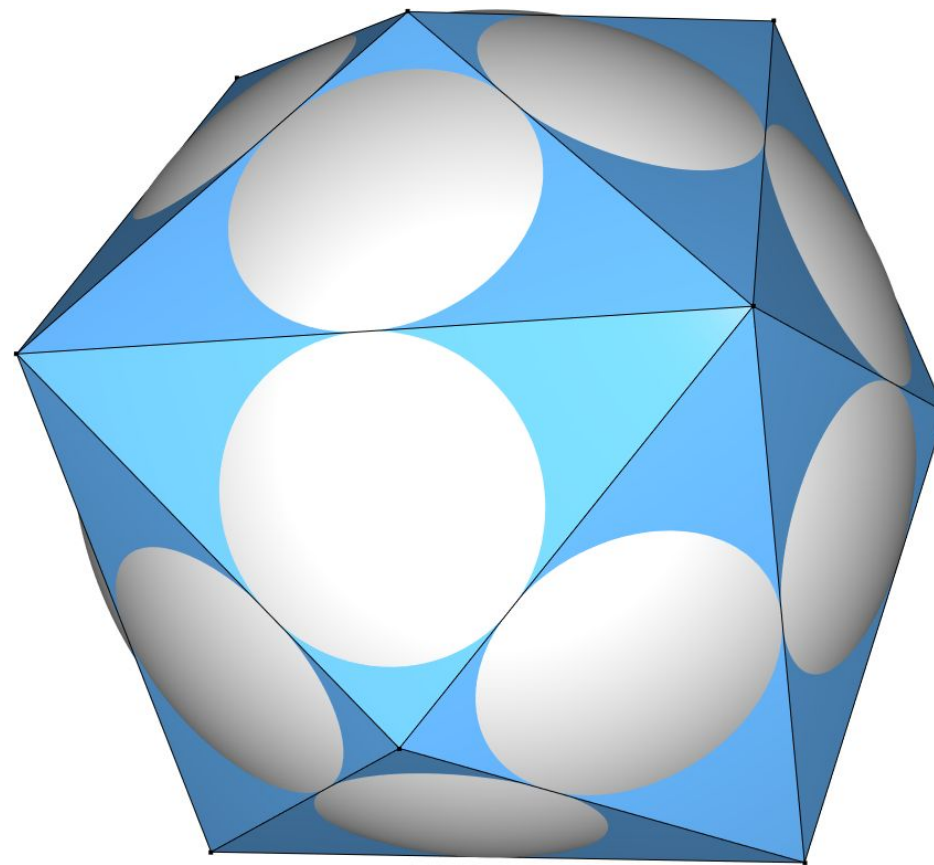
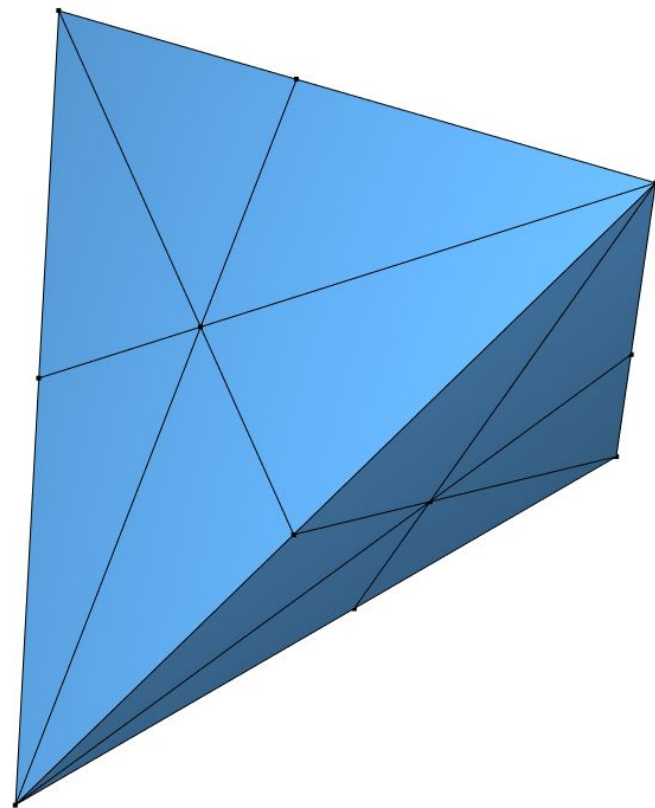
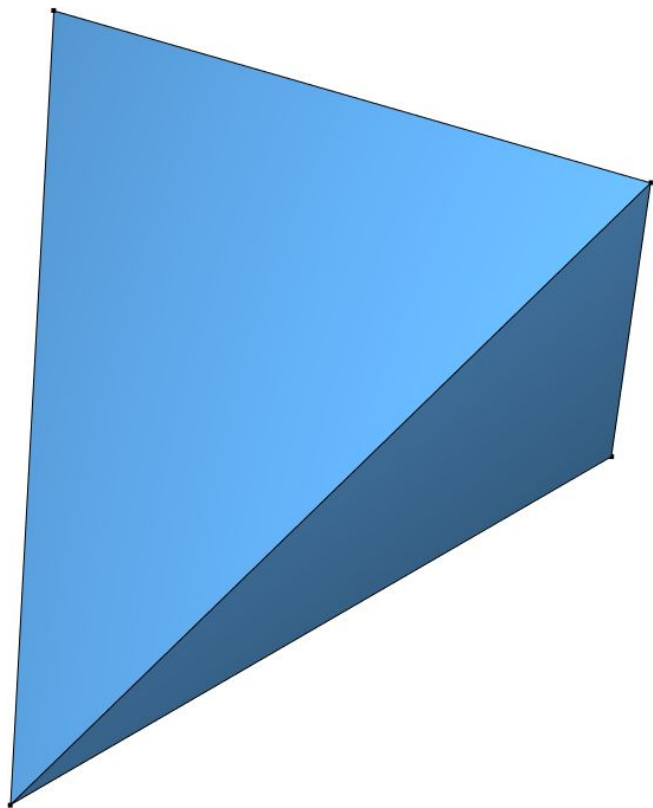




Etude d'autres subdivisions du polyèdre

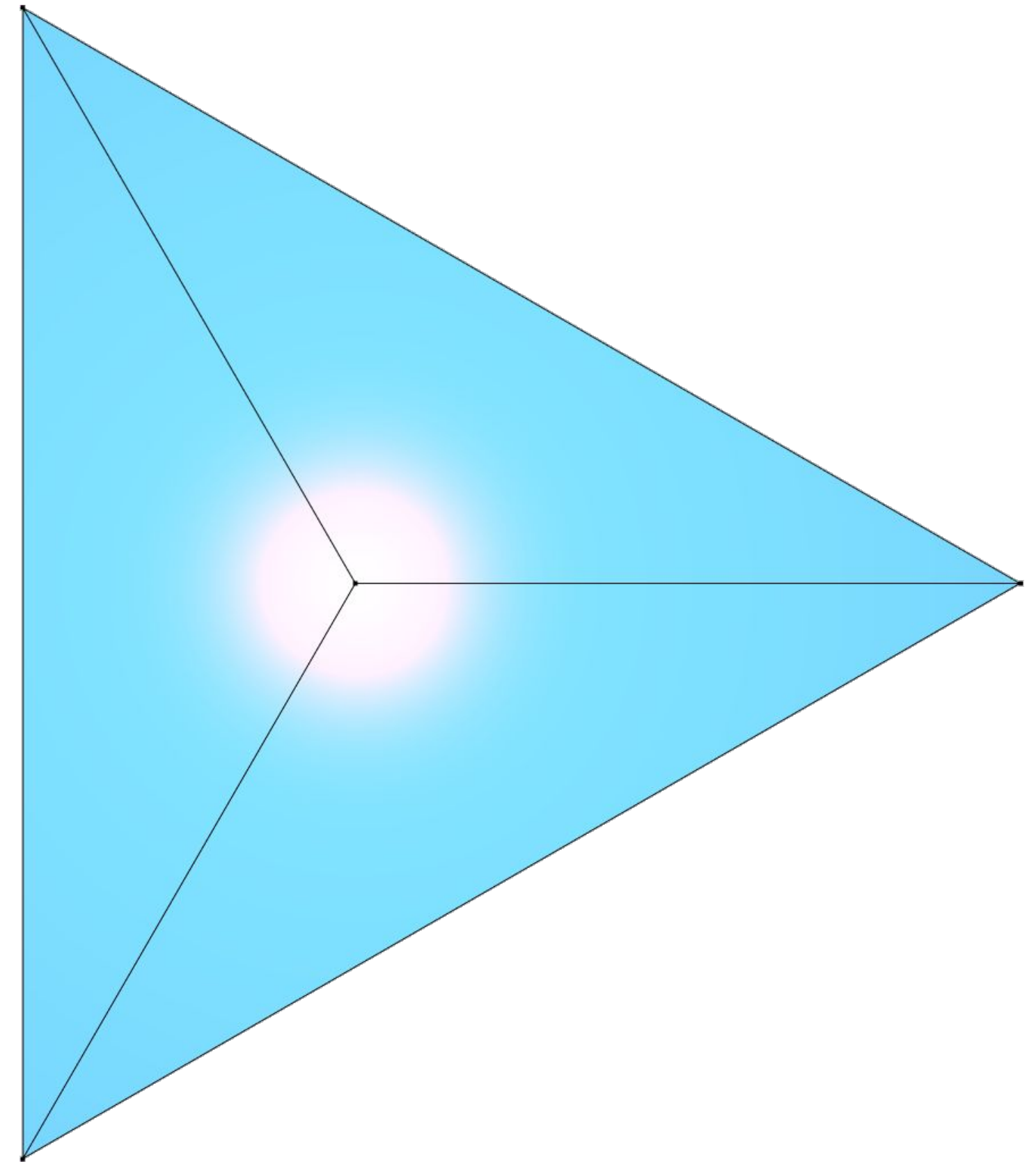
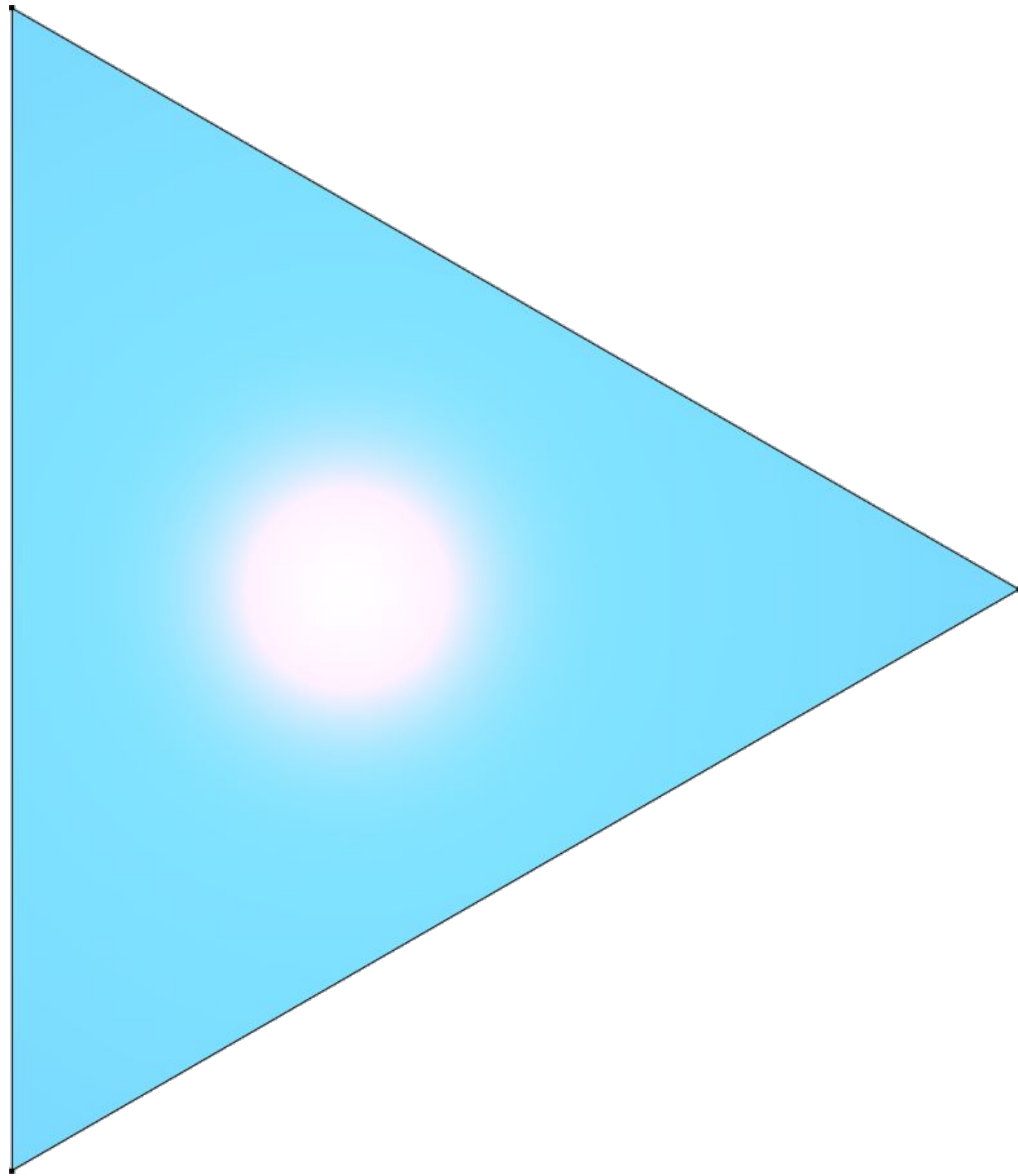


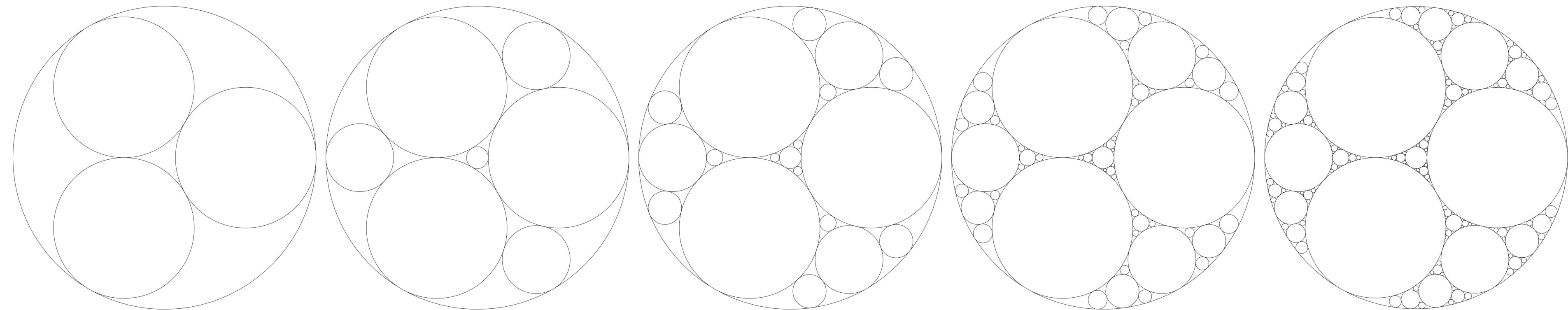
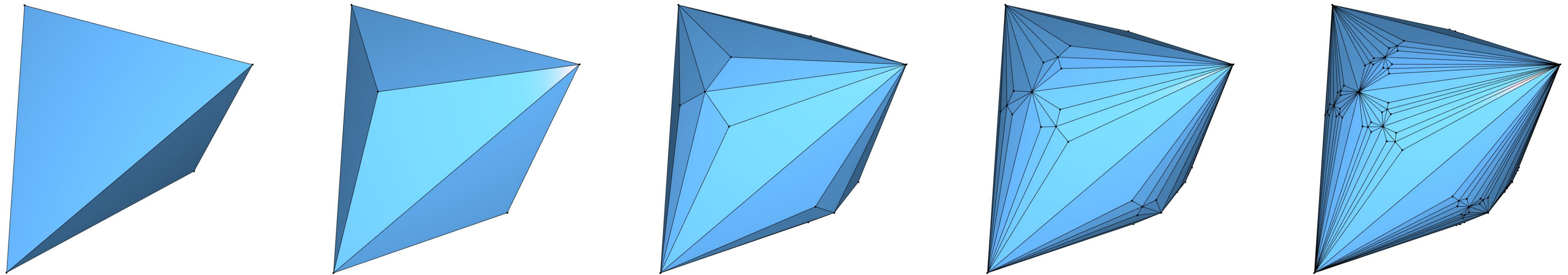
- Choix d'un polyèdre de départ
- Itérativement
 - Subdivision du polyèdre
 - Mise sous une forme arête-inscrit
 - Projection des cercles

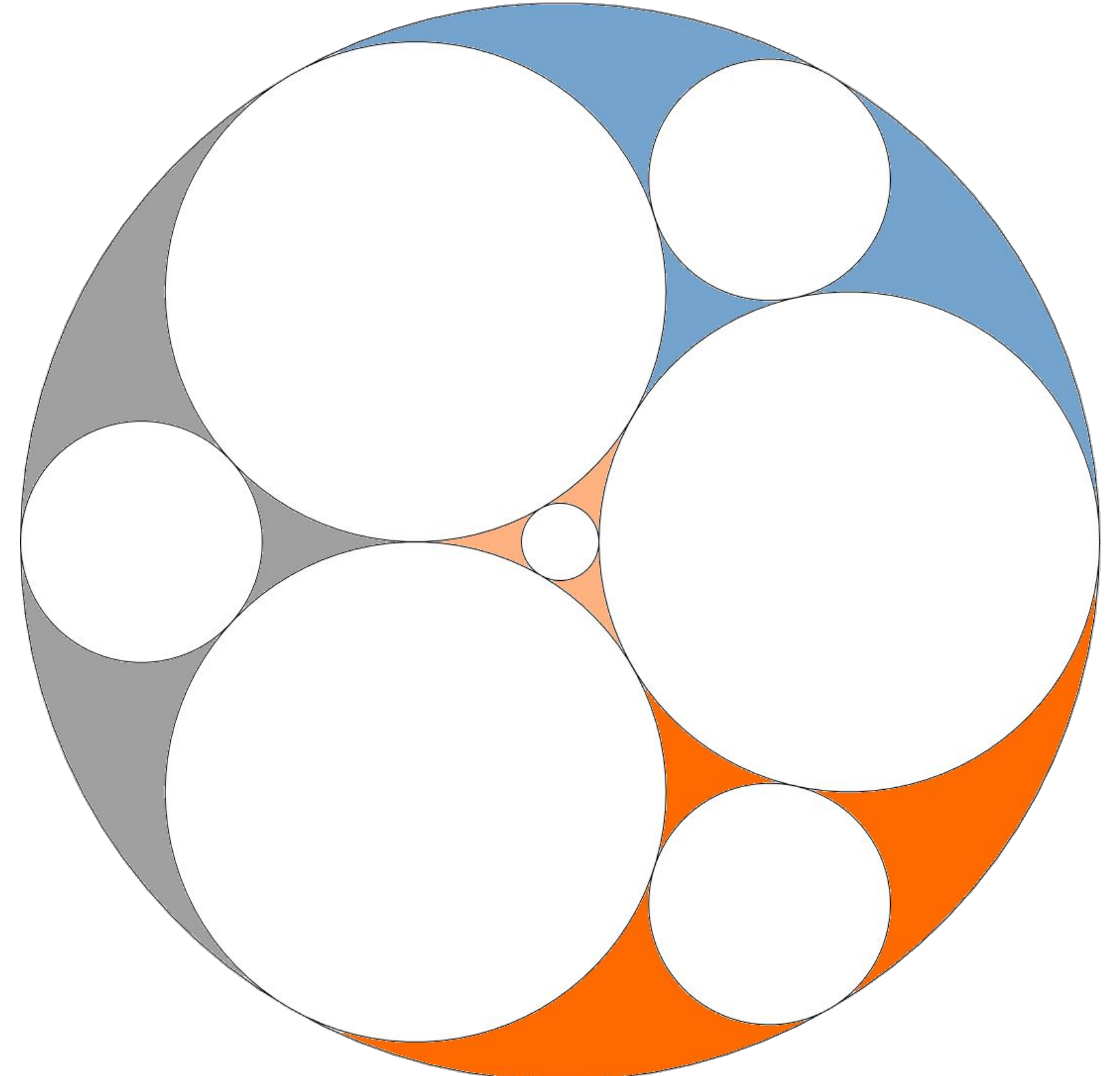
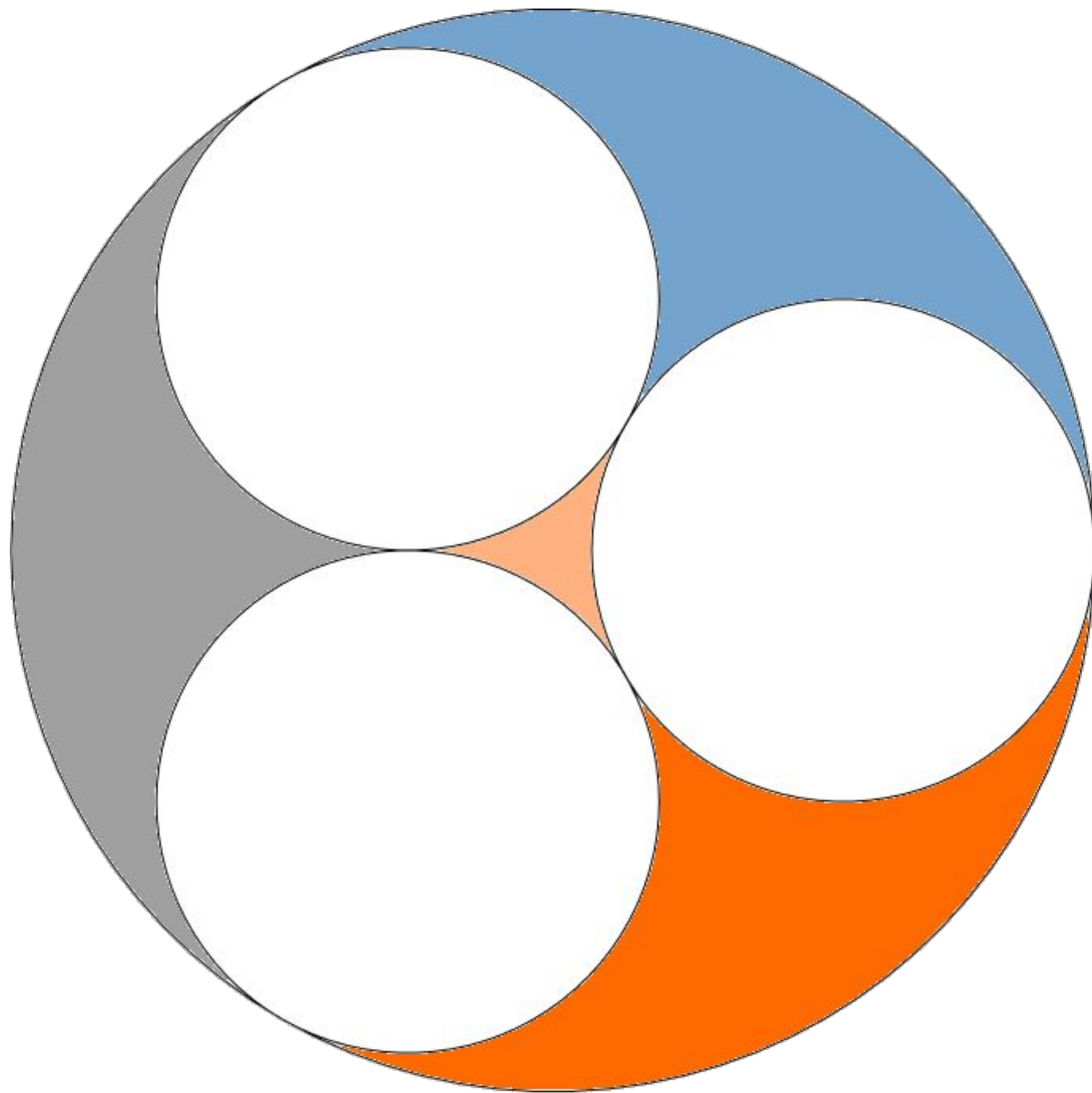


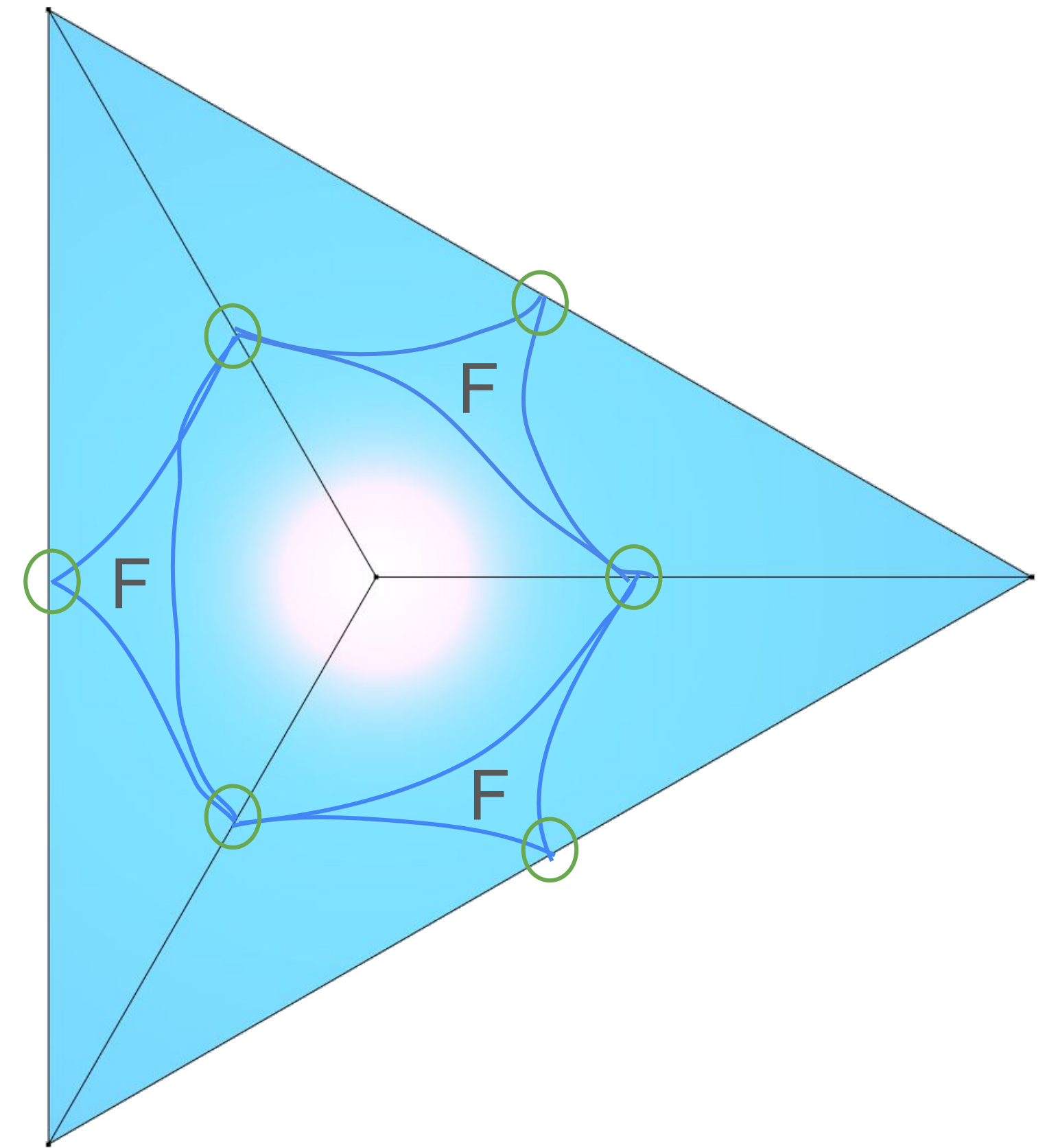
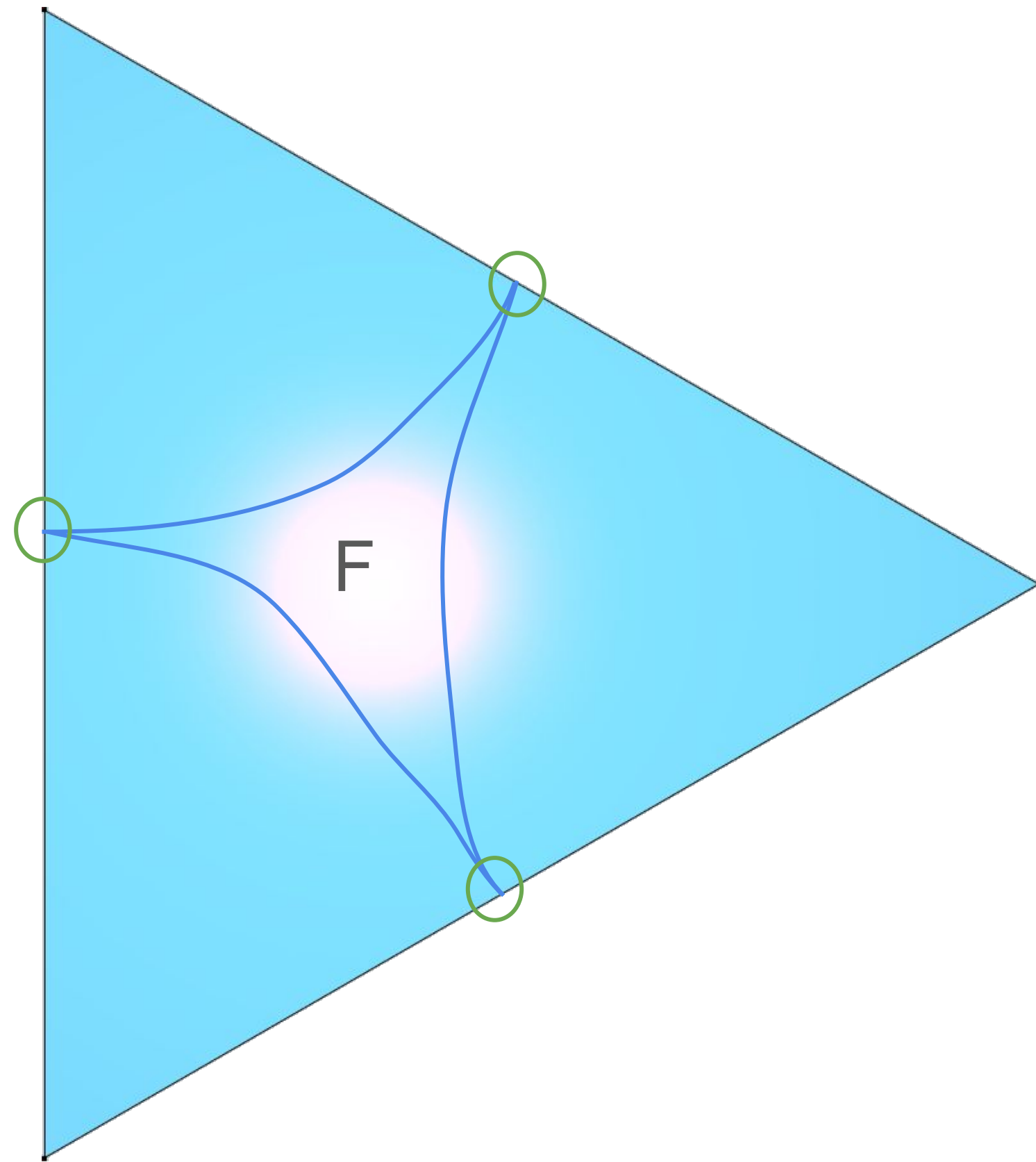
Subdivision d'Apollonius

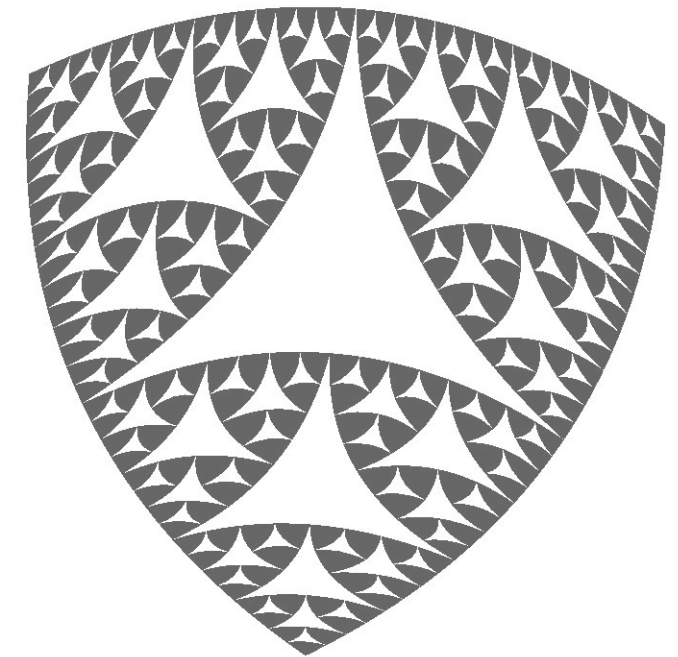
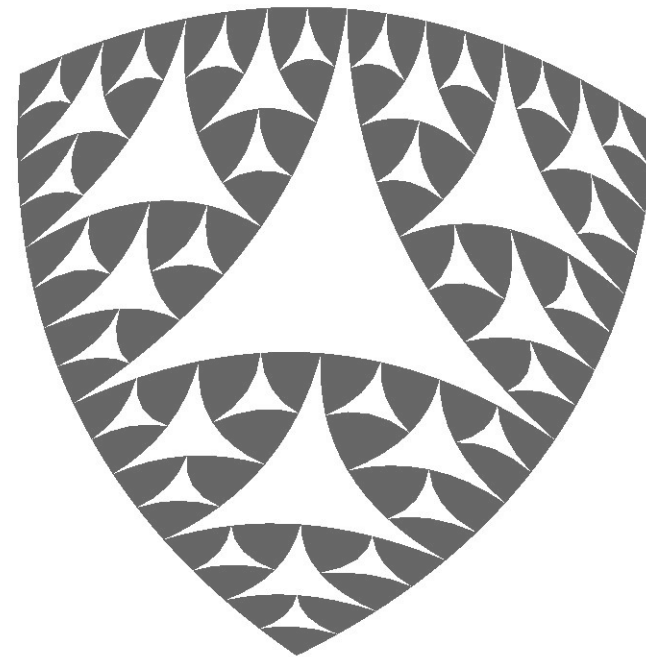
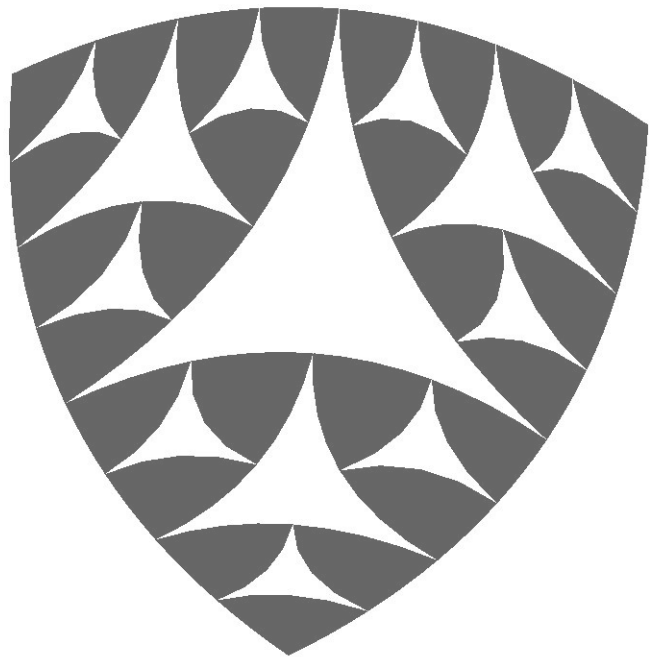
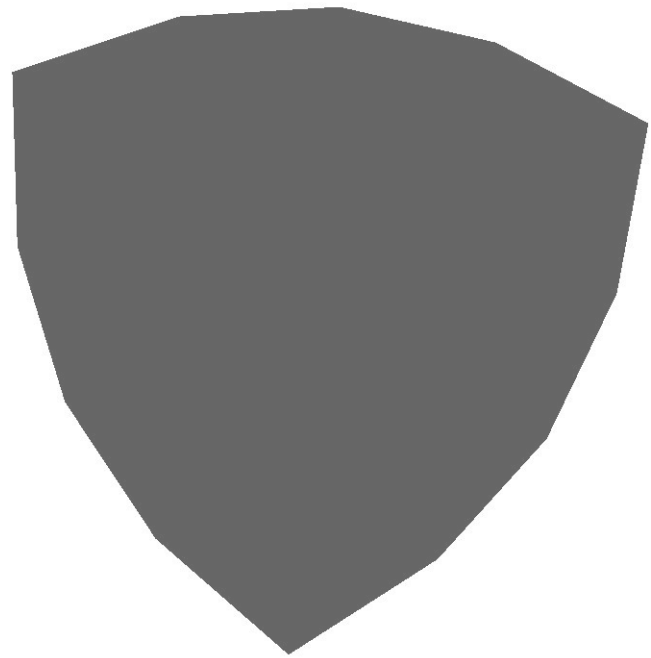






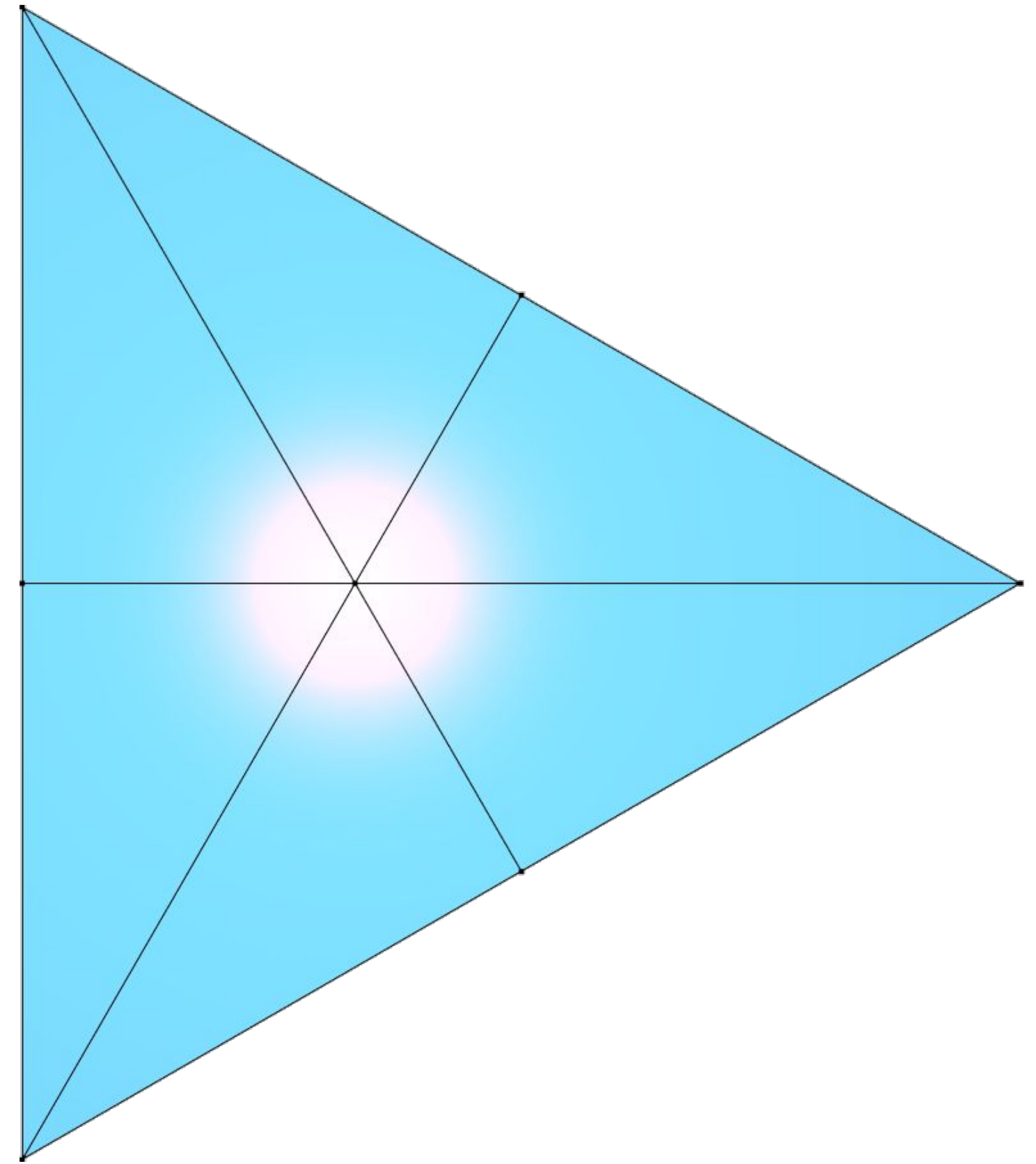
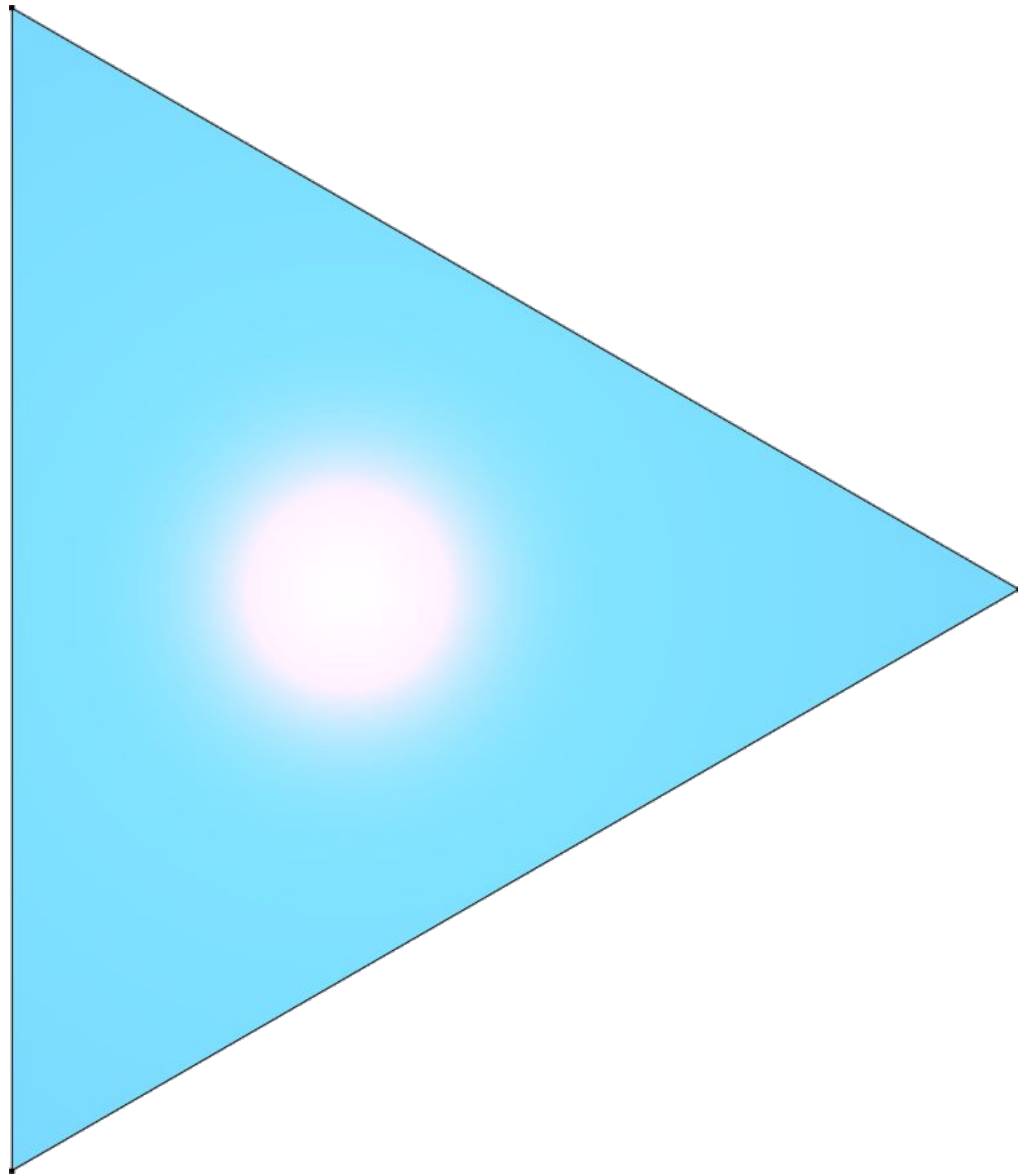


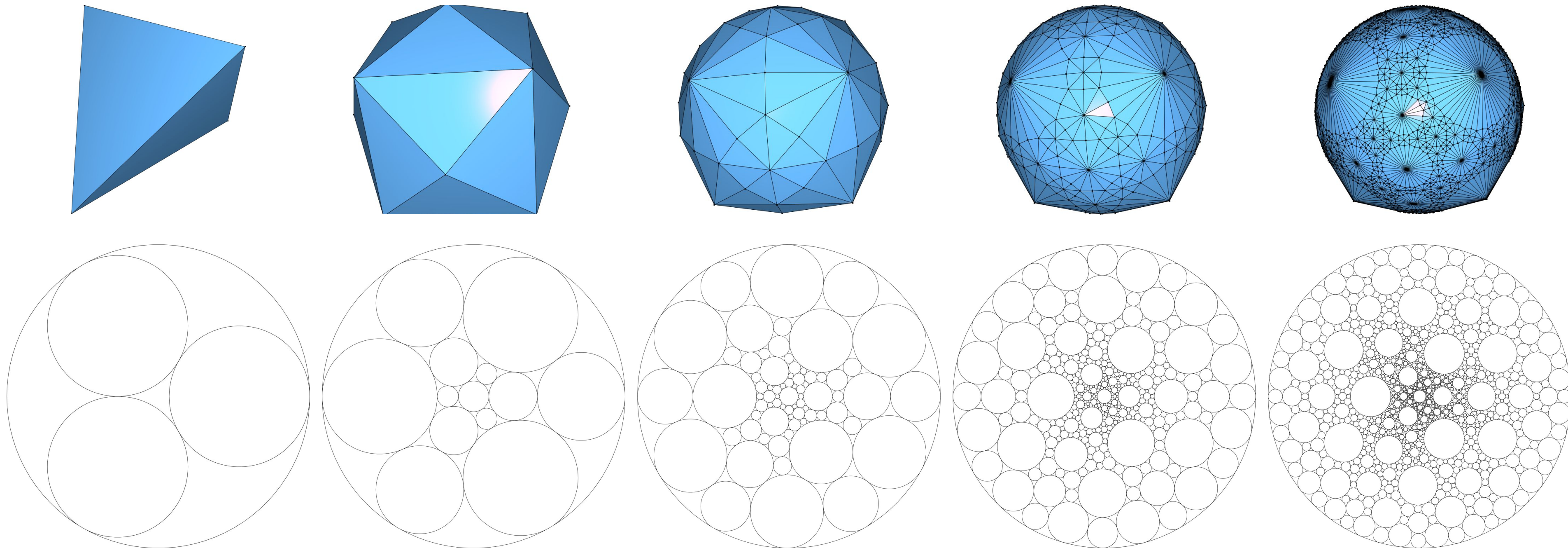


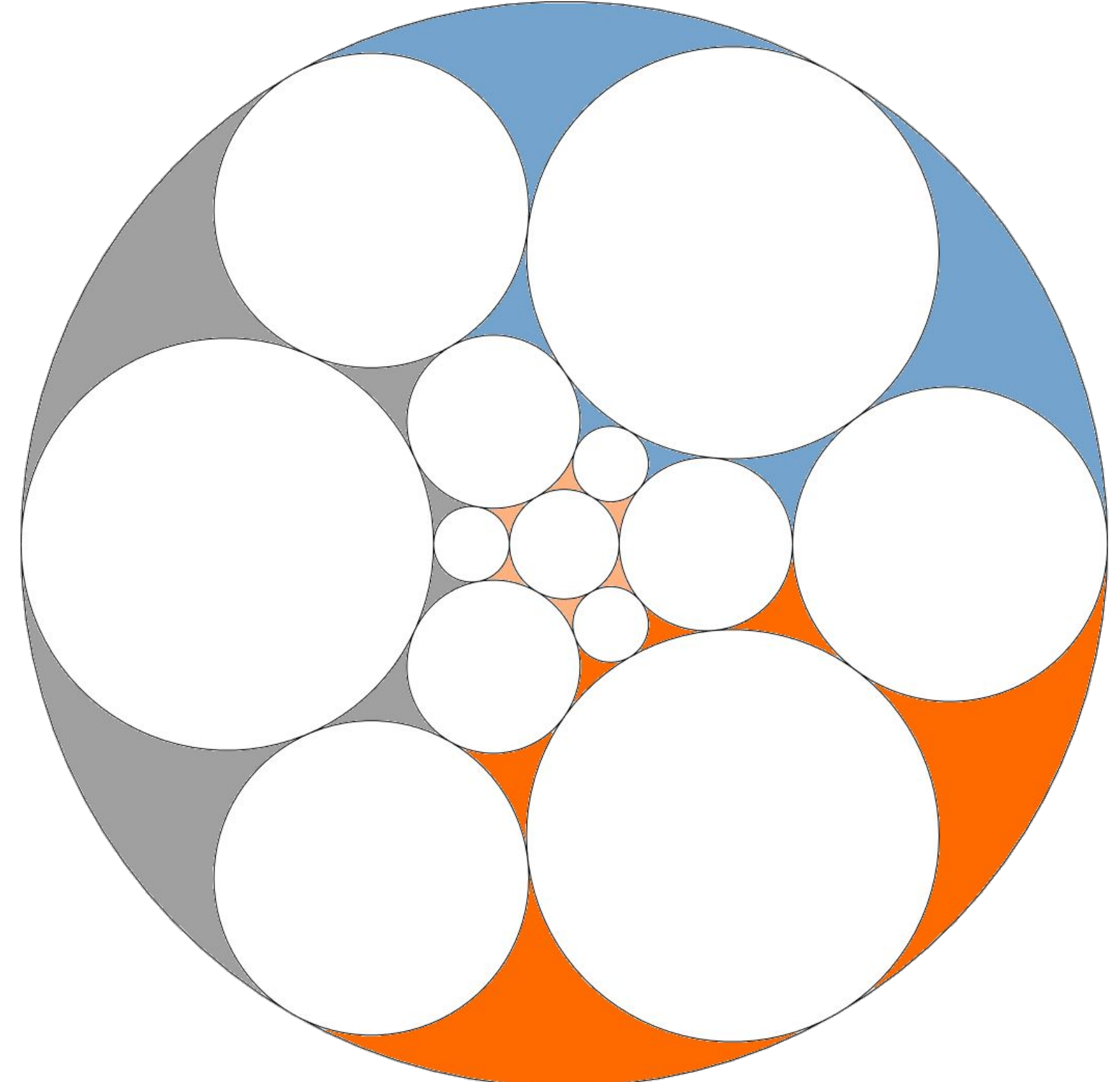
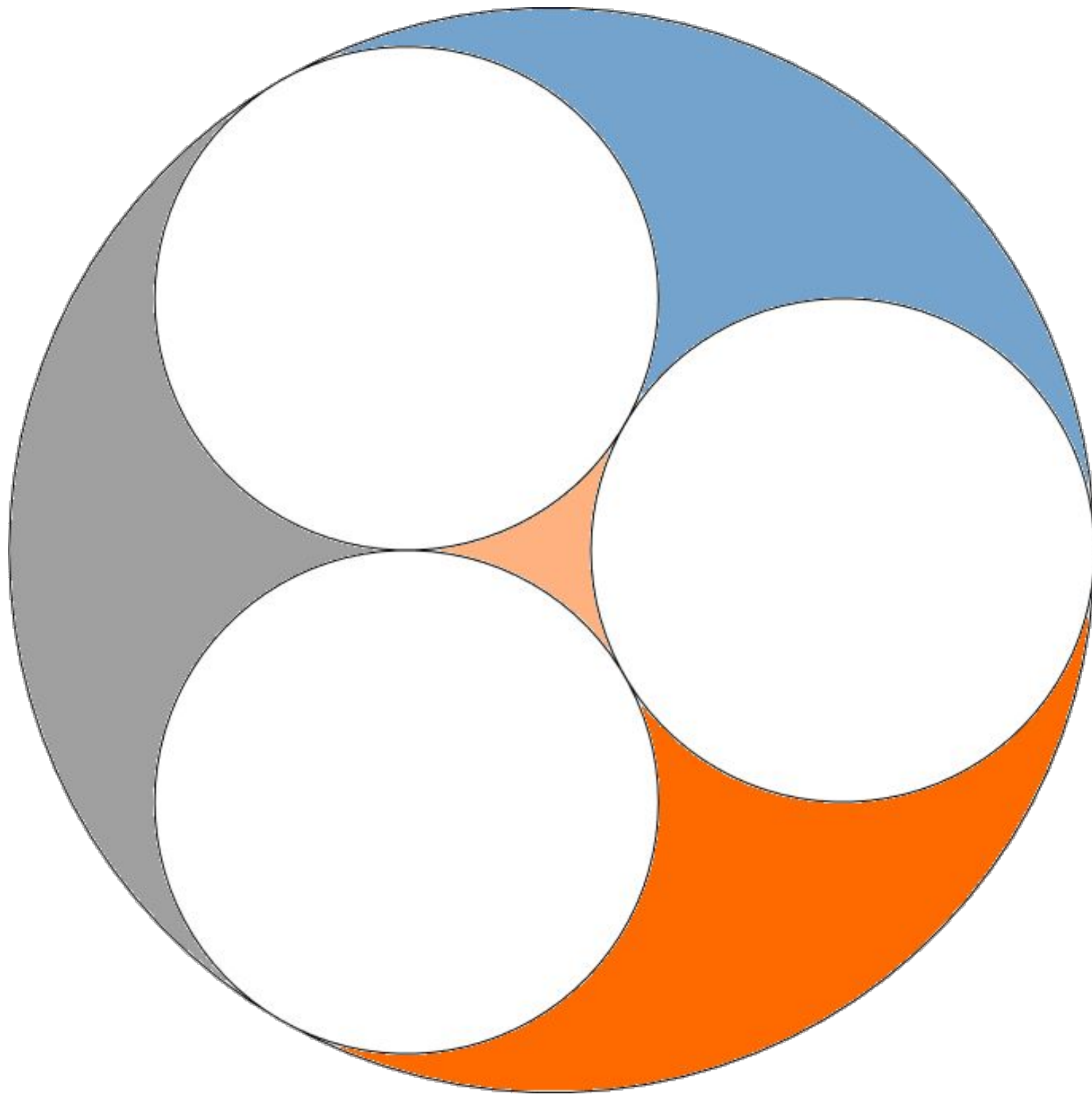


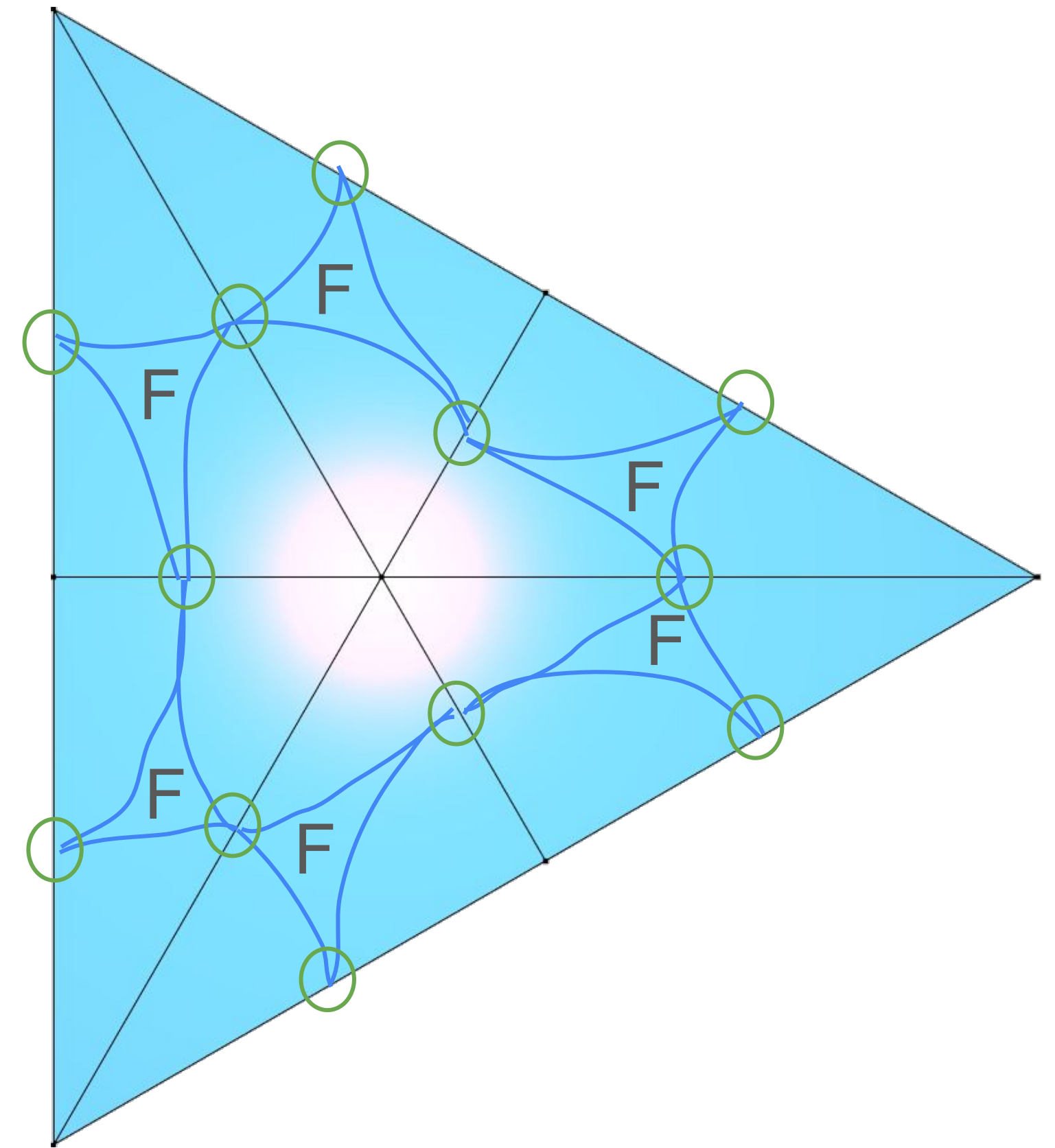
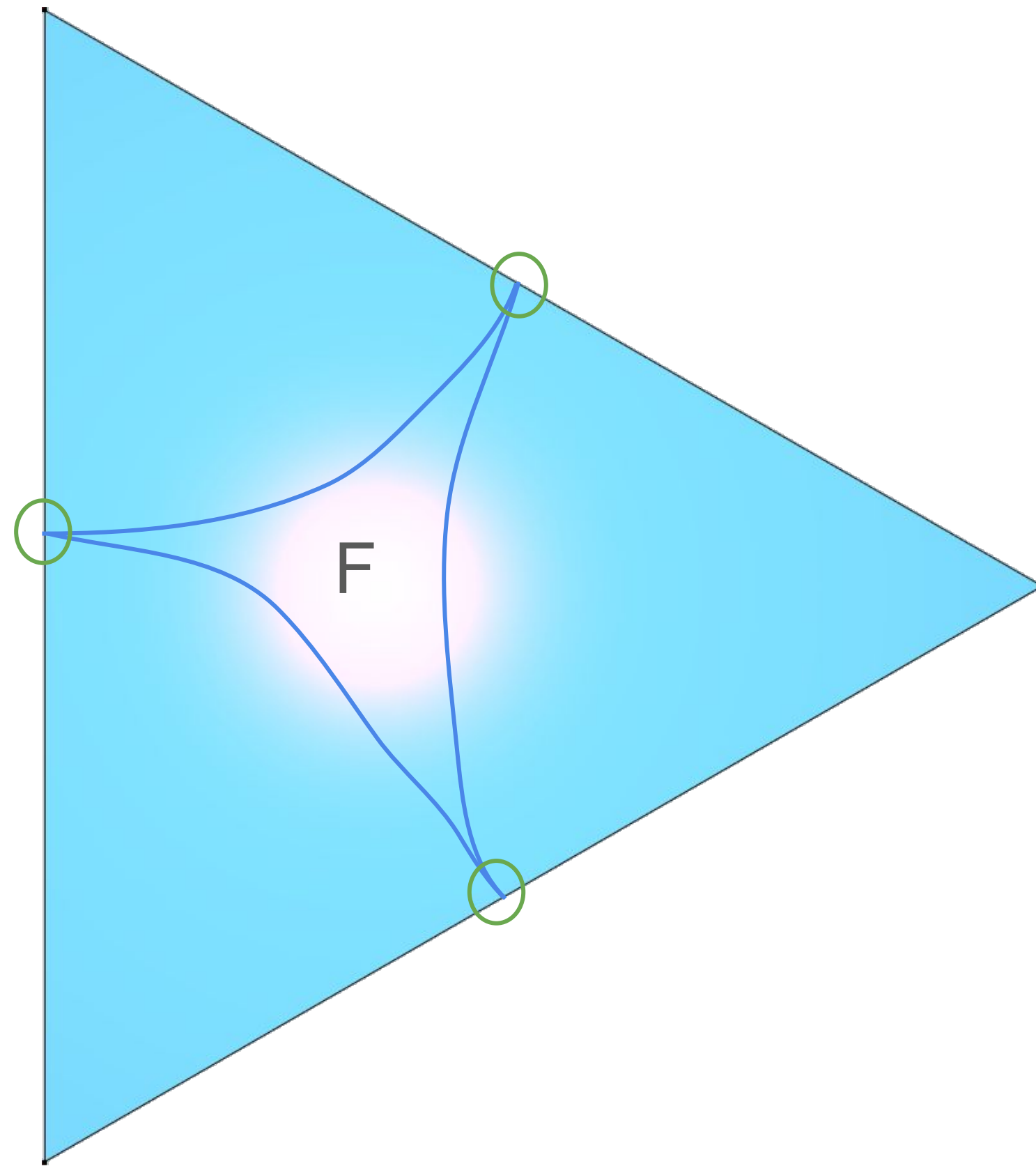
Subdivision barycentrique

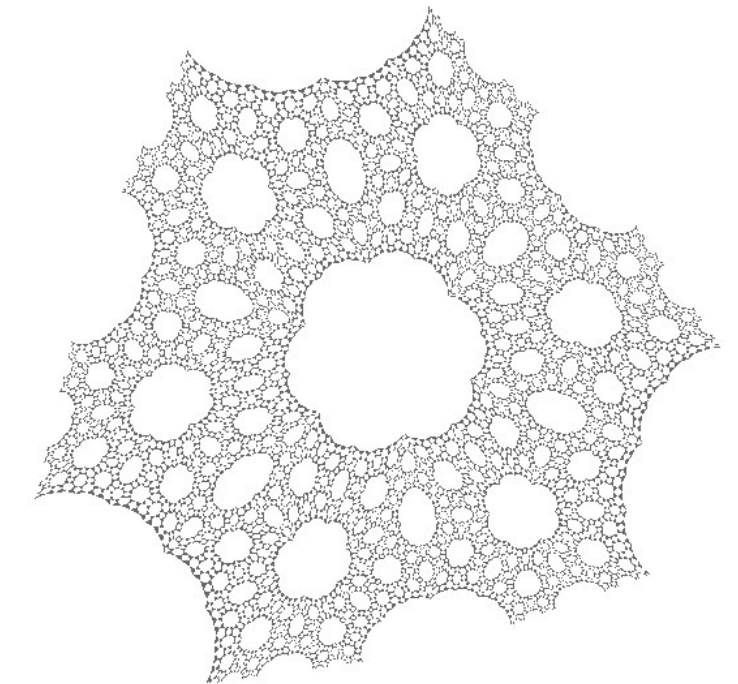
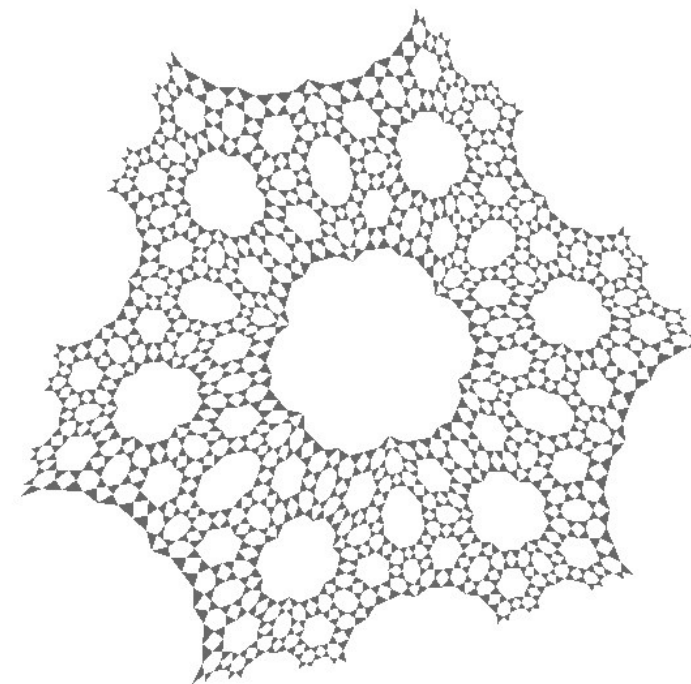
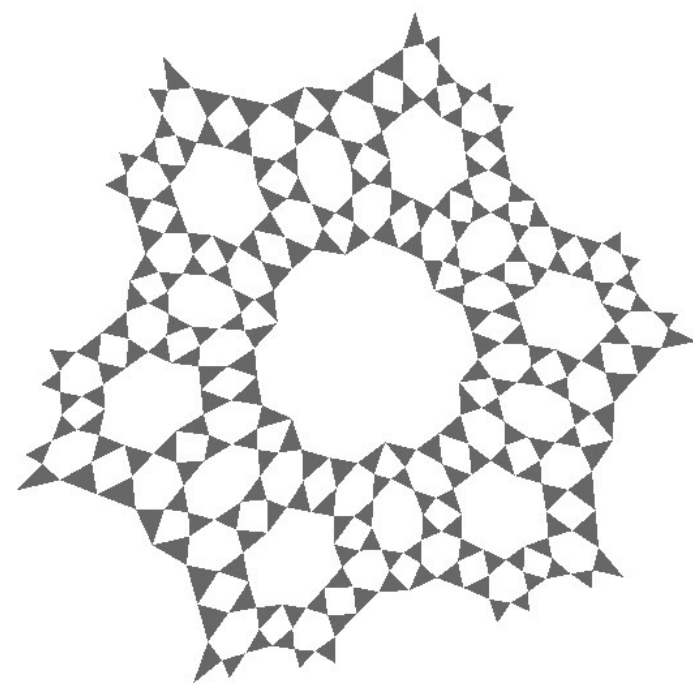
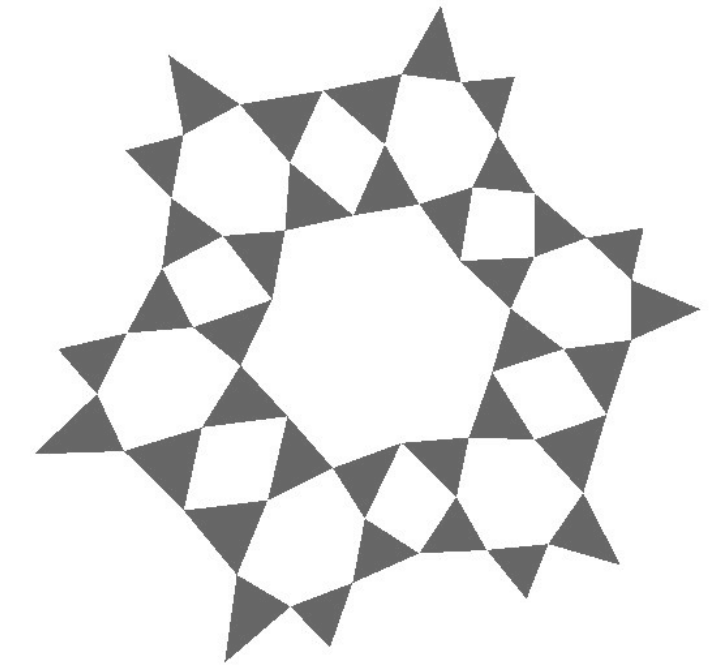
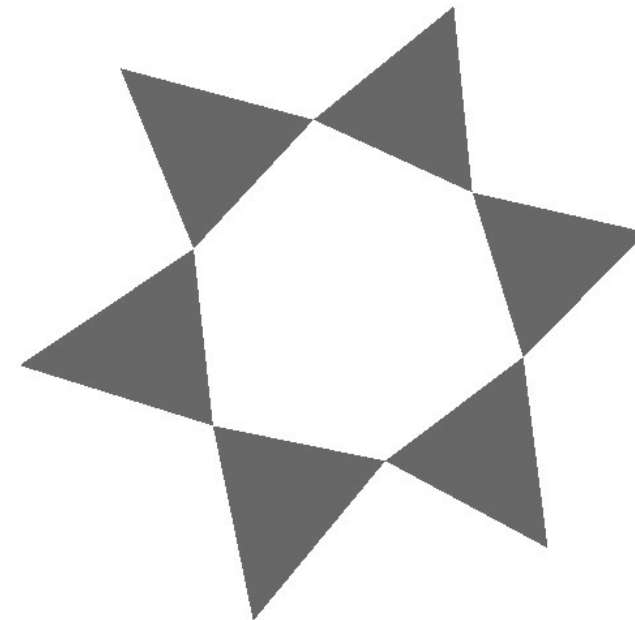
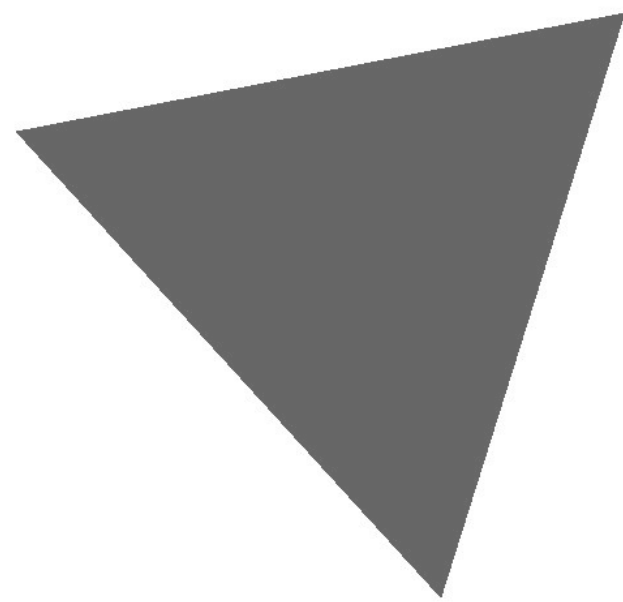






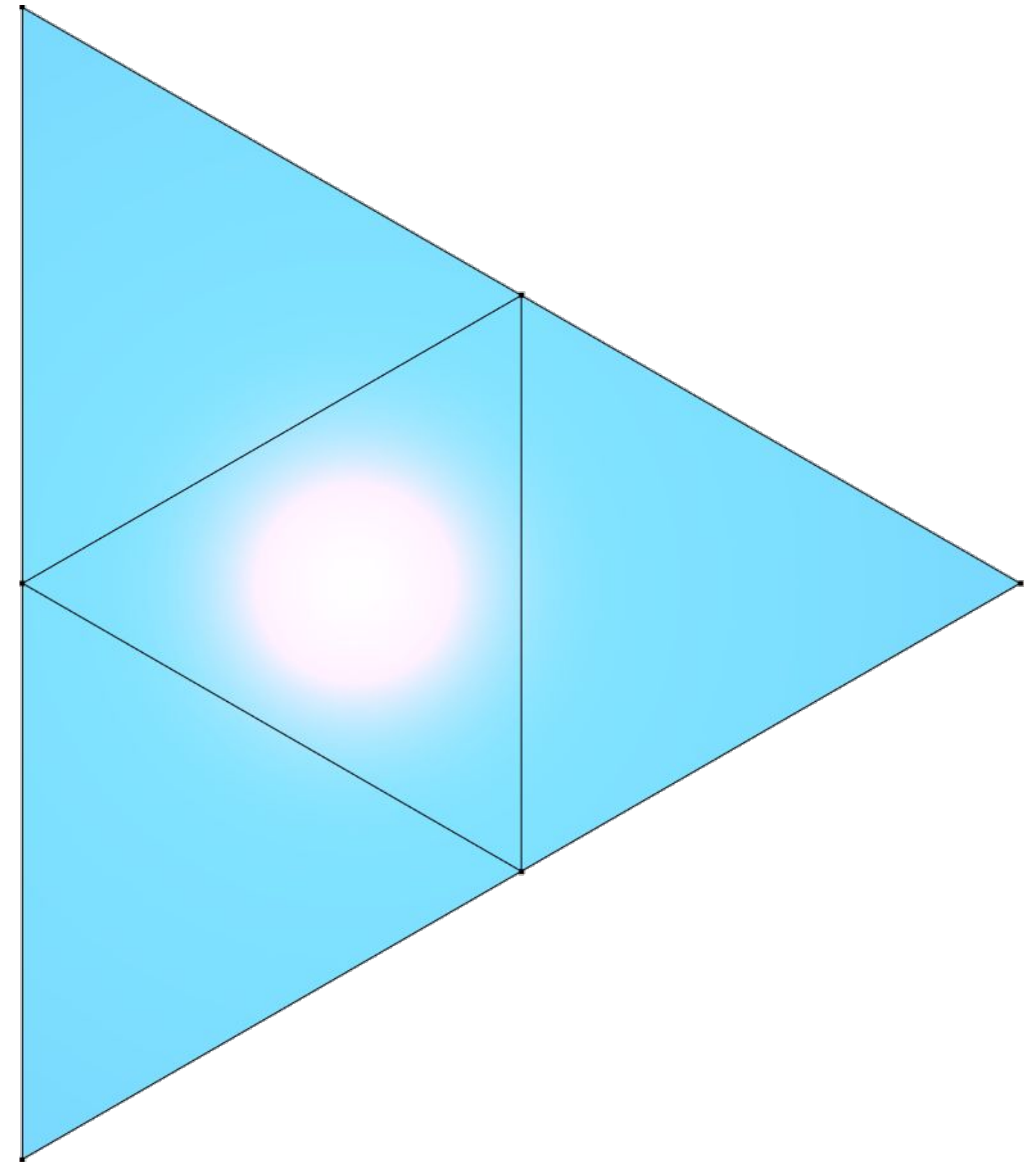
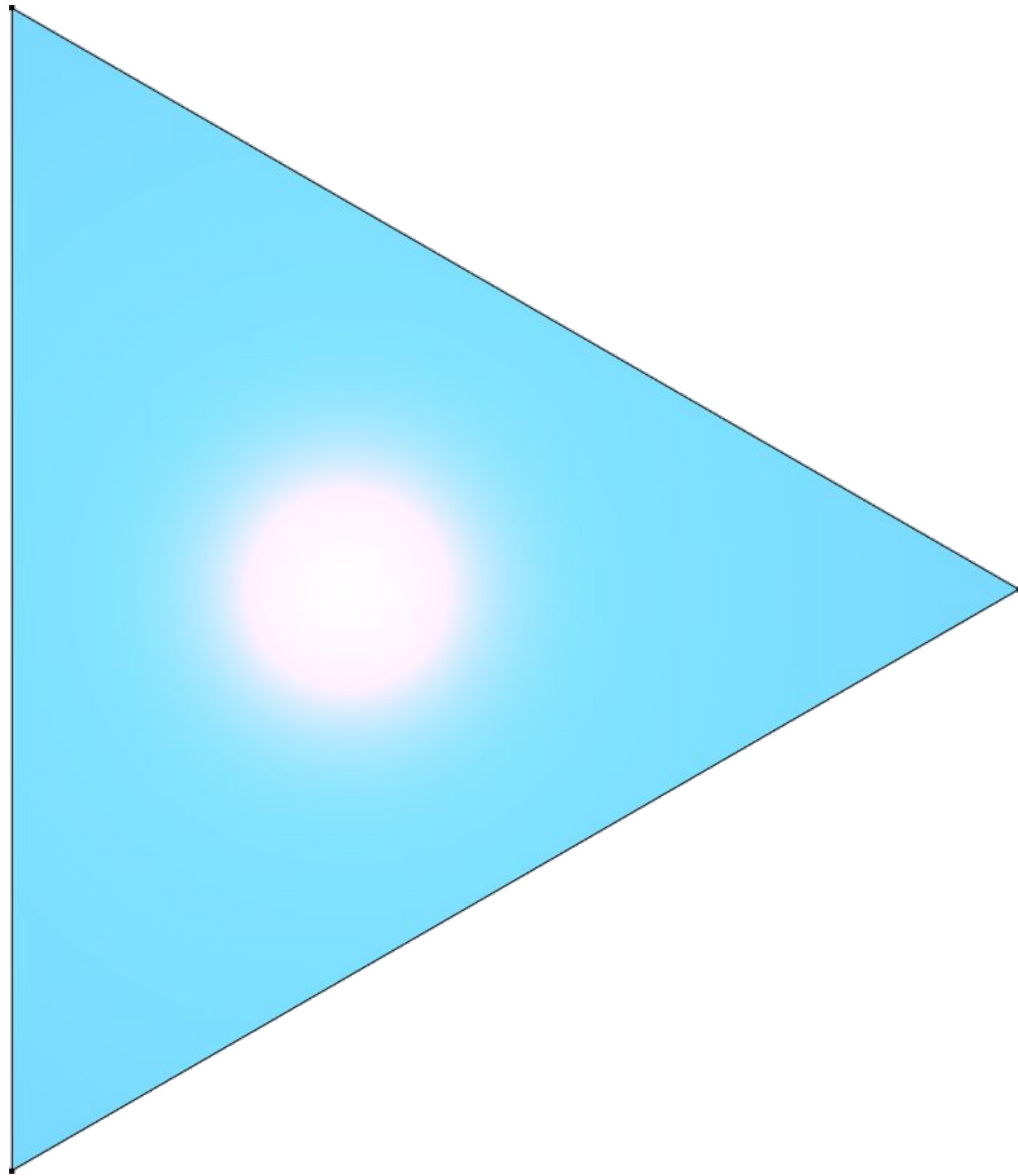


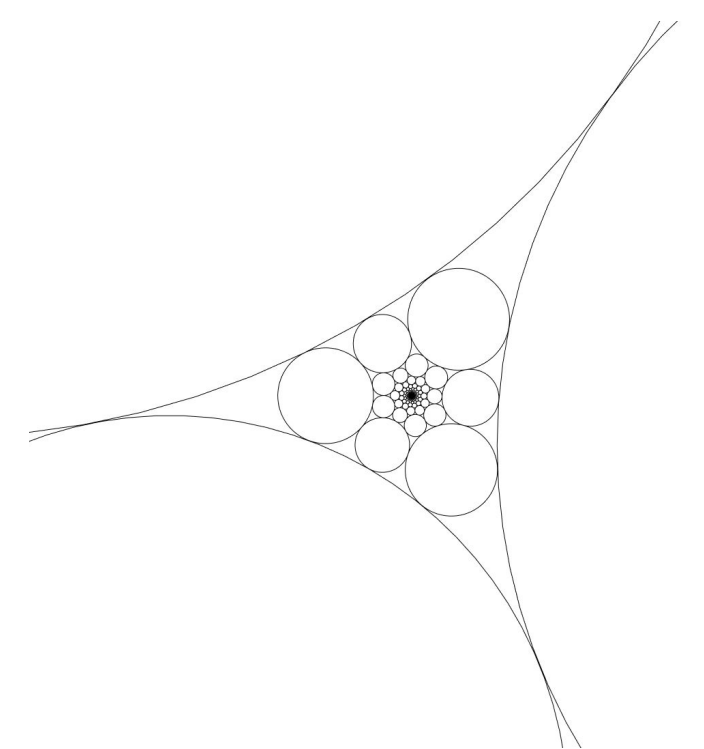
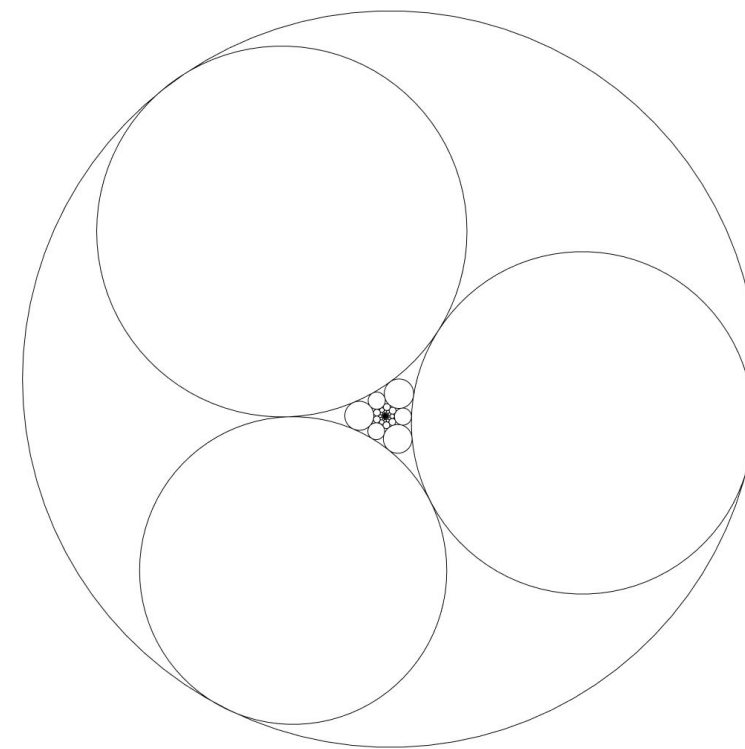
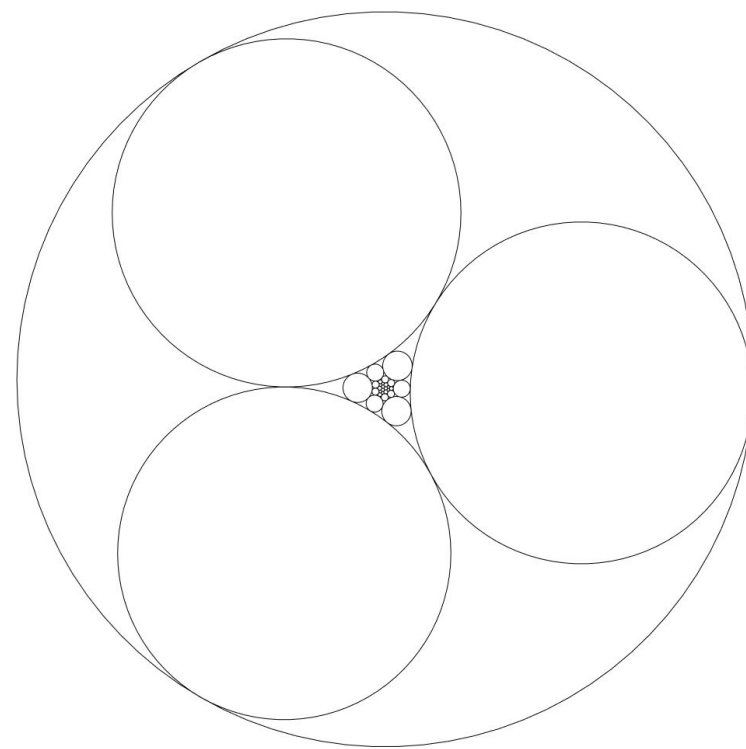
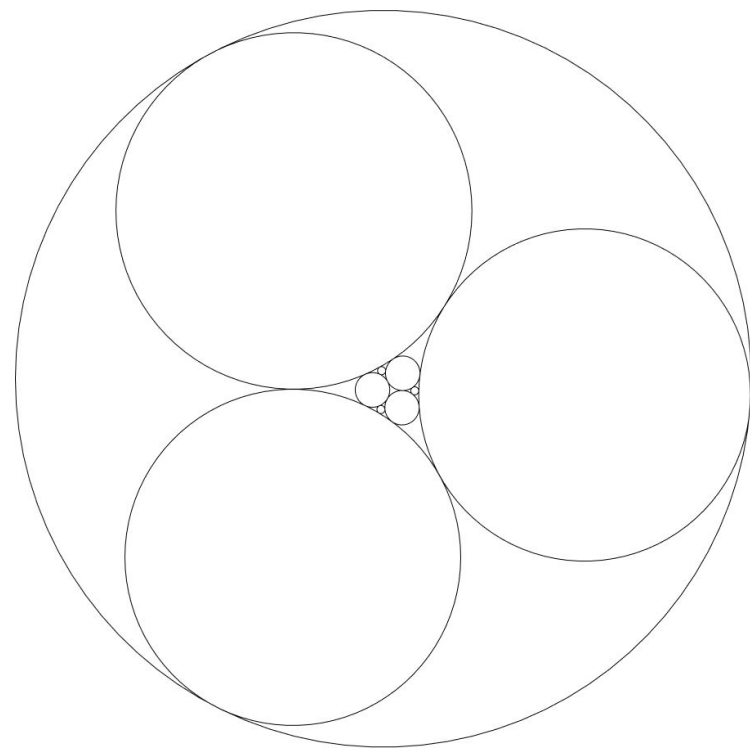
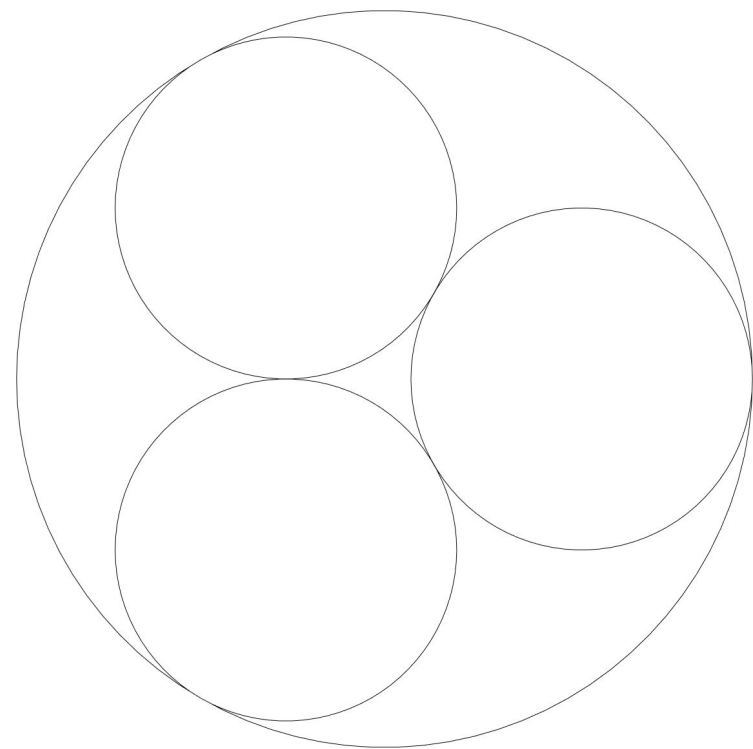
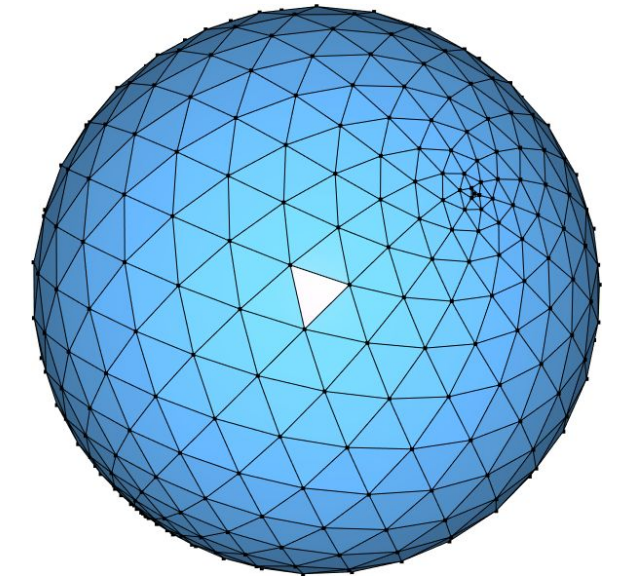
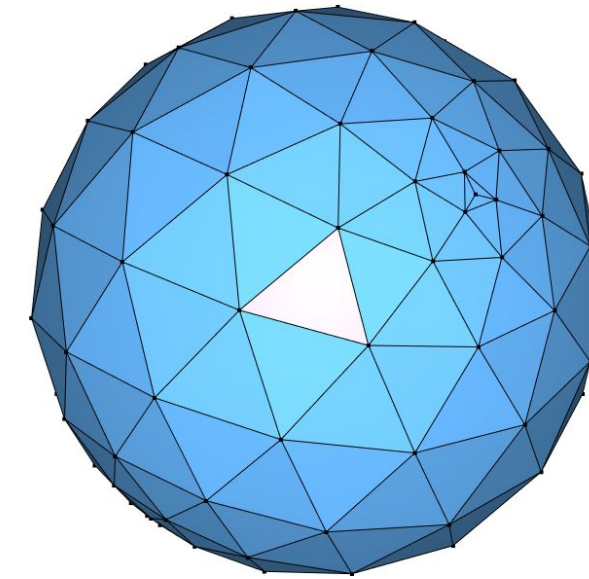
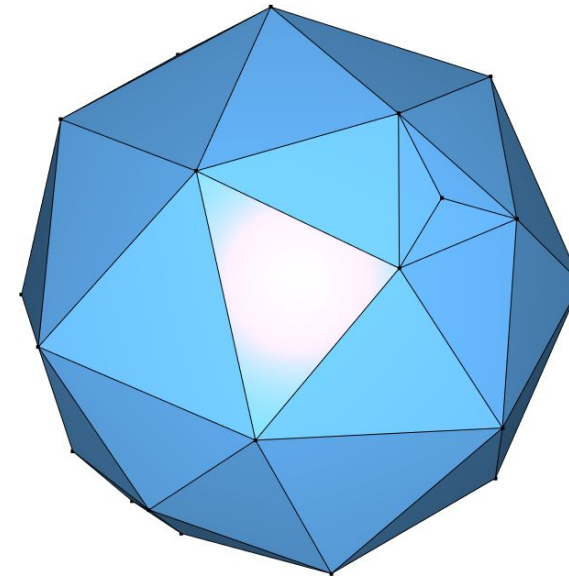
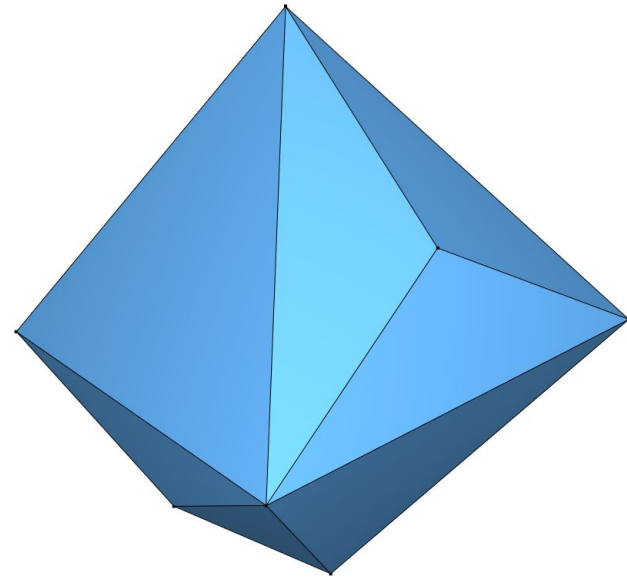
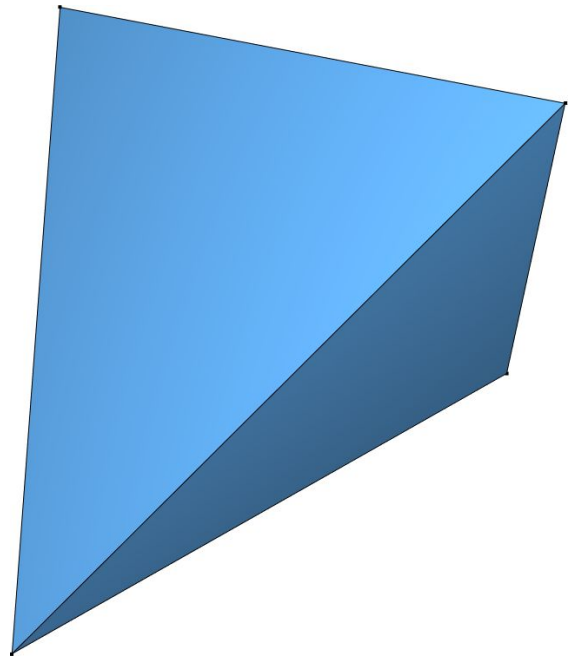


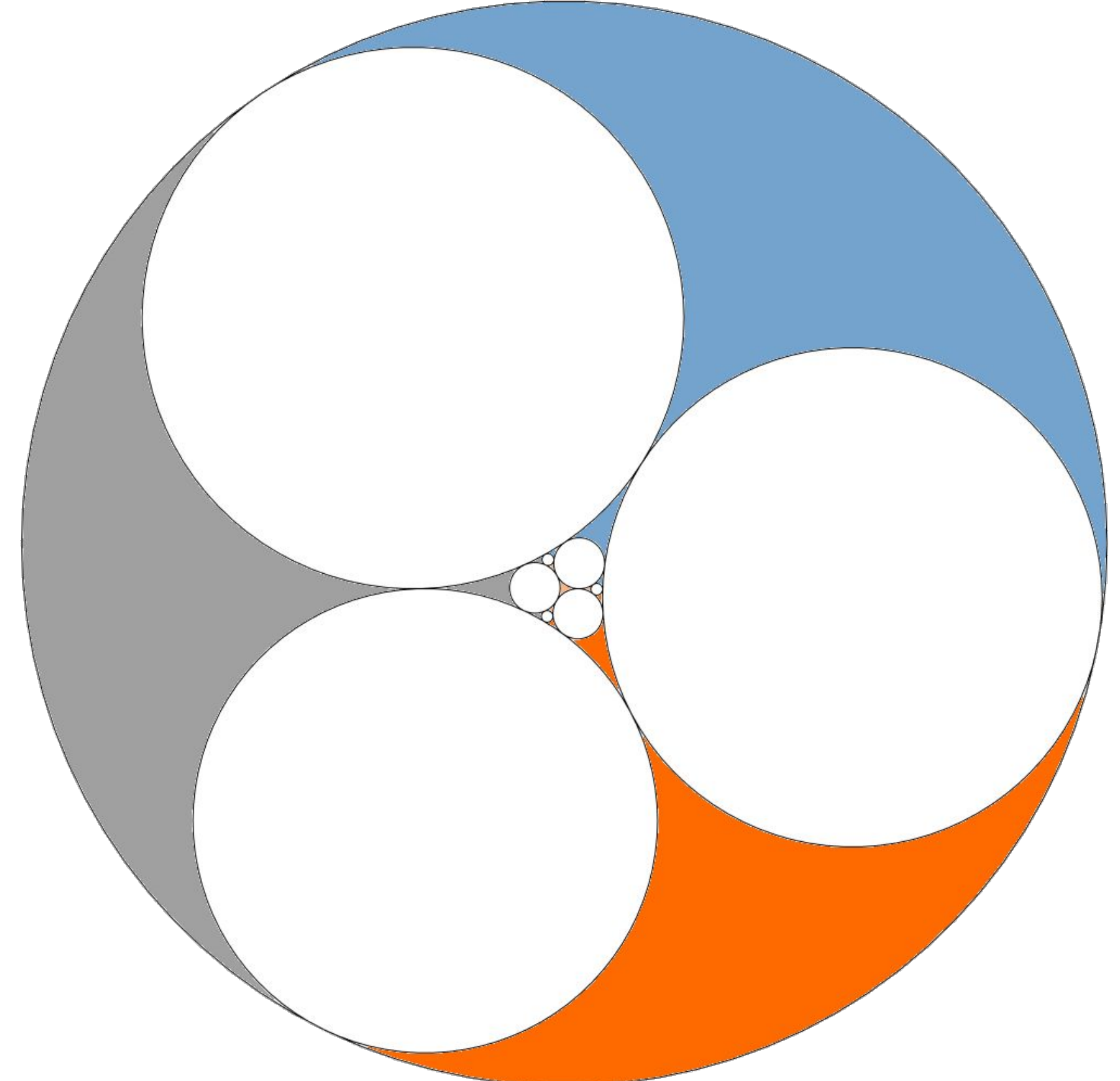
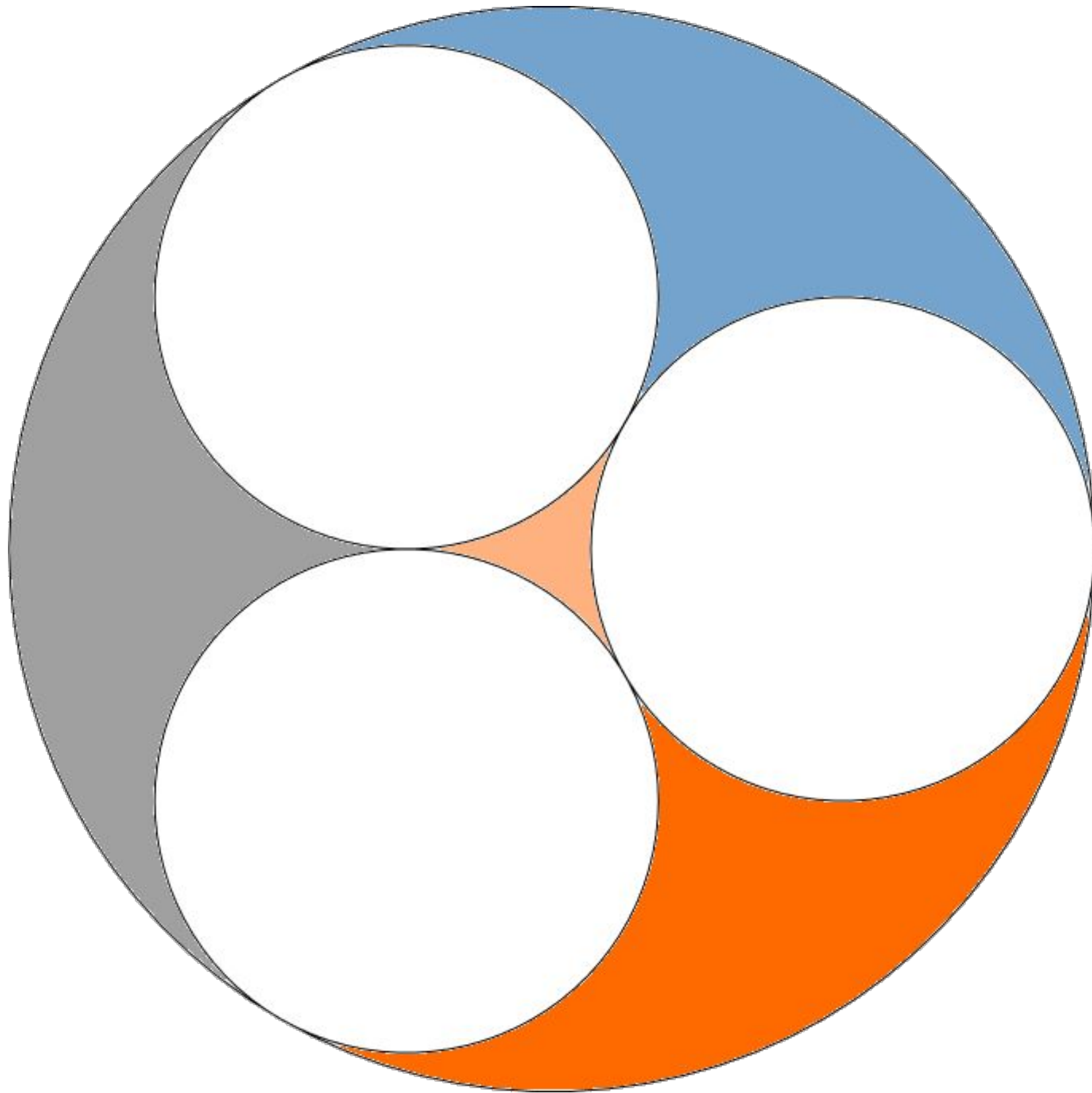


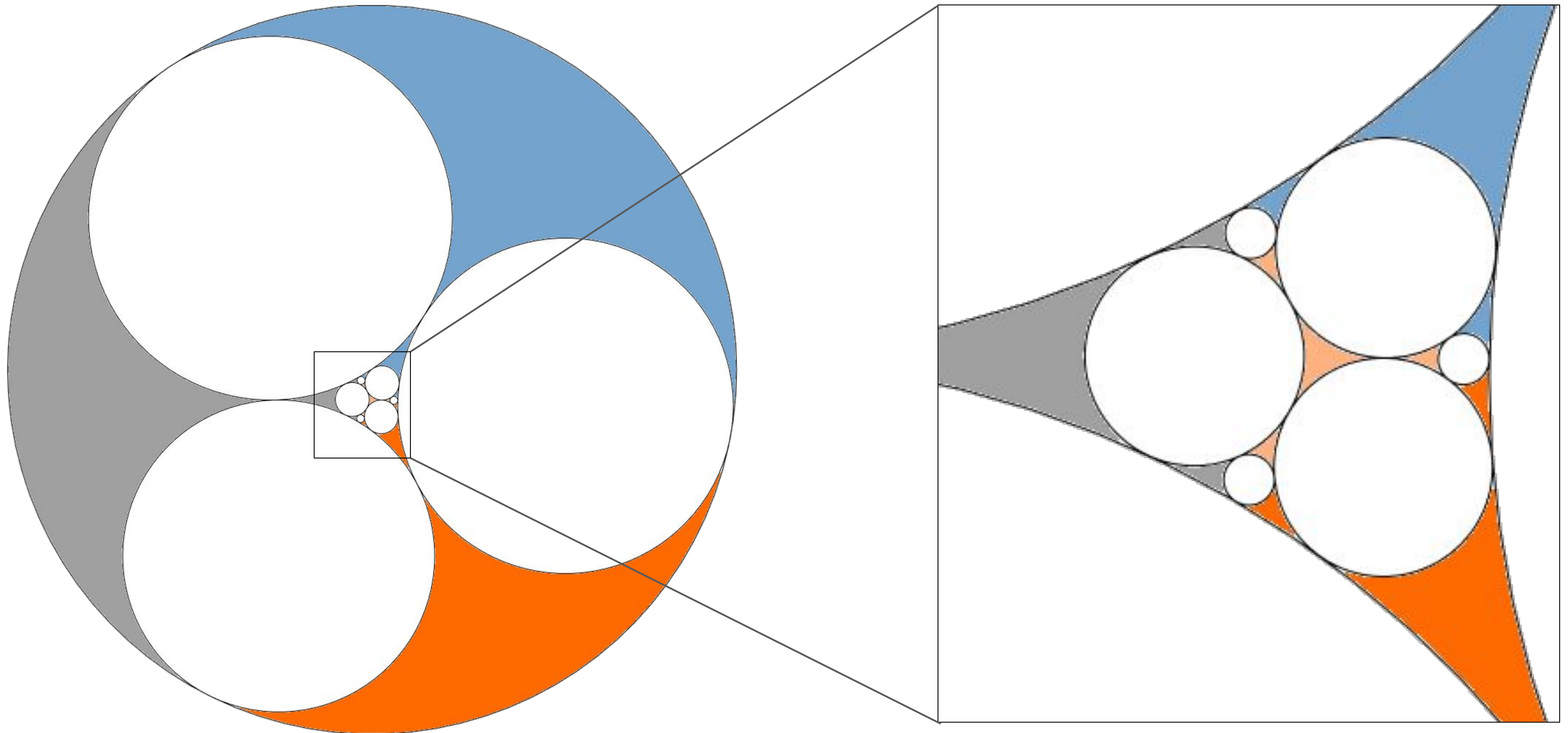
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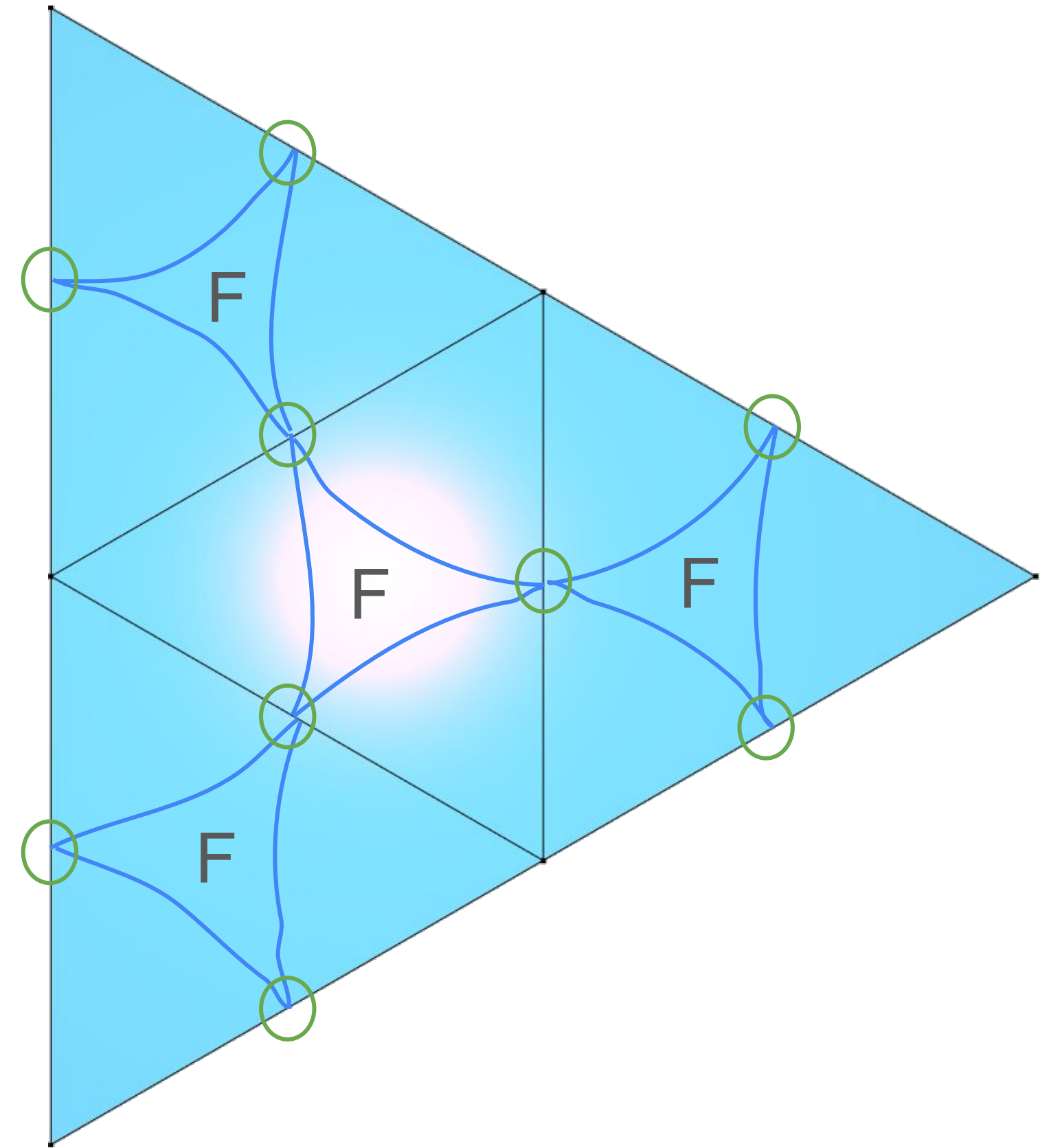
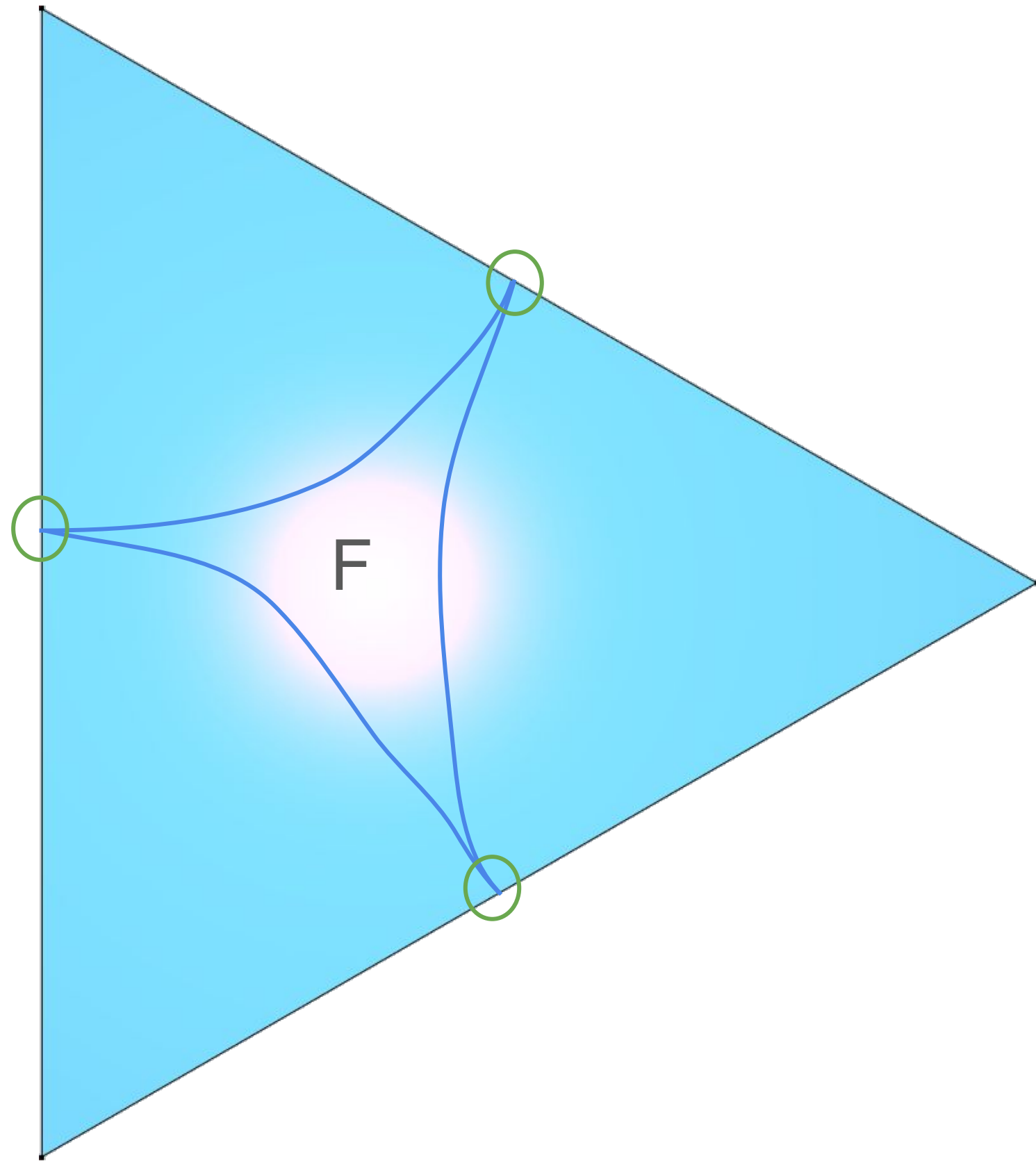


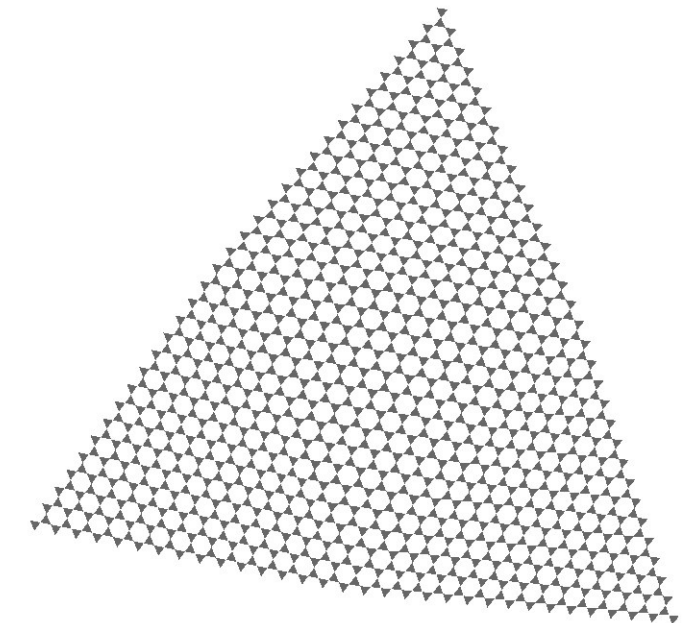
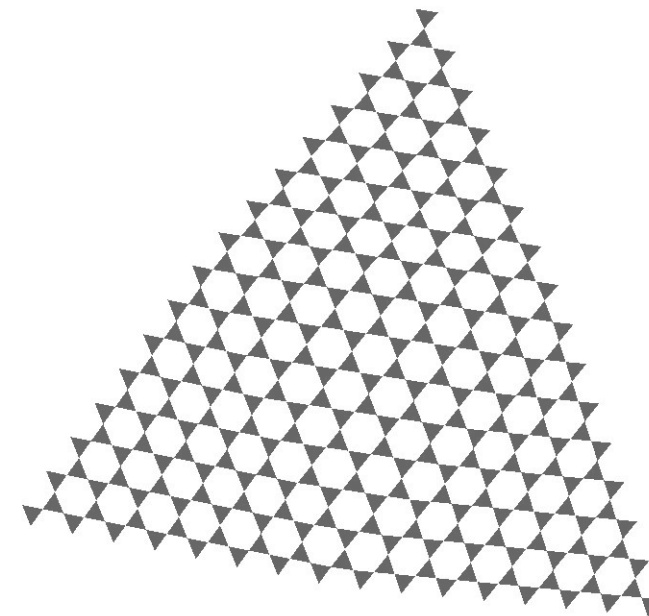
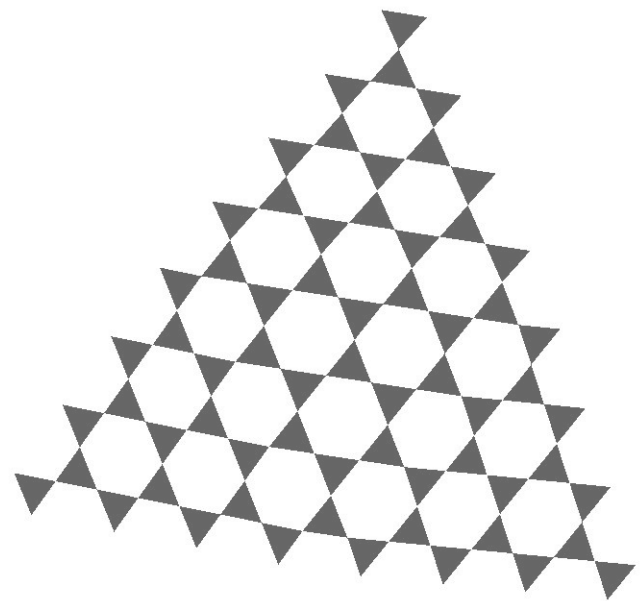
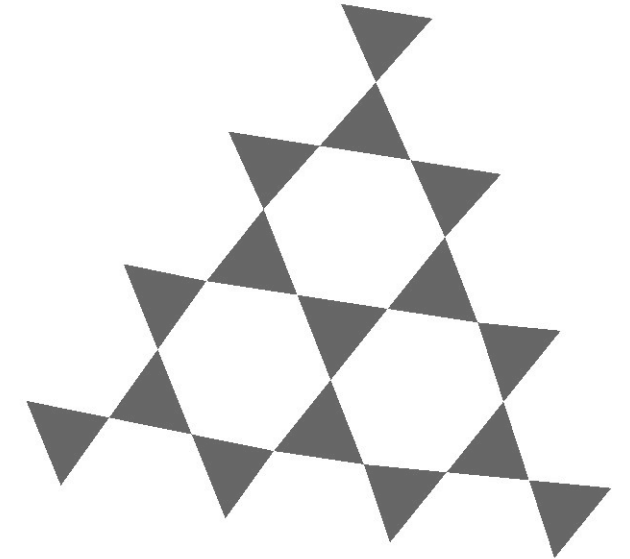
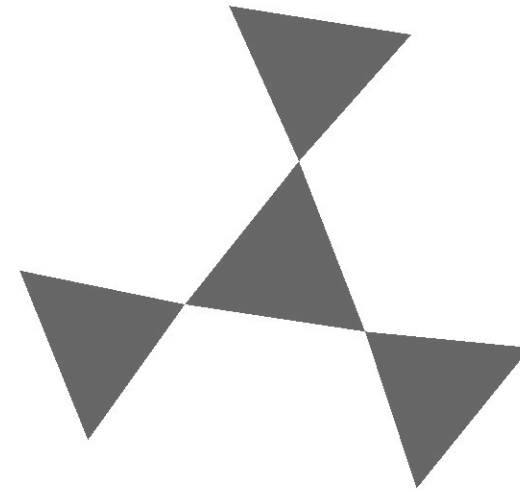
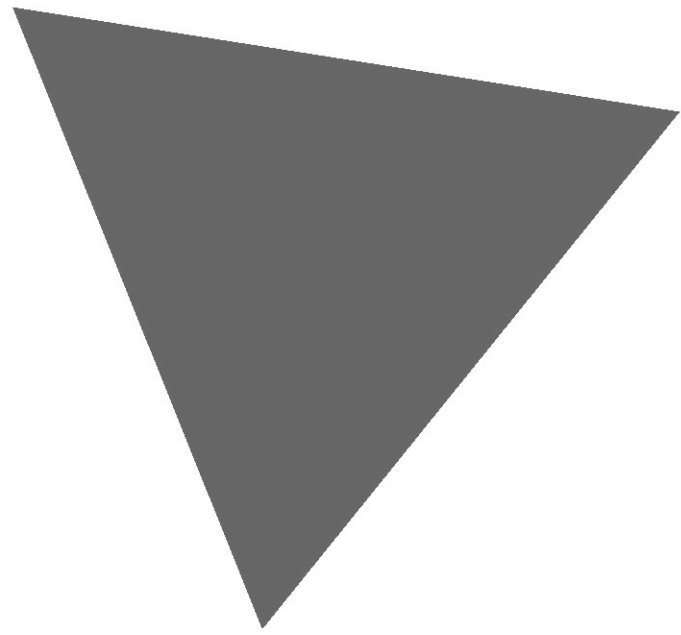








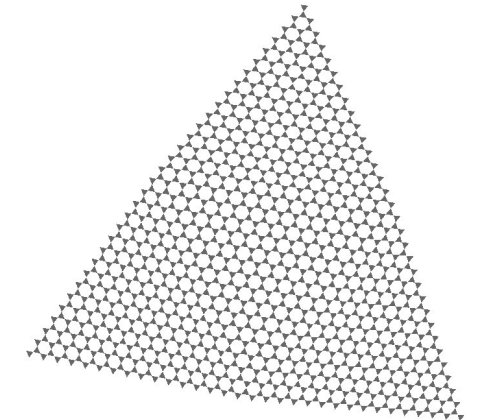
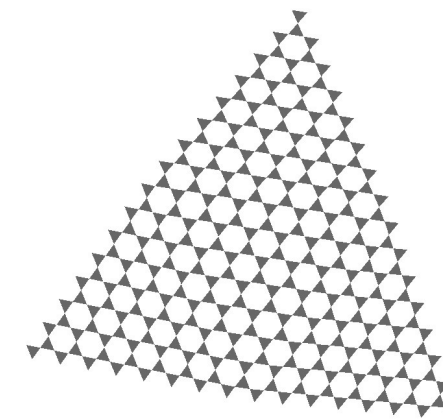
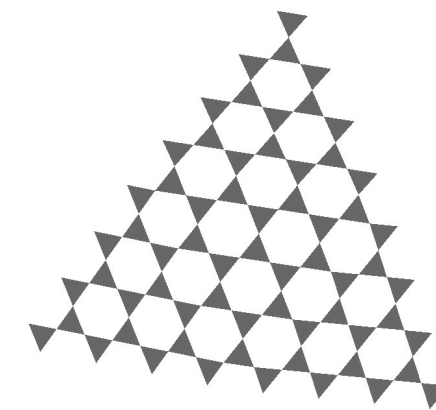
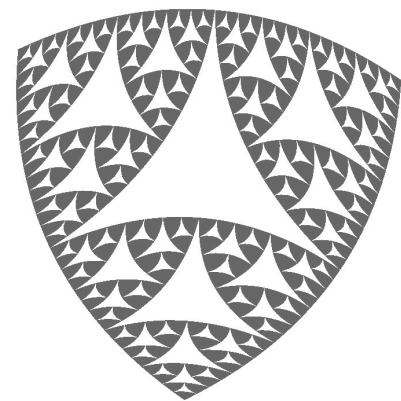
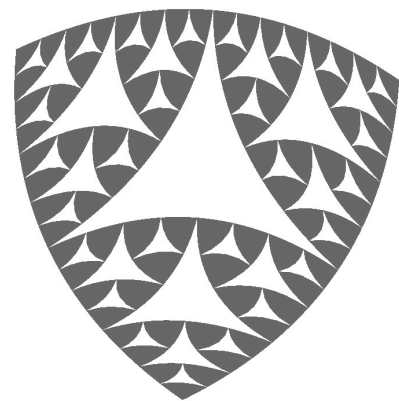
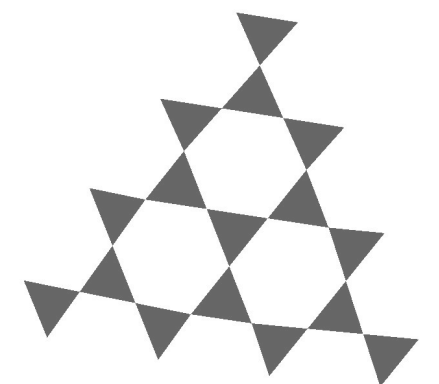
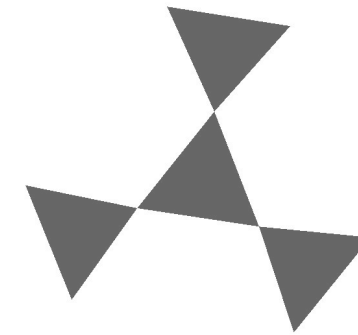
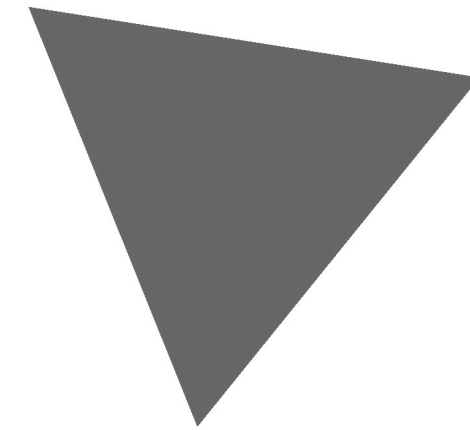
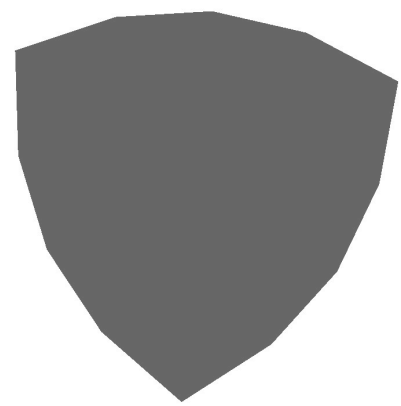




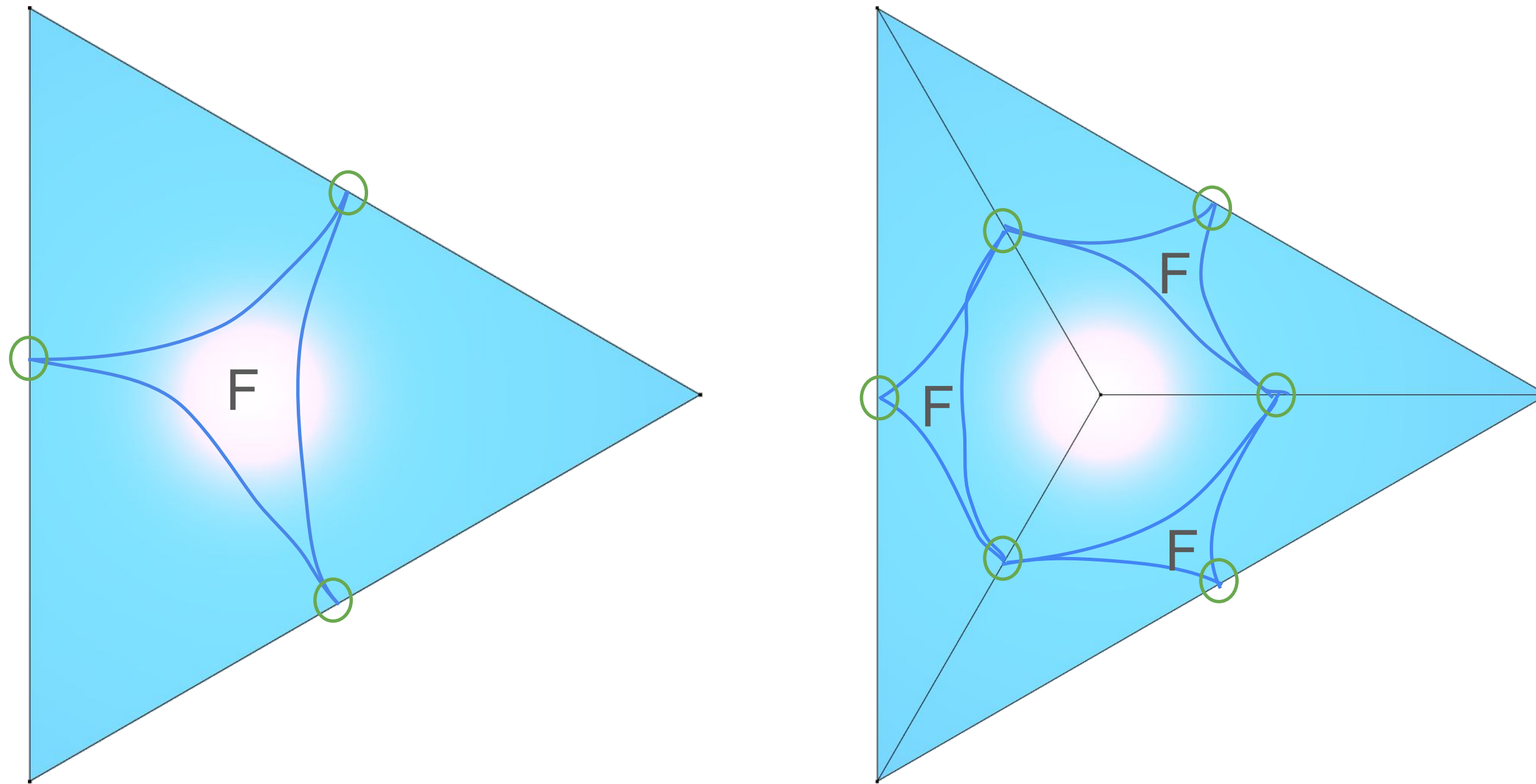
Étude du comportement des lacunes



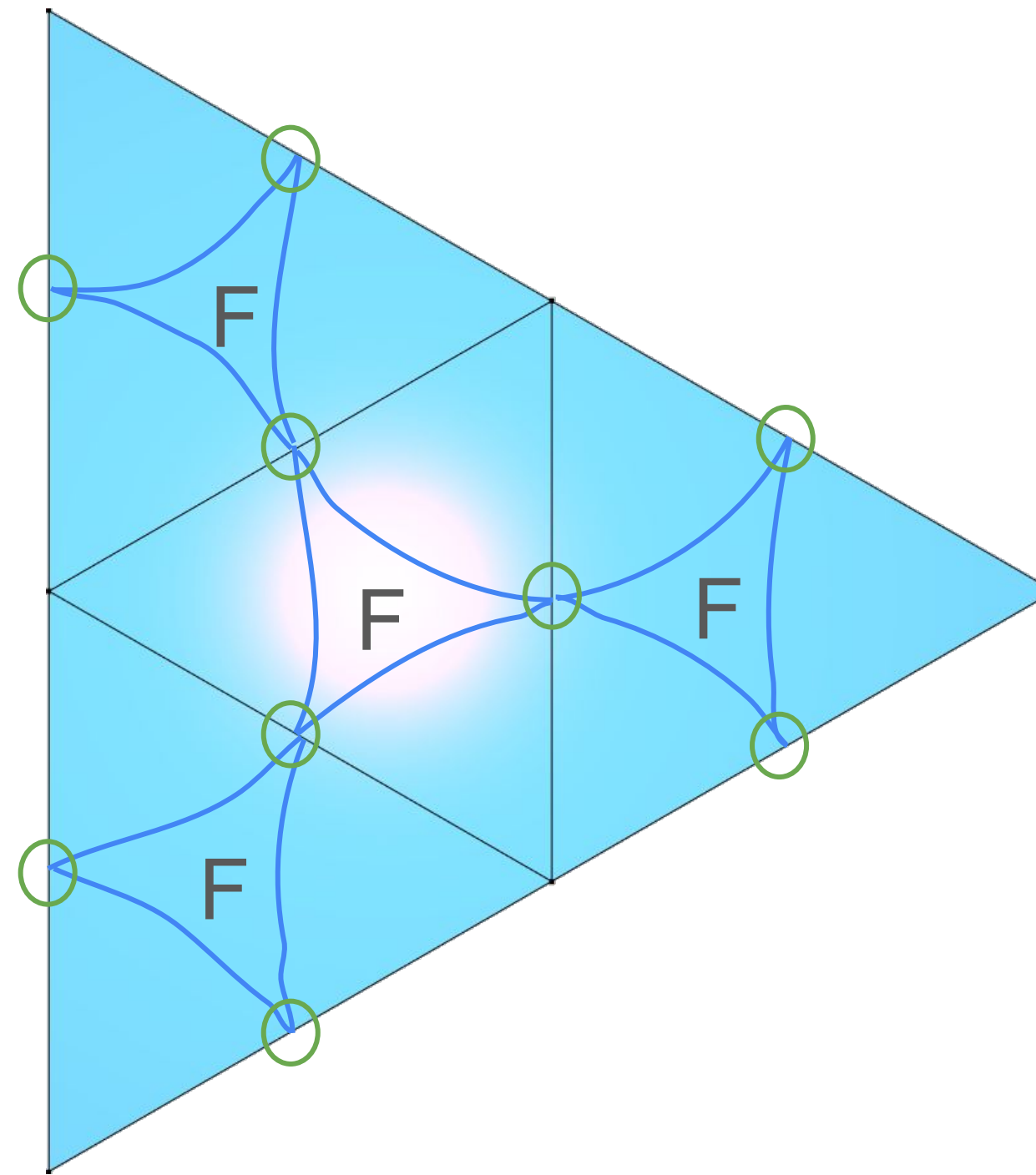
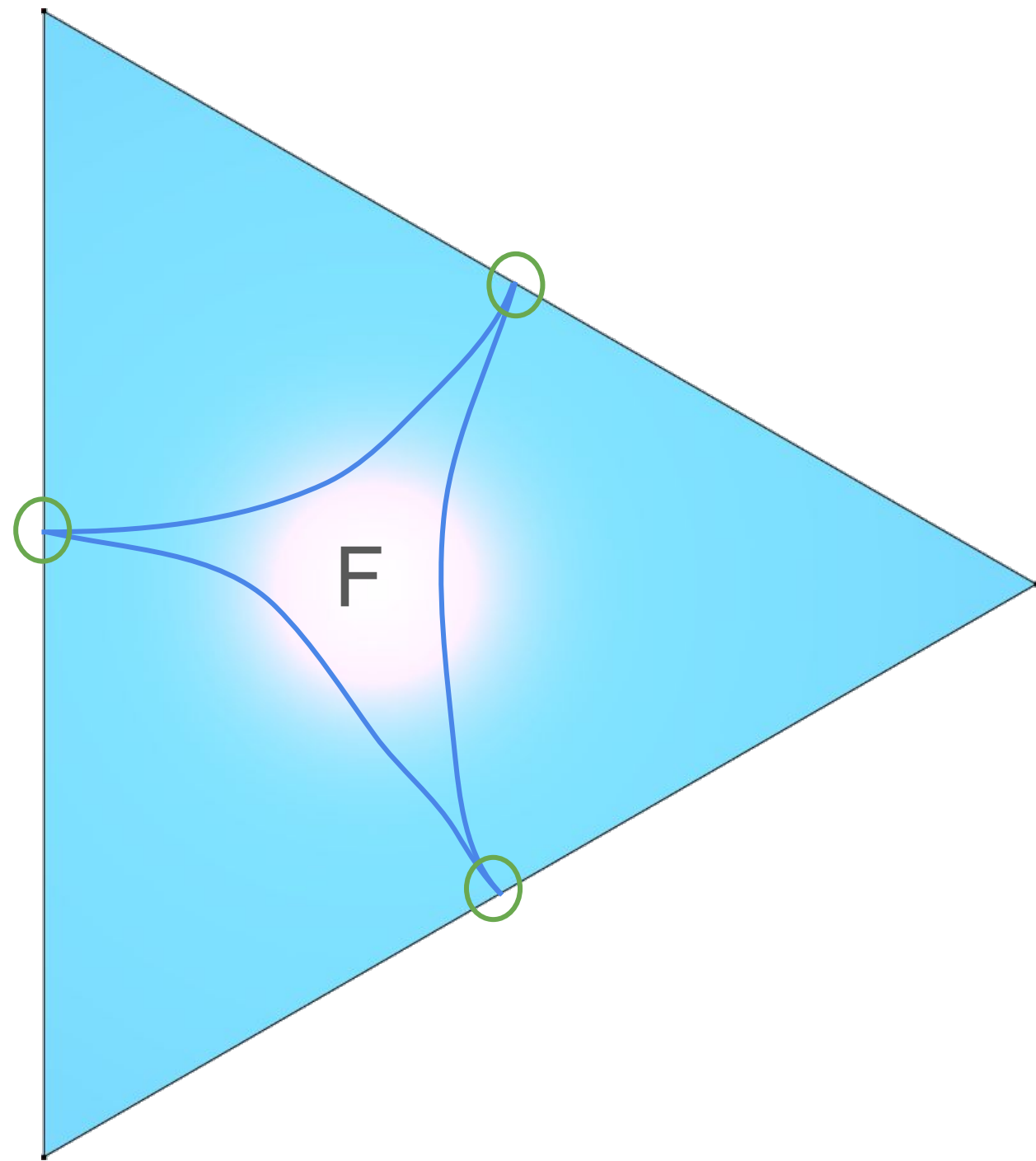
- Pourquoi les lacunes disparaissent dans certains cas ?



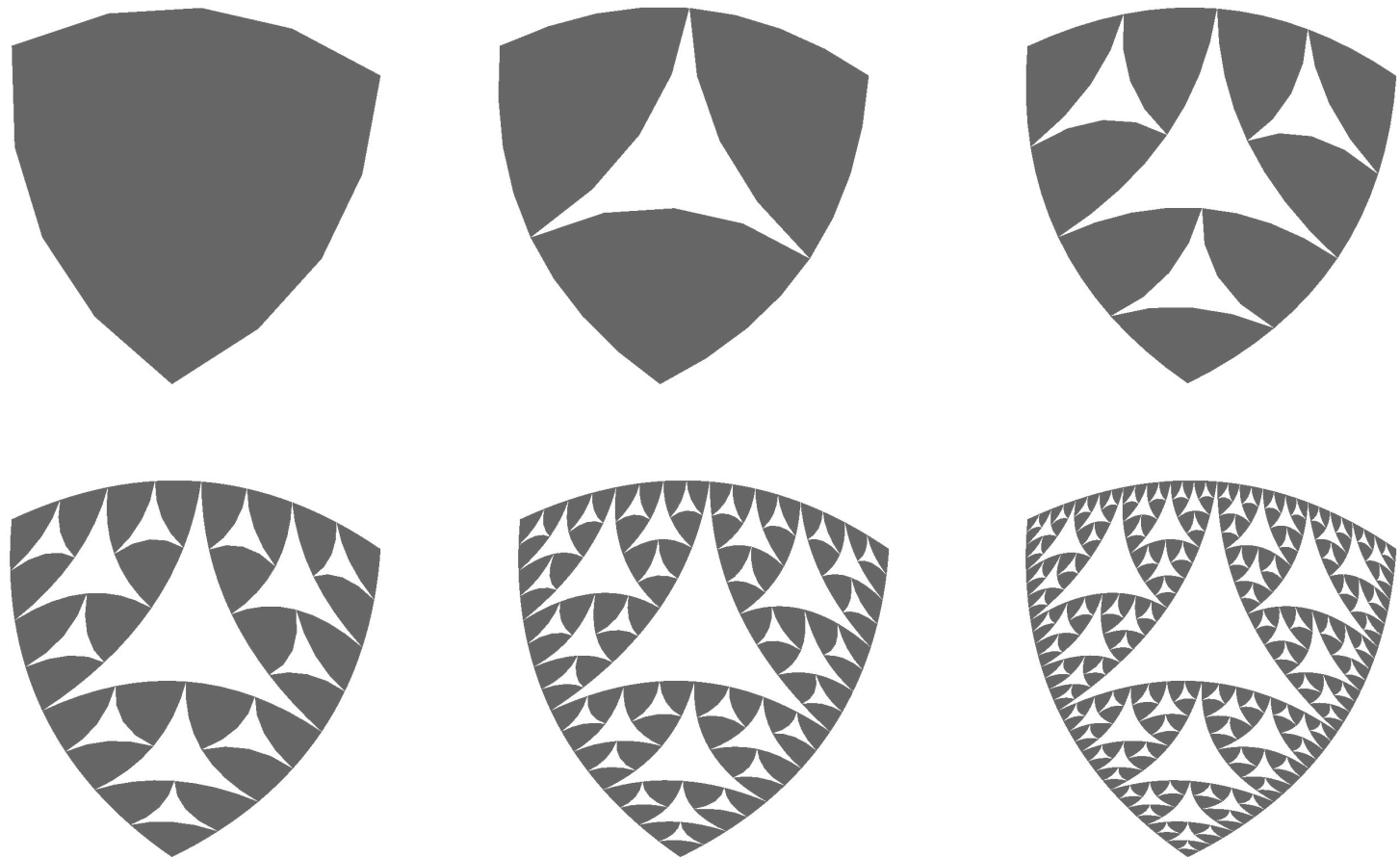
- Angle sectionné = courbe = conservation des lacunes



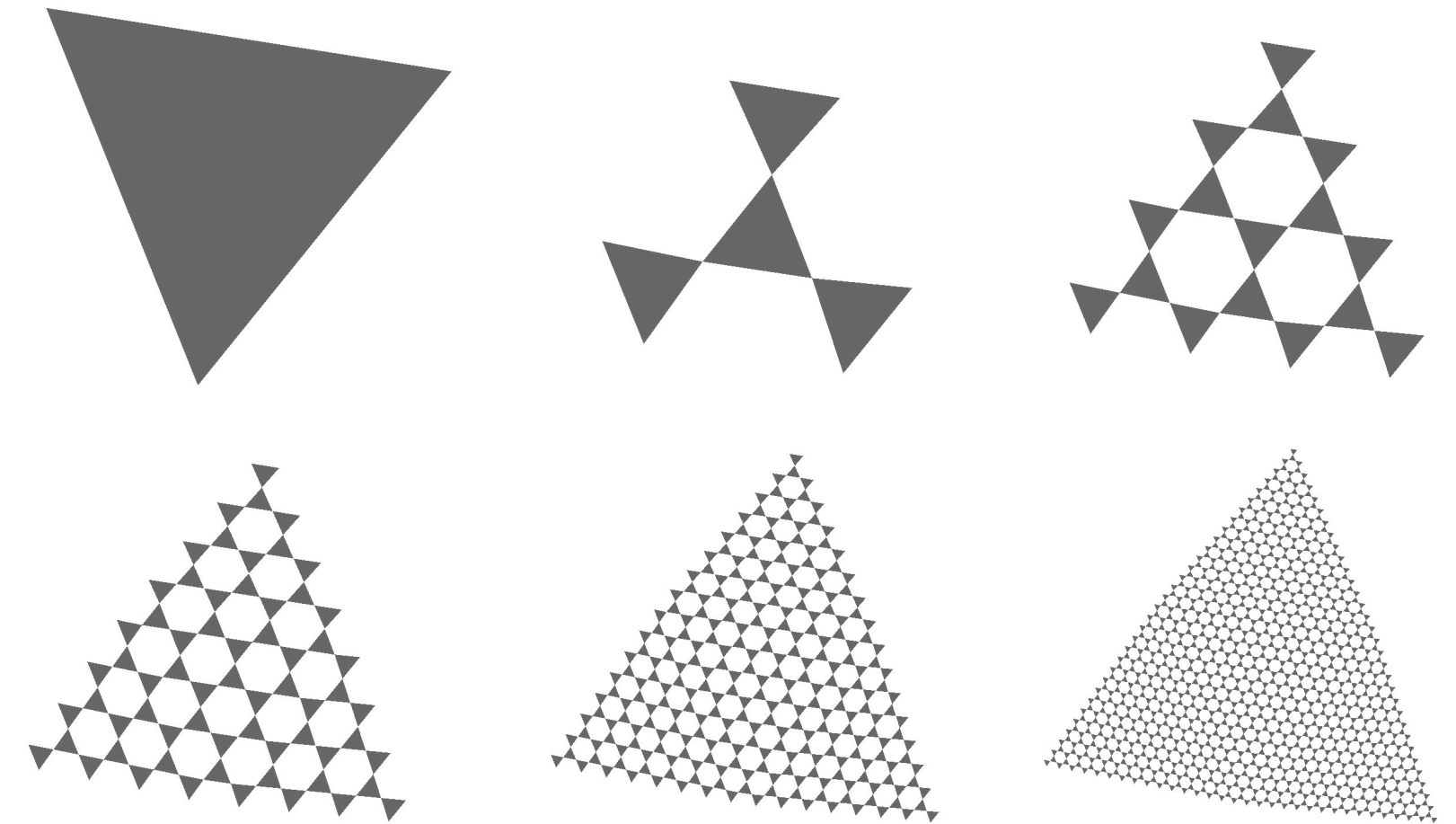
- Angle pas sectionné = point = disparition des lacunes



- Les lacunes dépendent de l'attracteur et de la primitive utilisée



Lacunes dans l'attracteur

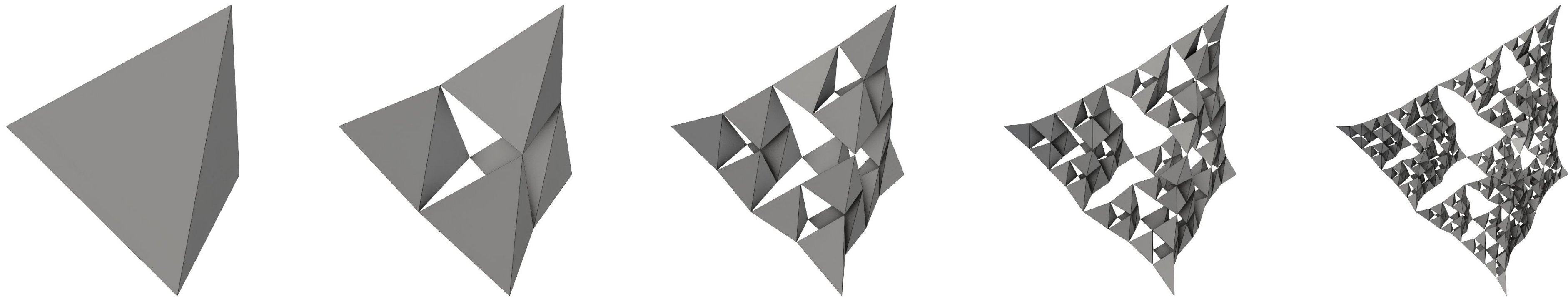
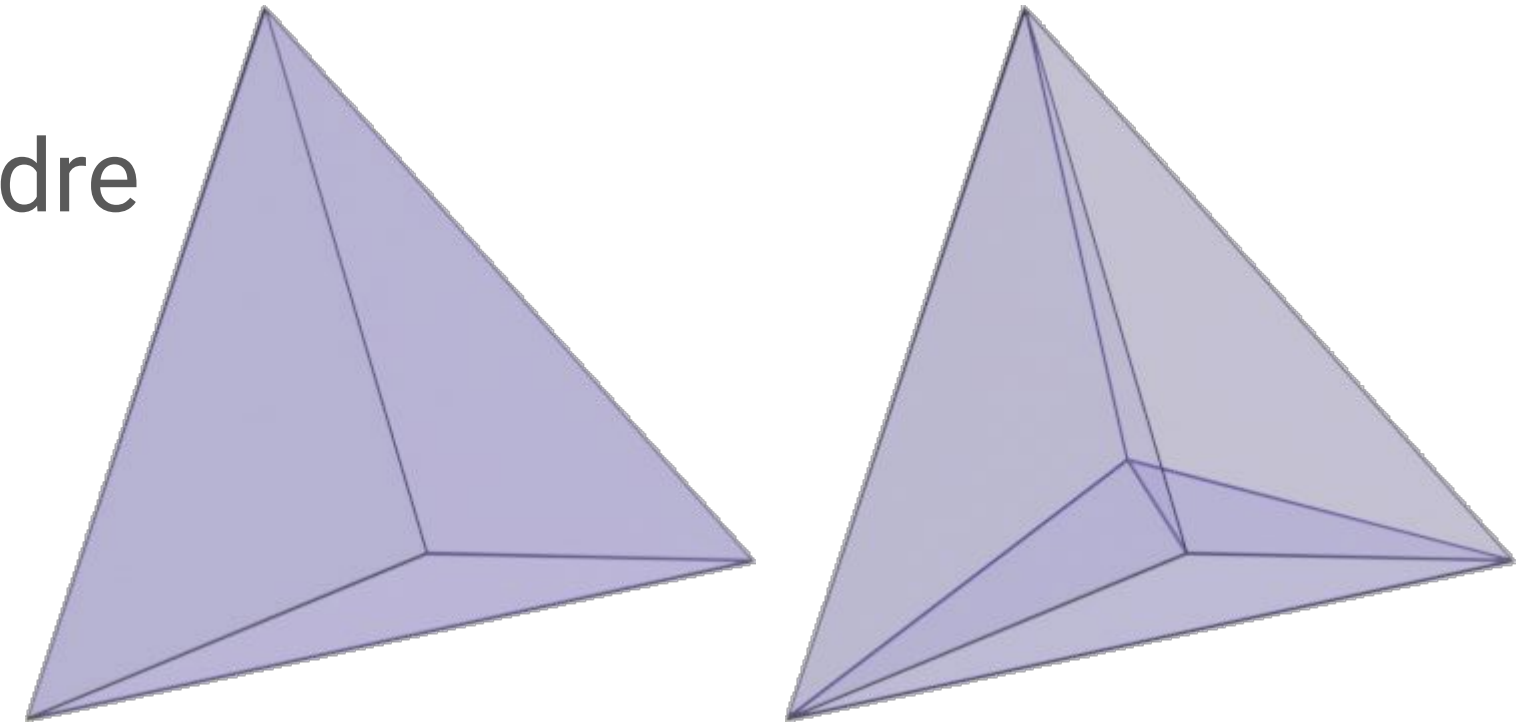


Lacunes de primitive

Et en 3D ?



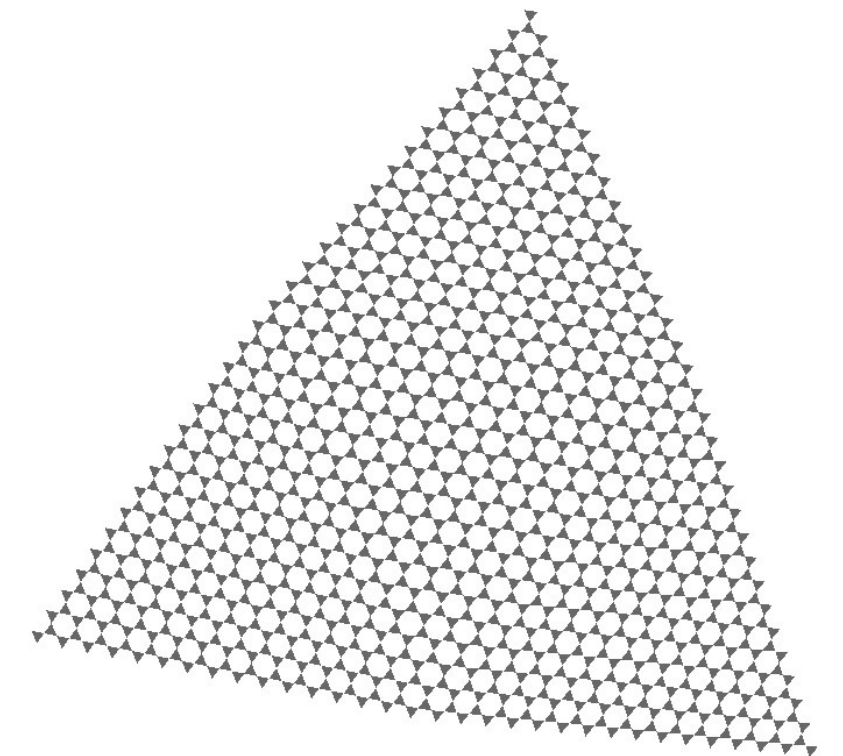
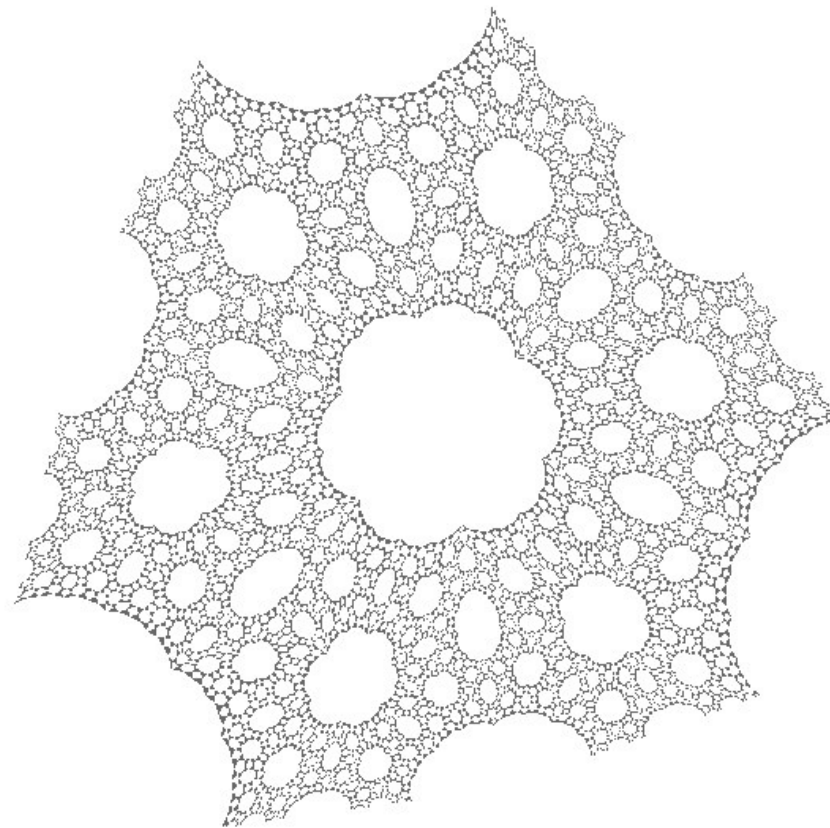
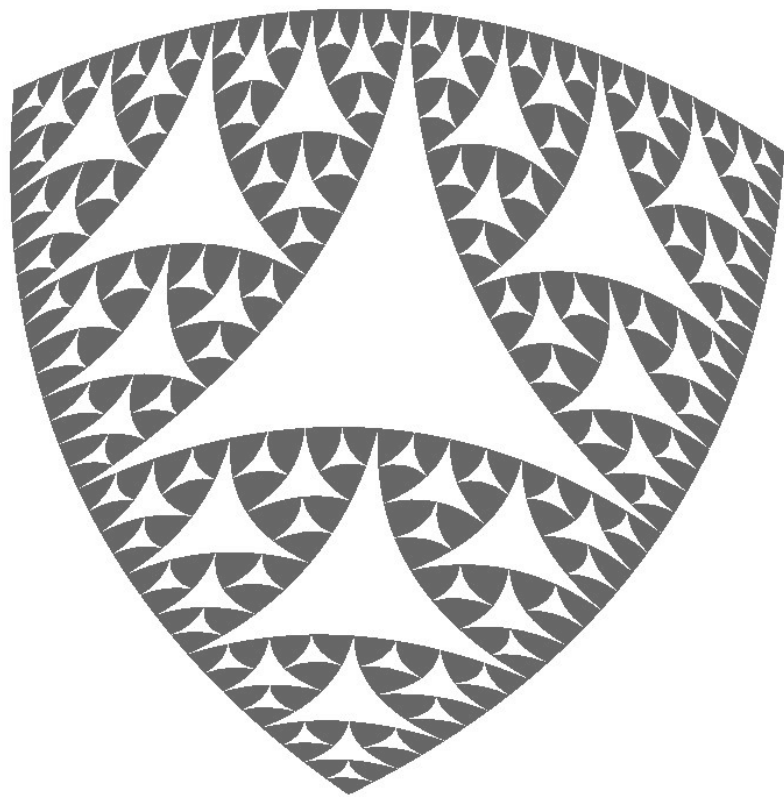
- Règle de subdivision du 4-polytope
- Étude sur une face du polytope = sur un polyèdre
 - Règle de subdivision d'un volume
- Codage des adjacences uniquement
 - Sur les faces en commun de 2 volumes



Conclusion



- Étude des empilements de sphères
 - Subdivision d'Apollonius, Barycentrique, Loop
- Procédé de modélisation avec un BC-IFS
 - Basé sur le polyèdre et une règle de subdivision



■ Automatisation de conception

■ Input

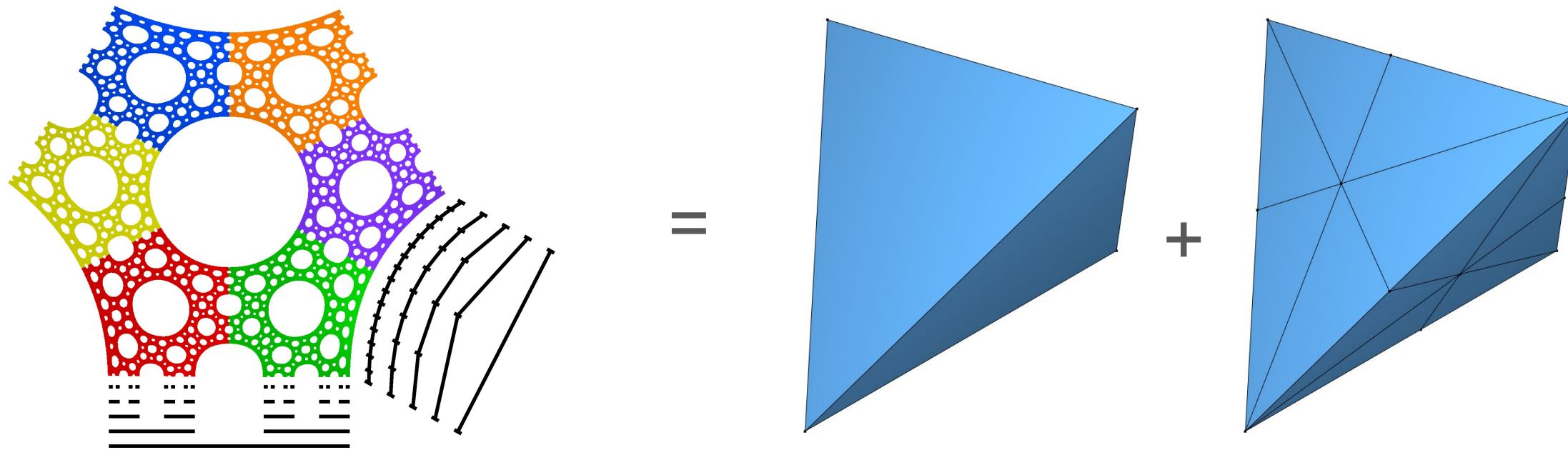
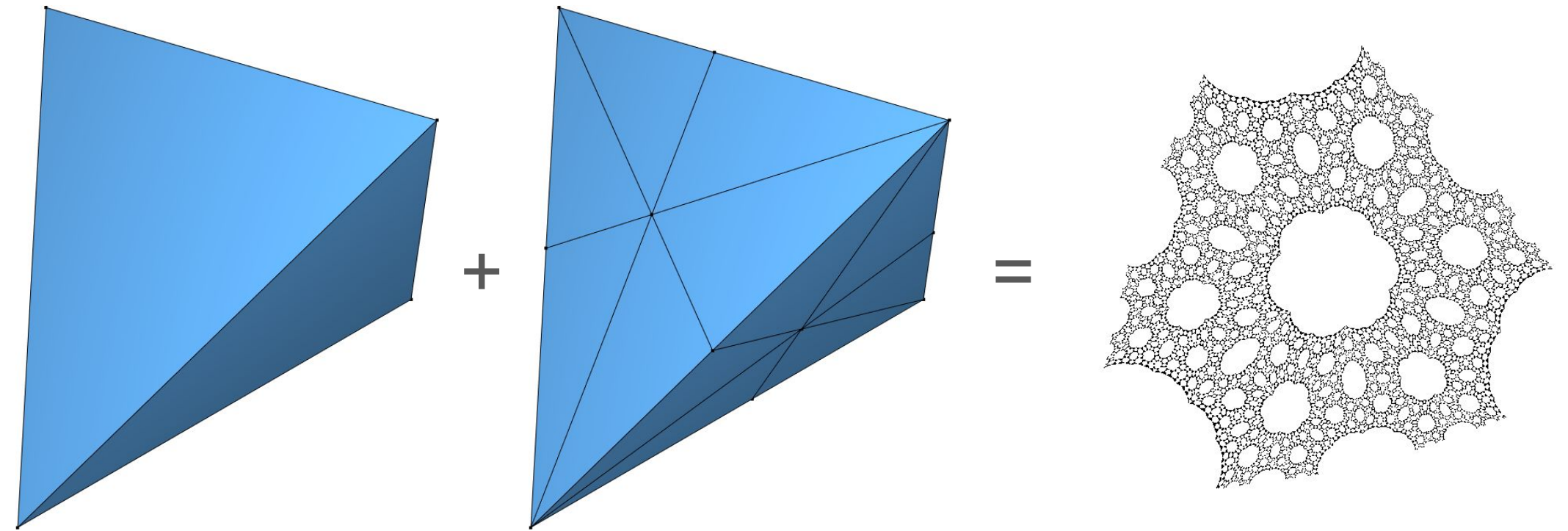
■ Polyèdre

■ Règles de subdivision

■ Output

■ BC-IFS

■ Problème inverse ?



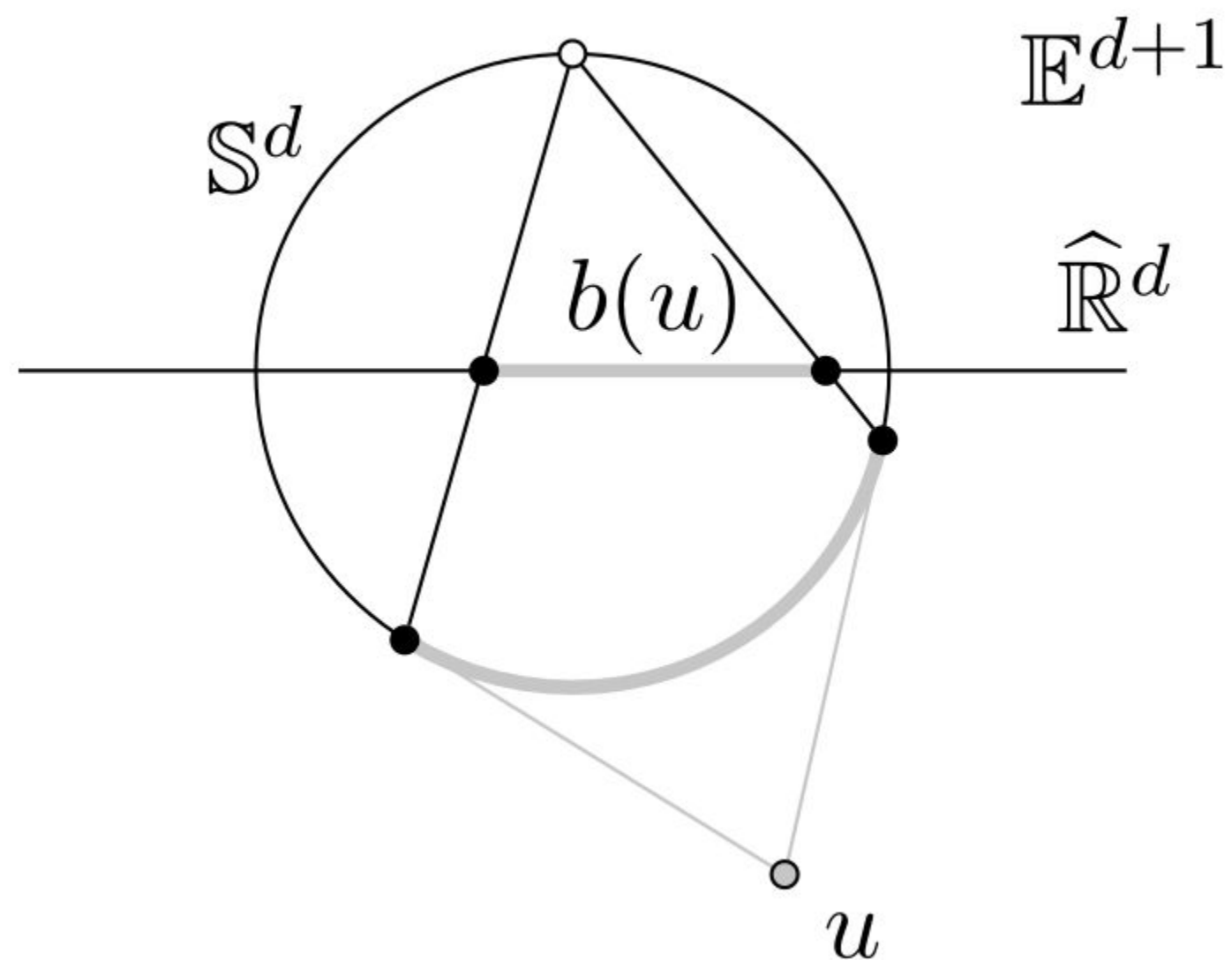
Merci de votre attention



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- Sokolov, D, Gouaty, G, Gentil, C, Mishkinis, A. Boundary controlled iterated function systems. In: Curves and Surfaces: 8th International Conference, Paris, France, June 12-18, 2014, Revised Selected Papers 8.3 Springer; 2015, p. 414–432.
- Rasskin, I. (2021). A polytopal approach to Apollonian packings and discrete knotted structures. (Doctoral dissertation, Université de Montpellier).
- Gentil, C., Gouaty, G., & Sokolov, D. (2020). Modélisation géométrique de formes fractales pour la CAO. (Vol. 3) ISTE Group.

En cas de besoins





$$\mathbf{i}(b(u)) = \frac{1}{\sqrt{\|u\|^2 - 1}} \begin{pmatrix} u \\ 1 \end{pmatrix}$$

$$u = \begin{pmatrix} e_0 \\ \overline{e_n} \\ e_1 \\ \overline{e_n} \\ \vdots \\ e_{n-1} \\ \overline{e_n} \end{pmatrix}$$

Rasskin, I. (2021). A polytopal approach to Apollonian packings and discrete knotted structures. (Doctoral dissertation, Université de Montpellier).

■ IFS

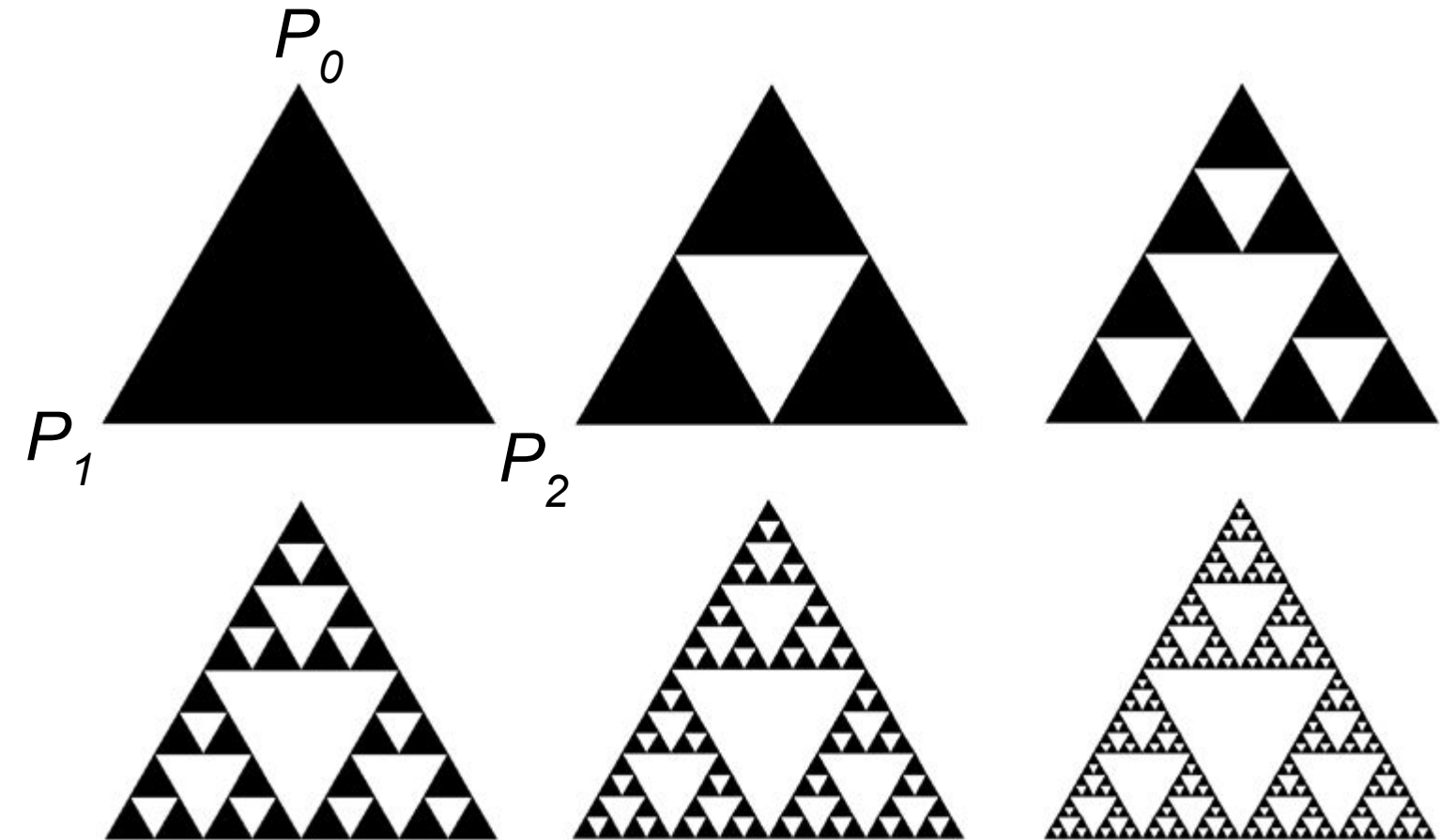
- Transformations contractantes
- Points de contrôles

$$T_0 = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 1 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$

$$T_1 = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 1 \end{pmatrix}$$

$$T_2 = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$

$$S = T_0(S) \cup T_1(S) \cup T_2(S)$$



■ IFS

- Transformations contractantes
- Points de contrôles

$$T_0 = \begin{pmatrix} \boxed{\frac{1}{2}} & 0 & 0 \\ \frac{1}{2} & 1 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$

$$T_1 = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 1 \end{pmatrix}$$

$$T_2 = \begin{pmatrix} 1 & \boxed{\frac{1}{2}} & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$

$$S = T_0(S) \cup T_1(S) \cup T_2(S)$$

